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Motivation, Objectives and Structure of the Thesis

The cost of computing is increasingly the cost of *moving data*, not the cost of arithmetic. In a conventional computer, the processor and the memory are physically separate, and every operation requires operands to be fetched across the channel that joins them. On modern hardware, a single off-chip memory access can cost three to four orders of magnitude more energy than the arithmetic it feeds [1], and the workloads that now dominate—large machine-learning models and continuous streams from physiological and environmental sensors—are precisely those that demand enormous numbers of such accesses [2]. As classical transistor scaling no longer hides this imbalance, the energy and latency of data movement have become the limiting factor of digital computing [3]. This thesis is motivated by that limit, and by one of the physical strategies proposed to relax it: building computation out of devices whose own internal dynamics carry and transform information, so that signalling, memory, and processing share a single physical substrate, as they do in the biological synapse [4, 5].

The device idea pursued here is the *volatile organic memristive device*: a simple two-terminal element, made from an ion-containing polymer composite, whose conductance retains a decaying record of its recent electrical history. Rather than treating this volatility as a defect to be

engineered away (as it would be for a non-volatile memory cell), the thesis treats the fading-memory timescale as the functional resource. A device that forgets at a controlled rate is a physical *leaky integrator*, and a collection of such devices with different timescales is a substrate for *temporal* (reservoir-style) computing on time-varying signals. The central claim of the work is that the chemistry of the polymer composite—above all its composition—sets that timescale, and that the resulting dynamics can be put to computational use.

The thesis builds this argument on four levels of evidence:

- **A fully characterised exemplar (Chapter 2).** A single proof-of-concept Super Yellow/Hybrane/lithium-triflate device is characterised with the complete synaptic measurement suite—current–voltage hysteresis, potentiation and depression, excitatory post-synaptic current, separated short- and long-term retention, and spike-timing-dependent plasticity—and given a mechanistic interpretation in terms of cation–anion mobility asymmetry.
- **A campaign-scale reproducibility and provenance layer (Chapter 3).** The subsequent Hybrane reproducibility collapse is diagnosed across the fabrication archive, separating finished-device shelf stability from batch-scale supply reproducibility and motivating the switch to PEO. The same provenance record and feature pipeline then become the methodological basis for the quantitative chapters.
- **A replicated composition series with a sample-limited chemical survey (Chapter 4).** A poly(ethylene oxide)-based corpus is built under a common protocol. The composition series is *replicated*; host, anion, and cation substitutions are sample-limited and are not read as material laws. In particular, the $\text{Li}^+ > \text{Na}^+ > \text{K}^+$ cation ordering is treated as a hypothesis to be tested.

- **An in-silico temporal-computing demonstration from measured dynamics (Chapter 5).** Compact behavioural models extracted from the Chapter 4 measurements are used in reservoir-computing simulations, with heterogeneity claimed only where the simulations support it and negative controls (including the affective-label null on real physiological streams) reported explicitly.

Open Questions and Objectives

The work is organised around three groups of questions, each answered by a stage of the evidence hierarchy.

First, realising and understanding an exemplar. Can a fully reversible two-terminal organic memristive device be made from a solution-processed polymer-electrolyte composite using standard organic-electronic fabrication? Can one device support analogue switching, distinguishable short- and long-term memory, an excitatory post-synaptic current response, and spike-timing-dependent plasticity? Can the fast and slow memory components be explained by the composite’s cation–anion mobility asymmetry?

Second, composition as the quantitative control variable. Within a common host and a common alkali-metal triflate family, how do the device dynamics depend on composition when the polymer-to-semiconductor and salt-to-semiconductor mass fractions are varied around a reference formulation? This composition dependence is the axis the thesis sets out to *replicate*. Host, anion, and cation substitutions pose hypotheses about the same dynamics—in particular, whether moving the cation along the Li^+ , Na^+ , K^+ series shifts the measured fading-memory timescale as coordination chemistry suggests.

Third, utility for temporal computing. Can the volatile response of these devices be exploited for temporal computation? Can composition supply

task-matched fading-memory timescales, while sample-limited chemistry and device-to-device spread populate a heterogeneous bank? And can measured variability contribute to computation rather than merely degrade it?

These questions translate into six ordered objectives:

1. Design, fabricate, and characterise a proof-of-concept two-terminal organic memristive device based on a Super Yellow, Hybrane, and lithium-triflate composite, processed from a single solution onto a standard ITO/composite/Ag stack, and demonstrate the basic memristive and synaptic functionality in a single platform.
2. Provide a mechanistic account of the device’s two memory regimes in terms of the displacement of mobile triflate anions and lithium cations within the oxygen-rich ion-conducting host, supported by hard-soft acid-base reasoning on the cation-oxygen interaction.
3. Build a comparative experimental corpus on a poly(ethylene oxide)-based platform loaded with alkali-metal triflate salts, fabricated under a common protocol, and characterise it through three common dynamical measurements: I–V hysteresis, variable-number-of-pulses potentiation, and variable-delay depotentiation.
4. Quantify the composition dependence through a replicated PEO and lithium-triflate concentration series, and survey host, anion, and cation substitutions on the same three measurements.
5. Construct experimentally grounded behavioural models from those measurements, using composition-indexed fading-memory times, nonlinear pulse-update rules, device-to-device variability envelopes, and read-transfer behaviour.
6. Demonstrate, in simulation, temporal-computing functionality from these models in a reservoir-computing framework: memory-capacity

and NARMA benchmarks, WESAD physiological-context reconstruction, and WESAD affective-computing tasks.

Structure of the Thesis

The thesis is organised in six chapters that follow this objective list.

Chapter 1 — Introduction. Provides the conceptual and materials background, moving from the limits of conventional computing, through the biological synapse and the memristor, to the polymer-electrolyte composite adopted as the device strategy for the rest of the thesis.

Chapter 2 — Proof of Concept. Reports the Super Yellow, Hybrane, and lithium-triflate device. It is the only device in the thesis for which the complete synaptic suite is treated as primary evidence, and it isolates the cation–host coordination chemistry that the comparative work then tests. The chapter is built on Prado-Socorro et al. [6].

Chapter 3 — Reproducibility, Degradation, and the Device-Provenance Methodology. Documents the reproducibility crisis that ended the Hybrane fabrication line and motivated the switch to a PEO host. Under explicit controls (per-device weighting, a standard-protocol corpus, sweep-amplitude matching), it establishes the batch-over-time collapse of the Hybrane switching window, offers a moisture-driven hydrolytic-aging hypothesis for the host stock, and introduces the fabrication-provenance record, the normalized characterization procedure, and the automated feature pipeline that the remaining chapters rely on.

Chapter 4 — Compositional and Chemical Control. Reformulates the proof-of-concept architecture on a PEO host loaded with alkali-metal triflates. Its quantitative core is the replicated PEO/lithium-triflate composition grid; host, anion, and cation substitutions are surveyed at smaller sample sizes. All claims rely on the same three common measurements.

Chapter 5 — Data-Driven Temporal Computing. Turns the experimental measurements into temporal-computing functionality. Compact behavioural models extracted from the Chapter 4 dynamics drive reservoir-computing simulations: memory-capacity and NARMA-style benchmarks, WESAD physiological-context reconstruction, and WESAD affective-computing demonstrations.

Chapter 6 — Conclusions and Outlook. Brings together the device-chemistry axis and the application axis, benchmarks the work against temporal, reservoir, and event-driven neuromorphic hardware, and states the limitations of the evidence base.

Chapter 1

Introduction

The performance and energy of a modern computer are increasingly limited not by how fast it can compute, but by how much data it must move between where information is stored and where it is processed [1, 3]. Two trends have made this limit acute. First, the dominant workloads have shifted toward large machine-learning models and continuous streams of sensor data, both of which are defined by the sheer number of memory accesses they require [2]. Second, the transistor scaling that once absorbed these costs has slowed, so the energy of data movement is no longer hidden by each new process generation. The response, across industry and research alike, has been to bring memory and computation physically closer together, and—more radically—to let the dynamics of the physical device itself perform part of the computation. The biological synapse is the canonical example of the latter idea: a single element that transmits a signal, modifies its strength according to recent history, and stores that history across multiple timescales [7].

This thesis examines organic memristive devices built from ion-containing polymer composites and explores how their internal dynamics can serve as exactly this kind of computational resource. The work lies at the

intersection of solid-state electronics, polymer physical chemistry, and computational neuroscience: a two-terminal device must behave like an electrical circuit element, its state must be set by ion motion in a soft host, and its usefulness is judged by whether that state provides the fading memory needed for temporal computation.

1.1 The Crisis in Modern Computing

Modern digital computers rest on an architecture that has been extraordinarily successful at executing arbitrary programs on general-purpose hardware. That success, however, was built on an assumption that no longer holds: that the cost of a computation is essentially the cost of its arithmetic. On contemporary hardware the dominant cost is instead the transport of operands to and from memory, and simply making transistors smaller and clocks faster no longer addresses it [1, 8]. The practical limits are therefore architectural as well as technological: the *memory wall*, the *energy wall*, and the slowing of classical transistor scaling all originate in the separation between stored data and the circuitry that acts on it [3, 9].

1.1.1 Von Neumann Architecture and Its Limitations

The architectural model of modern digital computers is the *stored-program* concept set out by von Neumann in 1945 [10]: instructions and data share one memory and are fetched word by word by a central processing unit across a connecting bus. This is what makes a single machine general-purpose, and engineering refinements (registers, caches, branch prediction, memory controllers) have left the topology essentially unchanged. The same topology is the source of the limitation. Because the processor and the memory are *physically distinct* subsystems, every instruction

fetch, operand load, and result write-back must cross the channel between them; no arithmetic can begin before its operands arrive, and no result is preserved until it is moved back. This is the *processor–memory dichotomy*.

Because only a bounded amount of data can cross that channel per cycle—the *von Neumann bottleneck* [11]—performance is set by the slowest interface rather than by the arithmetic. Pipelining, multi-level caches, speculation, prefetching, and out-of-order issue all mitigate this, but they preserve the separation between computation and storage, so the bottleneck is shifted rather than removed. Wulf and McKee [9] named the resulting imbalance the *memory wall*: with processor speed long outpacing DRAM latency and bandwidth, an ever-larger share of each cycle is spent waiting on memory. Modern microprocessors accordingly devote much of their silicon area, and an even larger fraction of their power, to caches, prefetch logic, memory controllers, and interconnect—hardware whose sole purpose is to keep the arithmetic units supplied with data [8]. The bandwidth and energy aspects of this imbalance are summarised in fig. 1.1.

In 2014, Horowitz [1] provided a comparison of costs for elementary operations in a representative CMOS process where an addition of 32-bit integers costs on the order of 0.1 pJ, but an SRAM access can cost tens of picojoules, while a DRAM access could cost several hundred picojoules to nanojoules (table 1.1). These per-operation figures matter only in proportion to how often each operation is performed, and this is exactly where modern workloads make the asymmetry decisive. A single inference pass through a large neural network performs billions of multiply–accumulate operations, each of which must fetch a stored weight before it can do the comparatively cheap arithmetic; training repeats this over enormous datasets. The energy budget is therefore spent overwhelmingly on operand transport rather than on the arithmetic it enables. This is the *energy wall*, which comes from the parasitic capacitance of interconnects

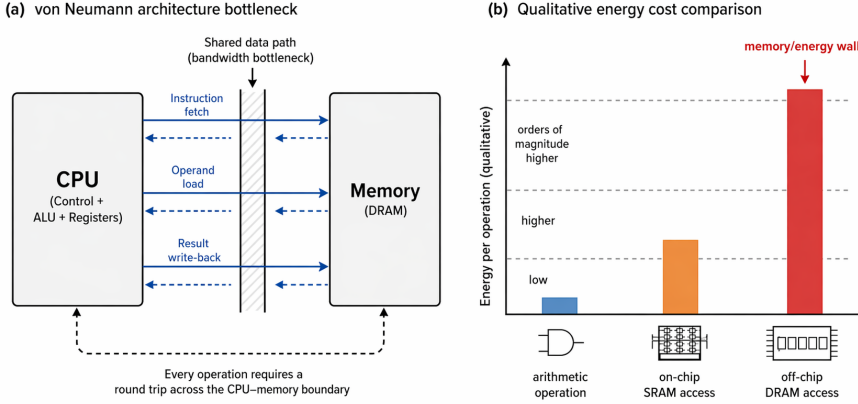


Figure 1.1: Schematic of the von Neumann architecture and the associated bandwidth and energy costs of data movement. (a) In the stored-program paradigm, the central processing unit and the memory unit are physically separated and communicate through a shared bus, so that every instruction fetch and every operand access must traverse that bus. (b) The energy cost of moving a word across this boundary exceeds, by orders of magnitude, the energy cost of the arithmetic operation performed on it, and this asymmetry has increased as technology has scaled [1].

and from the much greater total wire length of a memory subsystem than of an arithmetic unit. While computation and storage remain spatially separated, the architecture is limited by the physics of moving charge through wires.

For decades these inefficiencies were hidden by silicon scaling: Moore’s doubling of transistor counts [12] came paired with Dennard scaling [13], so each generation delivered more transistors that were also faster and cooler, and the memory and energy walls did not yet bind. Dennard scaling broke down around 2005, when leakage currents no longer allowed proportional voltage scaling [3]; transistor counts kept rising, but performance and energy gains slowed and power density became the limiting constraint—the *post-Dennard* or *dark-silicon* regime [14]. A new process node can therefore no longer hide the cost of the von Neumann model, and the problem must be confronted at the architectural and materials level. The pressure has

Table 1.1: Approximate energy cost per operation in a representative 45 nm CMOS process, after Horowitz [1]. Memory accesses cost between one and four orders of magnitude more energy than the arithmetic they enable, and the asymmetry widens with the size of the memory addressed and with the distance the data has to travel.

Operation	Energy	Class
8-bit integer add	0.03 pJ	Arithmetic
32-bit integer add	0.1 pJ	Arithmetic
32-bit floating-point add	0.9 pJ	Arithmetic
32-bit floating-point multiply	3.7 pJ	Arithmetic
Register-file read	0.1 pJ	On-chip storage
SRAM read (8 kB)	5 pJ	On-chip storage
SRAM read (1 MB)	100 pJ	On-chip storage
Cross-chip wire (~ 10 mm)	100 pJ	Interconnect
DRAM read	1.3 nJ	Off-chip storage

only grown as workloads shift toward machine-learning kernels dominated by repeated matrix operations on memory-resident parameters [2]: data movement worsens exactly as scaling stops compensating for it. Hennessy and Patterson [3] read this as an inflection point, arguing that the next stage of computer architecture will lean on domain-specific hardware in which memory and computation are brought much closer together.

1.1.2 In-Memory and Event-Driven Computing Paradigms

Two responses to this separation have emerged. The first keeps the digital workload largely intact but performs operations directly within, or very close to, the memory array; this route is described as *in-memory*, *near-memory*, or *processing-in-memory* computing. The second uses temporally structured event streams and treats the intrinsic dynamics of the physical device as part of the computation itself; this route is known as *event-driven* or *temporal* computing. The two responses, set against the von Neumann baseline they address, are summarised in fig. 1.2.

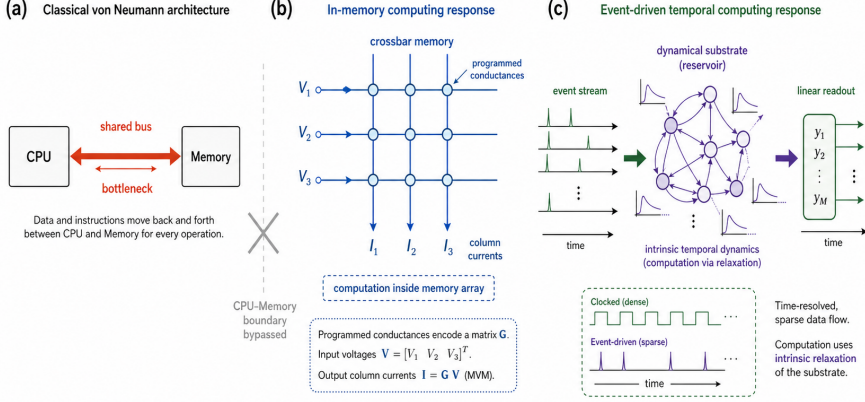


Figure 1.2: Two architectural responses to the von Neumann bottleneck. (a) In the classical stored-program architecture, every operation traverses a CPU–memory bus that fixes the bandwidth and energy ceilings of the system. (b) In the in-memory paradigm, computation is performed inside the memory array: a resistive crossbar with programmed conductances executes an analogue matrix–vector multiplication by Ohm’s and Kirchhoff’s laws, so the dot product no longer crosses the CPU–memory boundary [15, 16]. (c) In the event-driven paradigm, dense clocked data are replaced by a sparse stream of asynchronous events that drive a dynamical substrate whose intrinsic relaxation is itself the computation; only a linear readout layer downstream is trained [5, 17, 18].

In-memory computing bypasses the classical *fetch–compute–write-back* cycle by executing operations directly inside the memory array. This is demonstrated by implementing analogue matrix–vector multiplication using a resistive crossbar architecture, in three steps:

1. Each memory cell contains a conductance value that reflects a matrix entry.
2. Voltages are applied across the rows, representing the input vector.
3. The resulting column currents represent the sum of products, in accordance with Kirchhoff’s laws.

This changes the array from a passive storage device into an active computing tool. Demonstrations using metal-oxide resistive memories [15], phase-change materials [19, 20], and organic ionic devices [21–23] have achieved matrix–vector operations with energy efficiencies that surpass those of traditional digital accelerators, and similar principles can be extended to convolutional [24] and attention-like processes (the matrix-multiplication kernels at the core of transformer architectures [25]).

At the device level, however, this version of in-memory computing still requires memory properties. Each cell should hold a quasi-static conductance that encodes a trained weight, and which should be set once, read many times, and remain stable enough to meet the application’s numerical precision requirements. Non-volatility over seconds to years becomes a design goal.

Event-driven computing is motivated by the fact that many signals of interest carry information in time. Natural sounds, visual transients, motor commands, biomedical waveforms, and streaming sensor outputs are all non-static examples. For such signals, the computational model should be a system that receives events in continuous time and evolves in response, because information is not only present in the amplitude values but also in the waveform’s evolution and timing. At the algorithmic level, this is the domain of spiking neural networks [26]. At the circuit level, it corresponds to event-driven asynchronous logic. At the device level, which is the level of this thesis, it requires materials whose intrinsic relaxation dynamics are matched to the signals they process.

This framework mirrors biological nervous systems, which efficiently handle complex sensorimotor and cognitive tasks while consuming much less power than artificial systems. This efficiency partly results from architectural choices that diverge significantly from the von Neumann model. Neurons share information through sparse events, and synapses pass signals that change depending on recent activity. Information is

stored and altered through slow, gradual electrochemical processes, in which the ionic medium actively participates in computation. Although directly translating the brain into silicon isn't practical or useful, several principles do carry over: sparse-event coding [27], the use of slow device physics as a fundamental computing element, and viewing device heterogeneity as a feature for functional diversity rather than a flaw. These principles form the basis of neuromorphic engineering [4].

Boyd and Chua [28] showed that any time-invariant nonlinear operator with *fading memory*, in an appropriate function space, can be approximated by a finite Volterra series (a power-series expansion in time-shifted inputs). This means that a system whose current state retains a decaying record of past inputs can perform various temporal computations when paired with a trainable linear readout. This concept was later formalised through algorithms such as Jaeger's *echo state network* [17, 29] and Maass, Natschläger, and Markram's *liquid state machine* [18], which were combined under the umbrella of *reservoir computing* [30, 31]. In this framework, an untrained dynamical system—the reservoir—is driven by the input stream, with only the readout being trained. When the reservoir is realised in a physical substrate, the approach is known as *physical reservoir computing* [5], with implementations in optical, mechanical, spintronic, volatile-memristive, and organic electrochemical systems [32–35].

This redefines how a working device should be evaluated. If the goal is to maintain a trained weight permanently, a device that can't retain memory for long periods will be considered unsuitable. However, in a reservoir architecture, this finite memory retention *is* what defines the period during which previous inputs influence the current output, a concept referred to as the fading-memory timescale. Variability between devices can also play a different role: while it's a disadvantage for weight storage, a controlled variation in intrinsic timescales can enable a common readout

to interpret temporal structures over a wider range than any individual device could achieve [5].

1.2 The Biological Synapse as the Engineering Template

In neural circuits, a single physical element both transmits a signal and stores part of the history that shapes future transmissions: the synapse. The same molecular event that relays information from one neuron to the next also alters its future strength, so signalling and memory share a substrate. The device target is therefore synaptic signalling and plasticity rather than full neuron emulation.

1.2.1 Neural Signal Transmission: A Brief Overview

A neuron is a polarised cell specialised to receive, integrate, and transmit electrical signals [36]. Dendrites collect inputs from other cells, the soma integrates them, and when the membrane potential at the axon hillock crosses a threshold the axon propagates an outgoing signal whose terminals release a chemical messenger onto downstream cells. A mature animal neuron typically receives 10^3 – 10^4 such contacts [37], and each is a *synapse*. How a neuron computes is therefore governed far more by the number and strength of its synapses, and by how they change over time, than by the soma itself.

The axonal signal is the *action potential*: a stereotyped, millisecond-scale depolarisation that is all-or-nothing, so that information is carried not in its amplitude but in the timing, rate, and coincidence of spike trains [38, 39]. Neural computation is in this sense event-driven (section 1.1.2) and arises from the pattern of signals arriving at many synapses. The neuron's

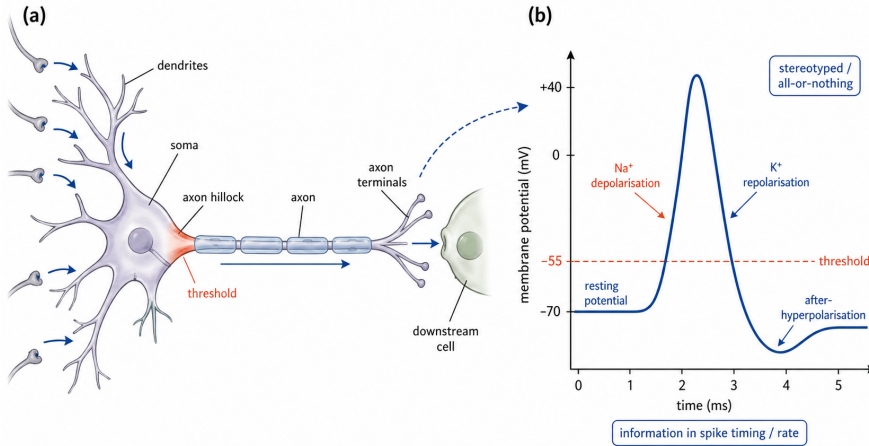


Figure 1.3: Anatomy of a neuron and the action-potential waveform. (a) A neuron collects synaptic inputs through its dendrites, integrates them at the soma, and fires an outgoing electrical event when the membrane potential at the axon hillock crosses a threshold; the resulting waveform propagates unidirectionally along the axon to the synaptic terminals, where it triggers chemical transmission onto downstream cells (the zoom shown in fig. 1.4). (b) The action potential is a stereotyped millisecond-scale depolarisation produced by the opening of voltage-gated Na^+ channels, followed by a K^+ -mediated repolarisation and a brief after-hyperpolarisation before the return to the resting potential; its all-or-nothing character means that information about the input is carried by the timing and rate of these events rather than by their amplitude, which is the cellular basis of the event-driven view of neural computation introduced in section 1.1.2.

anatomy and the action potential waveform are summarised in fig. 1.3.

Synaptic transmission is largely chemical. An action potential arriving at the terminal opens voltage-gated Ca^{2+} channels, and the Ca^{2+} influx triggers neurotransmitter-filled vesicles (glutamate at excitatory cortical synapses) to fuse with the presynaptic membrane [40]. The transmitter diffuses across the ~ 20 nm cleft and binds postsynaptic receptors [37], producing an excitatory or inhibitory postsynaptic current (EPSC or IPSC) whose amplitude and kinetics define the *strength* of the synapse at that moment. Crucially, the same molecular machinery that carries the

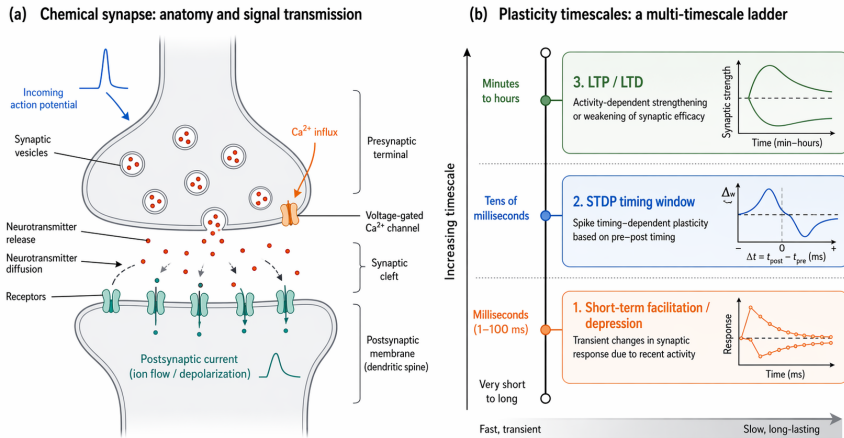


Figure 1.4: Anatomy and plasticity regimes of the chemical synapse. (a) A presynaptic terminal releases neurotransmitter vesicles into the synaptic cleft in response to a calcium-triggered action potential, and postsynaptic receptors convert the released signal into an excitatory postsynaptic current. (b) The same synapse supports a graded hierarchy of plasticity regimes — short-term facilitation and depression on millisecond timescales, spike-timing-dependent plasticity on tens of milliseconds, and long-term potentiation and depression on timescales from minutes to hours — each associated with distinct microscopic mechanisms.

signal also sets the strength of future transmissions.

The synapse is a transmission element, a modulator, and a memory element at the same time: it transmits a signal, adjusts the amplitude and time course of that signal, and stores lasting changes produced by previous activity. This co-localisation of transmission, modulation, and memory makes the synapse, more than the neuron as a whole, the most useful biological template for post-von-Neumann electronic devices. The relevant anatomy and plasticity regimes are summarised in fig. 1.4.

1.2.2 Synaptic Plasticity: The Physical Basis of Learning

Experience leaves a physical trace in the nervous system, stored mainly at synapses. Hebb’s 1949 postulate—that a presynaptic cell which repeatedly helps fire a postsynaptic cell strengthens its connection to it [41] (“cells that fire together, wire together”)—captures two features that matter for hardware: the rule is *local* to the two cells a synapse joins, and *sensitive to correlation*. Synaptic plasticity implements Hebbian or Hebbian-like behaviour across several timescales.

On the shortest timescales, *short-term plasticity* [42, 43] makes the immediate strength of a synapse depend on its recent activity. *Facilitation* is a transient increase in EPSC amplitude when a second spike follows the first after a short delay, usually attributed to residual presynaptic Ca^{2+} ; *depression* is a transient decrease caused by vesicle depletion or receptor desensitisation. With time constants from tens of milliseconds to several seconds, short-term plasticity makes the synapse a nonlinear, history-dependent temporal filter, so that identical spike sequences produce different responses depending on their timing.

On longer timescales, *long-term potentiation* (LTP) provides persistent, activity-dependent strengthening. First demonstrated by Bliss and Lømo in the rabbit hippocampus [44] and later shown to last days to weeks [45, 46], hippocampal LTP relies on NMDA receptors that open only when the postsynaptic membrane is already depolarised (a voltage-dependent Mg^{2+} block [47, 48]), making them Ca^{2+} -permeable coincidence detectors; the resulting Ca^{2+} influx drives additional AMPA receptors into the postsynaptic membrane [49, 50]. The complementary *long-term depression* (LTD) [51, 52] weakens synapses under low-frequency or anti-correlated activity, principally through reverse AMPA-receptor trafficking [53]. Together, LTP and LTD give the bidirectional, long-lasting amplitude

1.2. THE BIOLOGICAL SYNAPSE AS THE ENGINEERING TEMPLATE

changes considered a primary physiological basis of learning.

Spanning these regimes, *spike-timing-dependent plasticity* (STDP), described in cortical pyramidal cells [54] and quantified in hippocampal cultures by Bi and Poo [55], makes the sign and magnitude of the change depend on the delay between pre- and postsynaptic spikes within a window of a few tens of milliseconds: a presynaptic spike before the postsynaptic one potentiates, the reverse depresses, and large delays in either direction produce no lasting change. The synapse thus holds a short memory of recent presynaptic activity, compares it with postsynaptic firing, and converts that timing relationship into a longer-lasting amplitude change.

Across these mechanisms the synapse is not a passive store of a single “weight” but operates over many timescales at once: milliseconds for individual spikes, tens to thousands of milliseconds for short-term facilitation and depression, and minutes to hours and beyond for LTP and LTD. Driven by partially overlapping but distinct molecular mechanisms, these layers led Abbott and Regehr [7] to argue that a synapse is better understood as a computational unit than as a memory store. For hardware, both the long-lasting and the decaying state changes matter.

1.2.3 Why Hardware Emulation of the Synapse?

The motivation for hardware synapses is directly derived from section 1.1. In traditional computing, a software neural network performs calculations on data retrieved from a memory hierarchy physically separate from the arithmetic units. Each forward pass incurs data movement and energy costs as described earlier, and these costs increase with the number of simulated synapses. A typical human brain contains approximately 10^{14} – 10^{15} synapses [56] and consumes about 20 W [57]. A standard von Neumann architecture that consumes orders of magnitude more energy cannot easily bridge this gap through improved design alone, since

the cost of fetching a weight before multiplying it by an activation is limited by interconnect physics. Thus, in nature, the synapse exemplifies the most compact, low-power form of co-localised analogue memory and computation, making it an optimal physical template. The order-of-magnitude envelopes of these implementation regimes are compared in fig. 1.5.

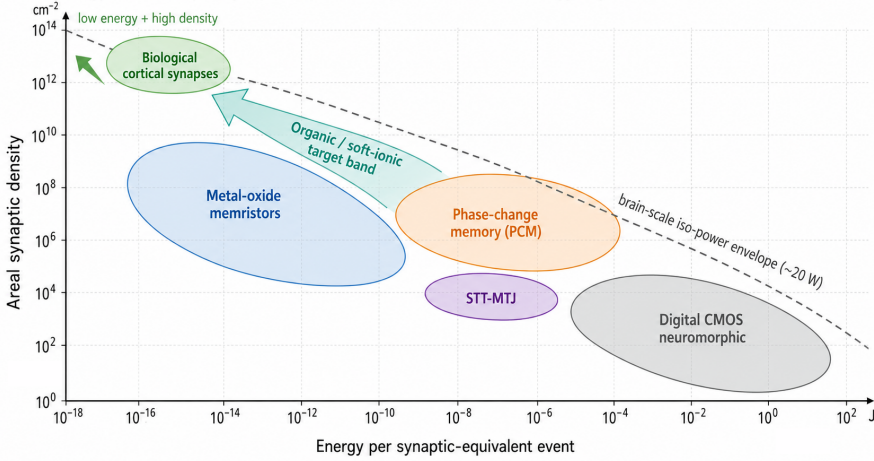


Figure 1.5: Approximate energy and density envelopes of synaptic-equivalent operations across implementation regimes. Each cluster represents a class of physical realisation: biological cortical synapses, digital CMOS neuromorphic implementations in which a synapse occupies an SRAM cell plus an arithmetic unit, the inorganic memristor families surveyed in section 1.4, and the soft-ionic composite target region forward-referenced to section 1.6. The diagonal reference line marks the brain-scale iso-power envelope (~ 20 W for $\sim 10^{14}$ – 10^{15} synapses) introduced at the start of section 1.2.3. Numerical bounds are conservative orders of magnitude drawn from the reviews cited in the surrounding text; individual devices may sit outside the envelopes shown.

The research programme built around this idea is *neuromorphic engineering*, whose programmatic origin is usually attributed to Mead [4]. Its goal is not merely to accelerate neural algorithms but to build circuits in which device physics, topology, and dynamics themselves embody computational characteristics like those of the nervous system, and within it the synapse

has become the main device focus.

A hardware synapse should satisfy several requirements:

- It should provide an analogue, multi-state conductance that can be set reversibly by electrical stimuli and read non-destructively with a small probe current.
- It should also be compact enough for large arrays, so that many thousands of elements can be addressed with a manageable number of input and output lines [27, 58].
- Its response should be nonlinear in a meaningful way, for example by displaying a pulse-number or pulse-amplitude threshold below which little change is induced [16, 58].
- Finally, its dynamics should match the signals it is meant to process, whose relevant timescales span from milliseconds to seconds and beyond.

Two of these requirements deserve emphasis. The first is *analogue*, multi-state conductance: a binary element stores one bit and would need many devices per connection to represent a graded weight, whereas an analogue element encodes several distinguishable states in one device, and for ionic devices the write energy can be low because the state is set by moving ions over short distances rather than charging large digital circuits [22, 23]. The second is *reversibility*: a device that can be potentiated but not reliably depressed cannot follow the bidirectional rules of section 1.2.2 and so cannot fully replicate synaptic learning [16]. The resulting specification—analogue, reversible, multi-state, low-energy conductance changes set by electrical pulses—is the one that motivates the devices studied in the rest of the thesis.

The biology of section 1.2.2 reveals that biological synapses incorporate quasi-stable memory via LTP and LTD, while providing temporal filter-

ing through short-term plasticity and STDP, allowing both functions to coexist within the same physical structure. A hardware system can then be useful in two complementary ways: as a persistent or quasi-persistent programmable weight (as seen in crossbar arrays for in-memory computing [16]), and as a dynamic fading-memory element. The second role is the primary focus of this thesis, following the reservoir-computing approach to event-driven temporal computing introduced in section 1.1.2 [5, 32–34].

This leads to a materials question. In biology, the timescales of short-term plasticity are determined by different processes such as Ca^{2+} buffering, vesicle recycling, receptor desensitisation, and other local mechanisms that vary across synapses [43]. A hardware alternative for this role should offer an analogous way to adjust its electrical timescale. Classical transistor-based memories do not provide that kind of analogue, history-dependent element; memristive devices are the natural two-terminal candidate.

1.3 The Memristor: Theory and History

A memristor or memristive device shows a current–voltage (i–v) response that depends on an internal state, which itself is influenced by the device’s previous electrical activity. The ideal form of this is the *memristor*; the broader experimental category is the family of *memristive devices*. The point needed here is simple but important: a device can be memristive even when its internal state relaxes after the input is removed. Volatility does not disqualify the device; it defines the fading-memory behaviour exploited later in the thesis.

1.3.1 Memristive Systems as History-Dependent Devices

A memristor is, intuitively, a resistor with memory: a two-terminal element whose resistance depends on the charge that has previously passed through it. Chua [59] introduced it in 1971 on symmetry grounds, completing the set of fundamental passive elements (resistor, capacitor, inductor) with a fourth that relates charge $q = \int i \, dt$ to flux linkage $\varphi = \int v \, dt$. It is described by $v = M(q) i$, where the *memristance* $M(q)$ has the units of resistance but, unlike an ordinary resistor, carries a record of the device's past.

If M is constant the element is an ordinary resistor; if M varies with q , the present response depends on the history of the device. Under periodic excitation this produces a *pinched hysteresis loop* in the i - v plane [60]: the loop passes through the origin (zero voltage gives zero current instantaneously) but its shape reflects the internal state left by prior stimulation. Pinched hysteresis is the standard experimental signature of memristive behaviour [61], and it is collected together with Chua's constitutive argument in fig. 1.6.

Few real devices satisfy the ideal relation $\varphi = \hat{\varphi}(q)$. The broader and more useful notion, introduced by Chua and Kang [62] in 1976, is the *memristive system*: an input-driven internal state $\mathbf{x}$ evolves under generally nonlinear dynamics and gates the instantaneous electrical response. For this thesis, the crucial allowance is that $\mathbf{x}$ may include relaxation terms that drive the device back toward equilibrium once the input is removed. A volatile device whose state decays is therefore still a memristive system, even though it is not an ideal memristor. This is the framework adopted for the polymer-composite devices reported here.

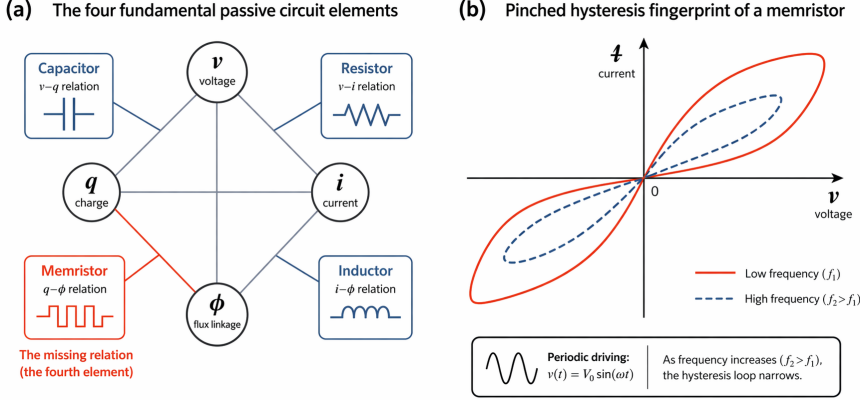


Figure 1.6: The memristor as a constitutive element and its experimental fingerprint. (a) The four fundamental two-terminal passive circuit elements, arranged by the pair of constitutive variables they relate: the resistor ($v-i$), the capacitor ($v-q$), the inductor ($i-\phi$), and the memristor ($q-\phi$), which completes the quadrant [59]. (b) A memristive device driven by a periodic voltage exhibits a pinched current–voltage hysteresis loop that narrows with increasing driving frequency and that passes through the origin at each zero-crossing; this loop is the canonical experimental fingerprint of memristive operation and is observed in all of the device families surveyed in the remainder of the chapter [60–62].

1.3.2 The First Physical Realisation: TiO_2 Memristor

For more than three decades after Chua’s proposal, the memristor remained mostly a theoretical construct. Thin-film metal-oxide resistive switches with hysteretic current–voltage characteristics had already been reported since the late 1960s [63–65], and many of them can now be recognised as memristive systems. At that time, however, they were not interpreted within Chua’s formalism. The field still lacked a standard physical device that connected the theoretical idea to a tangible thin-film structure.

In 2008, that connection was made possible by Strukov, Snider, Stewart

and Williams [66] at HP Laboratories. Their device consisted of a thin titanium dioxide (TiO_2) layer sandwiched between two platinum electrodes. Within the film, a reduced TiO_{2-x} region acted as an oxygen-deficient conductive layer, while the neighbouring TiO_2 region served as a resistive barrier. When a voltage was applied, oxygen vacancies moved, shifting the boundary between these two regions [67]. Given that the total resistance depends on the relative thickness of the conductive and resistive parts, the device resistance could then be modulated continuously by moving this internal boundary back and forth. The boundary position is the internal state, oxygen-vacancy drift updates it, and the resulting i - v response reproduces the pinched hysteresis predicted by the theory.

This result transformed the memristor from a formal circuit element into a concrete experimental object, thus giving life to a large body of work on resistive switching and memristive devices. A two-terminal resistance that can be programmed by electrical pulses, tuned over several conductance values, and read at low voltage is close to the analogue weight element required by neuromorphic hardware [68]. Crossbar demonstrations of in-memory matrix-vector multiplication followed soon after [15, 16]. The 2008 TiO_2 device is, however, only one mechanism—oxygen-vacancy drift in a transition-metal oxide—whereas the memristive formalism is far broader.

1.3.3 Classification of Memristive Mechanisms

In the field of device physics, the term *memristor* is usually considered a phenomenological concept, not a description of a particular microscopic mechanism. A device is considered "memristive" if its current-voltage response exhibits the pinched hysteresis that Chua described and if its internal state can be both set and read electrically. This can be achieved through many different physical processes, commonly grouped into broad

mechanistic families [16, 68].

These taxonomies combine two different classification axes. The first is the *spatial pattern* of the state change, whether filamentary or interface/distributed, depending on whether the conductance change is localised to a narrow conductive path or extended across the active region. The second is the *physical nature* of the state variable itself: ionic distribution, structural phase, magnetic configuration, or redox population. The two axes are orthogonal: an ionic device can also be non-filamentary, so the families considered here locate a device along both axes.

Filamentary switching. This is historically the most visible family. When a voltage is applied, a narrow conductive path forms through a previously insulating thin film, and said path can later be ruptured or reformed. In *conductive-bridge* random-access memories (CBRAM), the filament is built from metal atoms displaced electrochemically from an active electrode (like silver or copper) through a solid electrolyte [69, 70]. In *valence-change* memories, which are generally based on binary transition-metal oxides, the filament is instead associated with oxygen vacancies or partially reduced metal-cation states inside the oxide lattice [67].

These devices offer high ON/OFF ratios, fast switching, and non-volatile retention; their main limitation is stochasticity, since filament formation, positioning, and rupture vary across devices and cycles [71]. They therefore work well as binary or coarse multi-level memories but are poorly suited to fine analogue representation. Panel (a) of fig. 1.7 contrasts the CBRAM and VCM variants of these devices, while panels (b)–(e) display the other four families.

Interface (non-filamentary) switching. In these devices, conductance changes over a distributed region rather than through a single localised filament. Some implementations modulate the height or width of a Schottky barrier by accumulating or depleting mobile ions near a metal–

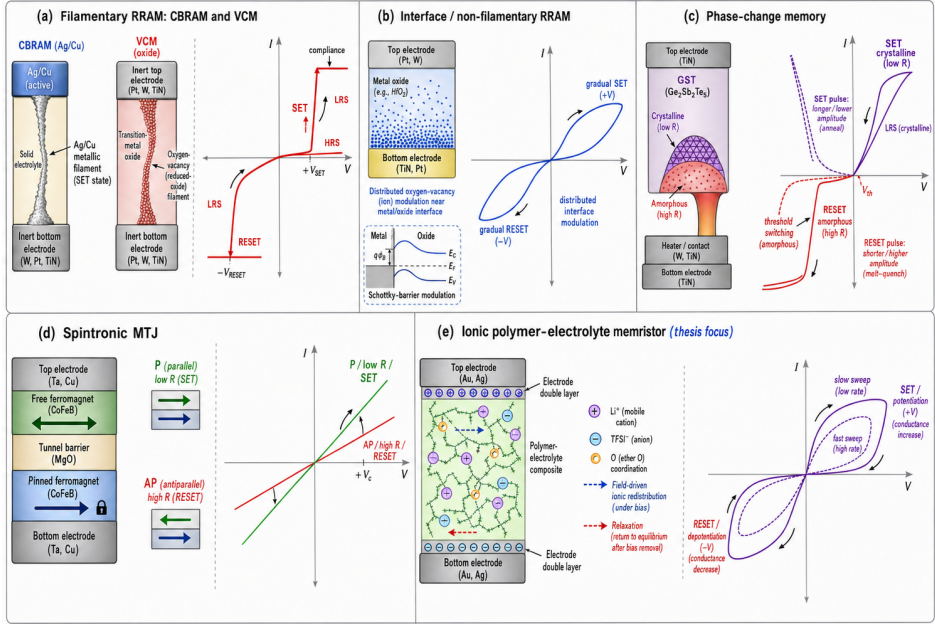


Figure 1.7: Mechanism gallery for the five memristive device families. (a) Filamentary devices, in their conductive-bridge (active-electrode metal filament) and valence-change (oxygen-vacancy filament) variants, switch abruptly between a high-resistance and a compliance-limited low-resistance branch. (b) Interface or non-filamentary devices switch through a spatially distributed change in the active layer, producing smooth, gradual, area-scaling hysteresis without an abrupt set or reset. (c) Phase-change devices toggle between amorphous and crystalline states of a chalcogenide alloy through Joule-heating melt-quench and anneal pulses, with a sharp threshold-switching event preceding the crystalline branch. (d) Spintronic magnetic tunnel junctions switch the relative orientation of two ferromagnetic layers, producing two nearly-linear branches separated by a small (order of two) ON/OFF ratio. (e) Ionic devices, in their polymer-electrolyte composite form most relevant to this thesis, redistribute mobile ions across the active layer under bias and relax back toward equilibrium when the bias is removed, producing continuous, sweep-rate-dependent hysteresis without abrupt transitions.

semiconductor interface [72]. Other implementations, especially in mixed ionic–electronic systems, drive a distributed electrochemical doping of the semiconductor [21, 22]. Given that the state change is spatially extended, these devices can support a quasi-continuous range of conductance values. The ON/OFF ratio is often smaller than that of filamentary devices, but variability is usually lower and the current tends to scale more predictably with device area. This one is the closest *spatial-pattern* class for several organic electrochemical devices, including the polymer-composite devices studied here, whose state variable is itself described under the ionic family below.

Phase-change memories. The active material is a chalcogenide alloy, usually $\text{Ge}_2\text{Sb}_2\text{Te}_5$ or a related composition, whose resistance depends on whether the material is in a crystalline or amorphous configuration. Joule-heating pulses are then used either to melt and quench the material into the amorphous state, or to anneal it back toward the crystalline state [19, 20, 73]. Phase-change memories offer excellent retention, large ON/OFF ratios, and high endurance. The cost lies in relatively large write energies and, unless the device is carefully engineered, a tendency toward binary or few-state operation. They are nonetheless mature storage-class memories and among the most important platforms for analogue in-memory computing demonstrations.

Spintronic (magnetic) devices. Their resistance depends on the magnetic configuration of a tunnel junction, typically determined by the relative orientation of two ferromagnetic layers [74]. Spintronic devices switch very quickly at low write energy and form the basis of spin-transfer-torque magnetic random-access memory. Their limitations for analogue synaptic use are a small ON/OFF ratio (often of order two) and few stable intermediate states. Even so, their intrinsic stochastic dynamics have been put to use as a computational resource within appropriately designed architectures [75, 76].

Ionic memristors. This family is the one most relevant to this thesis. Their internal state is the distribution of one or more mobile ionic species inside a solid matrix. The host can be an inorganic oxide, a chalcogenide, a halide perovskite, a mixed ionic–electronic conductor, or a polymer electrolyte. Likewise, the mobile species can be a small cation such as Li^+ , Na^+ , or Ag^+ , a proton, or a small anion; in halide perovskites it is typically a migrating halide vacancy, which has been used to build energy-efficient memristive and synaptic devices [77]. What sets this family apart is the fact that the state variable is itself a chemical distribution of mobile species, and not only a structural, morphological, or magnetic configuration of the host [70]. Its kinetics are then governed by ionic transport, coordination, and relaxation within the matrix.

Ionic systems are intrinsically history-dependent: the distribution of mobile species at any moment reflects the electric fields previously applied. They also relax once the field is removed, as intermolecular forces and concentration gradients drive the biased distribution back toward equilibrium, so volatility is intrinsic to their operation. The associated timescales are set by ion transport and binding, both tunable chemically through the host matrix and the mobile species—the link between chemistry and relaxation that leads this thesis to the organic polymer-electrolyte subfamily.

1.3.4 From Device Metrics to Fading Memory

Memristive devices are often compared through a compact set of practical figures of merit: high- and low-resistance states, ON/OFF ratio, set/reset voltage, write energy, switching speed, retention, endurance, update linearity and symmetry, and device-to-device or cycle-to-cycle variability [16, 68, 71]. These quantities are useful only after the computational role of the device has been fixed.

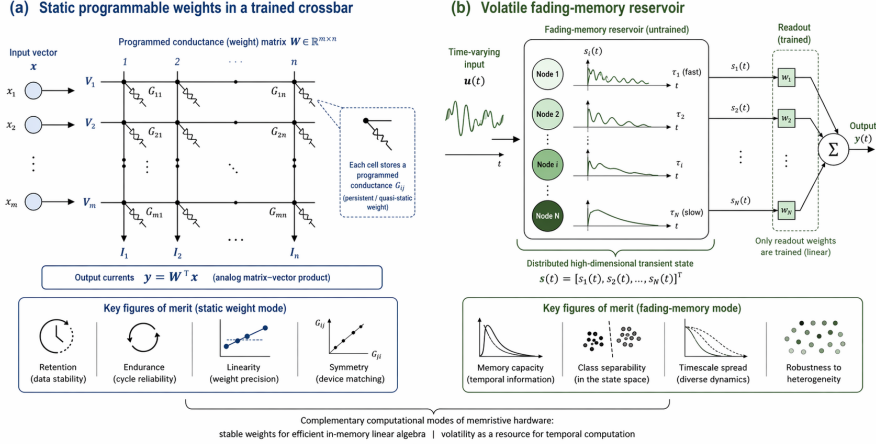


Figure 1.8: Two complementary computational roles for memristive devices. (a) A trained crossbar holds programmed conductances as non-volatile weights and performs matrix–vector multiplication by Ohm’s and Kirchhoff’s laws; here retention, endurance, linearity and symmetry are the relevant figures of merit. (b) A reservoir of volatile fading-memory nodes, driven by a time-varying input, projects temporal signals into a high-dimensional dynamical state; only a linear readout layer is trained, and the relevant figures of merit become memory capacity, class separability, and the spread of available time constants.

For analogue in-memory computing, the device stores a *trained weight*: a conductance programmed into a crossbar cell and read repeatedly during matrix–vector multiplication [15, 16]. Long retention, high endurance, low variability, and linear symmetric updates are therefore valuable because they preserve numerical accuracy. Judged by that standard, an organic or polymer-composite device with finite retention can look weak.

For temporal computing, the same finite retention becomes the operating principle. A fading-memory device should forget on a timescale matched to the input signal [18, 28]; nonlinearity can enrich the reservoir state, and device-to-device differences can supply a spread of temporal kernels. Figure 1.8 sketches the two modes schematically, and table 1.2 sets the corresponding parameter targets side by side.

Table 1.2: Direction in which each device parameter should be optimised under the trained-weight regime (section 1.3.4) and under the fading-memory regime (section 1.3.5). Several optima are not merely different in magnitude but opposed in direction, which is why a single device cannot in general satisfy both regimes simultaneously.

Parameter	Trained-weight optimum	Fading-memory optimum
Retention	As long as possible (s-years)	Matched to the input-signal timescale
Switching speed	As fast as possible	Matched to the input event rate
Write energy	Minimised (programmed once, read many)	Minimised within the fast-event budget
Conductance update linearity	Linear and uniform	Less critical; nonlinearity exploited as a feature
Update symmetry ($\pm$ pulses)	Symmetric	Less critical; asymmetry can be exploited
Cycle-to-cycle variability	Minimised	Minimised (intra-device drift remains a defect)
Device-to-device heterogeneity	Minimised	Useful as reservoir diversity [5]

Variability must also be separated into two kinds. Intra-device instability, where a device cannot reproduce its own response under identical conditions, remains a defect. Inter-device heterogeneity, by contrast, is harmful in a trained-weight array but useful in a reservoir, where devices with different time constants and nonlinear transfer functions provide a richer projection of the input than identical devices would [5].

1.3.5 Reservoir Computing with Volatile Memristors

Over the past decade, volatile memristive devices have been used as physical *leaky integrators*: their state rises under stimulation, decays after the stimulus is removed, and thereby carries a temporary record of recent input. The most important computational framework for that behaviour is physical reservoir computing [5].

In a volatile-memristor reservoir, the input is encoded as a stream of voltage pulses applied to an array of devices. Their internal states evolve in response to the stimuli and relax between them, so the array provides a nonlinear mapping of recent input history into a high-dimensional feature space. A linear readout trained on that space can then tackle temporal classification, regression, or prediction tasks. Demonstrations have been reported in diffusive metal-filament devices [32, 33], dynamic oxide memristors [34], and organic electrochemical systems [35].

Coincidence detection and multi-timescale filtering are closely related uses of the same physics. Pulses that arrive within the relaxation time of a device can accumulate, giving timing sensitivity and STDP-like responses [78]; banks of devices with different relaxation constants can process one input in parallel, with fast devices capturing sharp transients and slow devices integrating longer trends [27, 79]. In this thesis these ideas are folded into the reservoir-computing framework: coincidence

appears as one effect of the fading state, and the filter-bank idea becomes a heterogeneous reservoir whose spread of time constants is supplied by composition, chemistry, and device-to-device variation.

The central metric is therefore *memory capacity*: how well the present reservoir state reconstructs past inputs at various delays, complemented by class separability on benchmark tasks and robustness to measured variability [80]. These metrics are *task-matched*. They reward devices whose physical timescales align with the temporal structure of the signal, something that HRS, LRS, ON/OFF ratio, and retention alone do not capture.

The useful subset for this thesis is therefore narrower than the full memristor literature: ionic memristive systems whose state is a chemical distribution of mobile species, whose relaxation can be tuned, and whose volatility can be used as fading memory.

1.4 Inorganic Memristors for Neuromorphic Computing

The inorganic families that dominate neuromorphic hardware are metal-oxide resistive memories, phase-change memories, and spin-transfer-torque magnetic tunnel junctions. Each is technologically important, and together they have made memristive hardware a credible post-von-Neumann substrate. Other inorganic candidates — ferroelectric tunnel junctions and FeFETs, Mott-insulator devices, and threshold-switching selectors — are narrower in analogue synaptic operation and are left outside this comparison.

1.4.1 Metal-Oxide Memristors

The broadest and most developed inorganic family is resistive switching in transition-metal-oxide thin films, in the VCM and CBRAM variants of section 1.3.3. After the TiO_2 device [66], the field expanded to HfO_2 , TaO_x , NiO , and WO_x , many already familiar to silicon processing; back-end-of-line CMOS integration—the main practical reason for their prominence—was therefore feasible from the start and enabled large crossbar demonstrations [15, 69, 70]. As non-volatile memory their figures of merit are excellent: retention of years, nanosecond switching, endurance beyond 10^9 cycles, and sub-10 nm densities [16, 68], and the same family can also be made deliberately volatile in diffusive metal-filament form, where Ag or Cu clusters dissolve after programming [32–34, 81]. Asked to behave as an analogue synapse rather than a non-volatile weight cell, however, they are limited by filament stochasticity—an appreciable device-to-device and cycle-to-cycle spread in set/reset voltage and achieved conductance [71]—and by nonlinear, asymmetric conductance updates with only a few reliably distinguishable levels, which together cap the accuracy of in-place training on metal-oxide crossbars [16].

1.4.2 Phase-Change Memory

Phase-change memory encodes its state in the proportion of crystalline and amorphous material in a chalcogenide volume rather than in a localised filament (section 1.3.3) [19]. It is a mature storage-class technology and one of the strongest inorganic platforms for analogue in-memory computing: mixed-precision in-memory arithmetic accurate enough to solve practical linear systems [73] and neural-network training with PCM as the long-term weight store [82] have both been demonstrated, making it the most serious benchmark for trained-weight analogue applications [83]. Read as a synapse, its costs are those of a thermal-structural mechanism:

reset requires local melting, so write currents and thermal loads are large; the quenched amorphous phase drifts with time and must be compensated by reading, calibration, or algorithmic correction [20]; and crystallisation kinetics can be tuned through alloy and geometry but not with the degrees of freedom of soft matter. Its natural strength is therefore high-retention, high-endurance trained weights rather than the chemically tunable fading-memory regime pursued here.

1.4.3 Spin-Transfer-Torque Magnetic Tunnel Junctions

A spin-transfer-torque device is the magnetic tunnel junction (MTJ) of section 1.3.3, here with a thin MgO barrier and switched between parallel and antiparallel magnetisation by a spin-polarised current through a shared read/write port [74]. STT-MTJs switch in nanoseconds at femtojoule energies, exceed 10^{12} cycles, and integrate in the CMOS back end [84], and their intrinsic stochasticity has been turned into a computational resource for probabilistic computing [75, 76]. For analogue synaptic use the drawbacks are intrinsic: the parallel/antiparallel resistance ratio is small (often near two) and the stable intermediate states are few, improvable only through elaborate stacks (synthetic antiferromagnets, perpendicular-anisotropy or multilayer configurations) [84, 85]; retention and write energy trade off directly through thermal stability versus switching current [84]; and the crystalline ferromagnetic stack is incompatible with simple solution processing on arbitrary substrates. They are best viewed as fast, non-volatile, few-state stochastic elements rather than analogue, multi-timescale synapses.

1.4.4 General Limitations of Inorganic Memristors

The three families differ in mechanism and maturity but share the same limitations for the specific problem of this thesis. Their design parameters are mostly geometric, stoichiometric within narrow windows, or process-related [16, 71, 79]: there is no easy way to substitute one mobile ion for another with different binding chemistry, or to change a host so as to reshape its segmental motion. Each family can be made volatile in specific implementations—diffusive filaments [32–34, 81], amorphous-phase drift [20], superparamagnetic fluctuations [75]—but sweeping a relaxation time over orders of magnitude within one platform by tuning a single chemical variable is not natural, because vacancy drift, crystallisation, and magnetic relaxation are each fixed by the host’s intrinsic energy landscape. The fabrication routes (vacuum deposition, high-temperature annealing, top-down lithography [15, 82, 83]) are ideal for dense CMOS-integrated crossbars but restrict substrate and temperature, whereas solution-processable layers could be deposited on a far wider range of substrates and reformulated directly from solution [79]. And in these systems useful inter-device diversity and harmful intra-device instability are difficult to separate [16, 20, 71], whereas a family in which heterogeneity is introduced deliberately through composition—each device staying stable under repeated operation—would give a cleaner route to the controlled variability that reservoir computing can exploit [5].

In short, analogue, ion-mediated, chemically tunable, timescale-rich behaviour on soft, solution-processable substrates is not the natural strength of the inorganic families. That gap motivates the organic ion-mediated route. Figure 1.9 summarises the qualitative comparison, while table 1.3 provides the corresponding quantitative parameters.

Table 1.3: Typical operating parameters for the inorganic memristor families. Diffusive (volatile) metal-filament devices are listed separately because they are deliberately operated in the fading-memory rather than the non-volatile regime. Numerical ranges are conservative orders of magnitude drawn from the indicated reviews; individual devices may sit outside them. “cont.” = quasi-continuous conductance.

Family	Switching	Retention	ON/OFF	Endurance	Energy/write	Analogue states
VCM [67, 68, 71]	1–100 ns	years	10^{-10^3}	$10^9\text{--}10^{12}$	0.1–10 pJ	4–8
CBRAM [16, 68]	1–100 ns	years	$10^4\text{--}10^6$	$10^8\text{--}10^{10}$	0.1–10 pJ	2–4
PCM [19, 20, 73, 82]	10–100 ns	years (drifts)	$10^2\text{--}10^3$	$10^8\text{--}10^{10}$	10–100 pJ	8–16
STT-MTJ [75, 76]	1–10 ns	years	2–3	$> 10^{12}$	0.1–1 pJ	2
Diffusive (volatile) [32–34]	$\mu\text{s--ms}$	ms–s	10^{-10^3}	$\sim 10^6$	0.1–10 pJ	cont. (volatile)

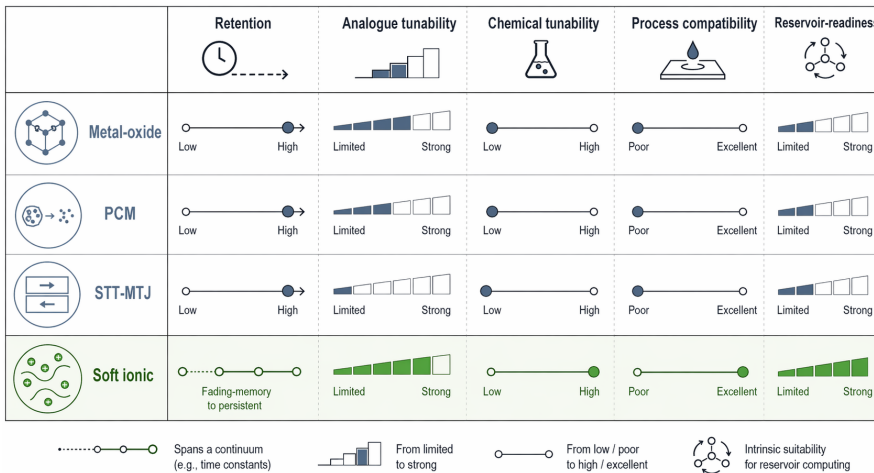


Figure 1.9: Comparison of inorganic memristor families with the soft-ionic composite route pursued in this thesis. For each family, qualitative assessments are shown along five axes: retention (the range of fading-memory to non-volatile behaviour that has been demonstrated), analogue tunability (the number of reliably programmable intermediate states), chemical tunability (the extent to which composition acts as an independent handle on the switching physics), substrate compatibility (vacuum/high-temperature workflows versus solution-processable and mild-temperature workflows), and reservoir-readiness (suitability as a fading-memory node in a temporal-computing scheme).

1.5 Organic Materials for Memristive Applications

The inorganic platforms each solve part of the problem, but none of them can be tuned by chemistry alone: reshaping the switching physics requires redesigning a rigid multilayer stack rather than changing a molecule, a host, a salt, or a mixing ratio [23, 79]. Organic and polymeric materials lift that constraint. A semiconducting polymer can be tuned through backbone, side chains, and molecular weight [86]; an ion-transport host through its chelating groups, topology, and glass-transition temperature [87, 88]; and a mobile ion through charge, radius, and Lewis acidity [89]. Most such

active layers are also solution-processable and processed below 150 °C, so a new formulation can be prepared by changing component concentrations in solution, and the mild processing is compatible with flexible, conformal, and biological substrates—as the stretchable artificial afferent nerve of Kim et al. [90] and the *nervetronic* systems reviewed by Park and co-workers [91] illustrate.

Two consequences matter most here. First, organic switching can be distributed over many molecular sites rather than confined to a single filament, which makes graded analogue states more accessible [21, 22]. Second, relaxation timescales can be engineered *chemically*: ion transport in a polymer electrolyte depends on how strongly the mobile ion binds the host, on host segmental mobility, and on the number of mobile species [87, 92], so timescales from milliseconds to many seconds—the range of biological short-term plasticity and slow temporal signals [42, 43]—can be reached within one material family by compositional change. Cost, reproducibility, biocompatibility, and lifetime still have to be shown case by case; the useful target is therefore not "organic" in general, but a reversible mixed ionic–electronic platform whose chemistry controls the fading-memory state.

1.5.1 Organic Mechanisms as Design Choices

As with their inorganic counterparts, the term *organic memristor* names a material family rather than a single mechanism. The relevant comparison is what each mechanism — each with its own state variable, reversibility, timescale, and degree of analogue control — offers for a chemically tunable, ion-mediated, two-terminal synaptic element.

Charge trapping and detrapping. Deep or shallow traps—molecular defects, embedded nanoparticles, or boundaries between incompatible polymer phases—are filled and emptied under bias, and the trap occupa-

tion, through its local space-charge environment, sets the conductance. Trapping is common in disordered organic semiconductors and operates at modest voltages, but its analogue control is limited: the state is often binary [93], the conductance can depend non-monotonically on occupation, the trap identity is hard to establish, and the retention (set by trap depth relative to thermal energy) is not easily tuned by composition. Panel (a) of fig. 1.10 sketches this trap-occupation mechanism.

Redox switching. A redox-active species—a discrete molecule in a polymer host, a redox polymer, or a unit on a conjugated backbone—is oxidised or reduced under bias, and the distribution of oxidation states sets the conductance. The state variable is then a chemical oxidation state with experimentally accessible potentials, reversible over many cycles in favourable cases; important examples include redox-active conjugated-polymer systems [94–96] and early electrochemical devices based on coordination complexes [97]. Not every redox process is cleanly reversible, though—side reactions, dimerisation, and slow degradation limit endurance—and purely redox devices tend to be bistable unless the response is distributed over many sites.

Conformational (structural) switching. In this case, the active layer contains a molecular species whose geometry, isomerisation state, or aggregation state can be modulated by an electric field, local heating, or coupled electrochemical processes. Examples include photochromic and electrochromic switches, as well as bistable molecular architectures whose conductance depends on molecular conformation [95, 96]. The state of these systems is related to a well-defined structural change, so they offer a high level of molecular design precision. However, the synthesis can be challenging, and the switching timescale is fixed by a particular molecular rearrangement.

Ionic redistribution in mixed ionic–electronic systems. This is the family most relevant to this thesis. The active layer combines an electronic

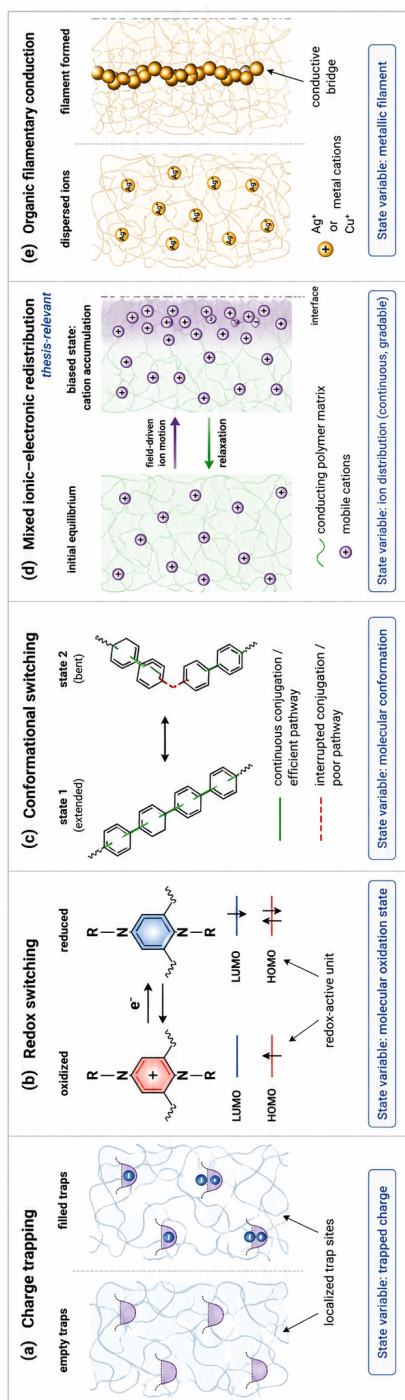


Figure 1.10: Molecular-level representation of the five mechanism families behind organic memristive behaviour. (a) Charge trapping: localised trap sites associated with defects, embedded nanoparticles, or interphase boundaries in a disordered polymer matrix, filled and emptied under bias. (b) Redox switching: a redox-active moiety alternating between its oxidised and reduced states, with the local electronic structure changing accordingly. (c) Conformational switching: a molecular unit toggling between two well-defined geometric or isomeric states with different conjugation or transport pathways. (d) Mixed ionic-electronic redistribution: a population of mobile cations within an electronically conducting polymer matrix, displaced and accumulated near an interface under bias and relaxing back toward equilibrium when the bias is removed. (e) Organic filamentary conduction: metallic cations (typically Ag^+ or Cu^+) from an active electrode migrating through the organic matrix to form a conductive bridge.

conductor—usually a conjugated polymer—with mobile ions held in an ion-conducting host. Under bias the ions move and accumulate near interfaces, modulating electronic transport through interfacial doping, electrostatic band bending, or electrochemical doping; the conductance change is gradual, multi-state, and directly tied to the amount and arrangement of displaced ionic charge [23]. The same physics underlies light-emitting electrochemical cells (LECs) [98–104], where redistribution forms a doped junction, and organic electrochemical transistors (OECTs) [105], where a gate drives the ion exchange. In the two-terminal memristive mode the driving bias itself plays the role of the gate, and the residual ionic distribution left after the bias is removed is what carries the memory of the stimulus.

The state variable is thus the distribution of mobile ions in the host, and relaxation after bias—governed by ion transport, the polymer binding environment, salt concentration, and morphology—sets the fading-memory process. The practical costs are that ionic motion is slower than electronic switching and that side reactions with the electrodes must be suppressed through chemistry and contact choice [106].

Organic filamentary conduction. Metallic species (typically Ag^+ or Cu^+ from an active electrode) migrate through the organic matrix and form conductive bridges similar to those found in inorganic CBRAM [69, 70]. These devices inherit the strengths of filamentary switching, such as large ON/OFF ratios and low switching voltages, but also its weaknesses, like stochastic filament formation and intra-device instability [107, 108].

The five families and their trade-offs are collected in table 1.4.

Table 1.4: Mechanism families behind organic memristive behaviour.

Family	State variable	Reversibility	Analogue gradation	Timescale tunability	Refs.
Charge trapping	Trap occupation	Partial	Limited (often binary)	Low	[93]
Redox	Oxidation state of a moiety	Variable	Bistable to few-state	Low–medium	[94, 97]
Conformational	Molecular geometry / isomer	Good (per design)	Bistable / few-state	Low (per molecule)	[95, 96]
Mixed ionic–electronic	Mobile-ion distribution	Good	Continuous	High (host chemistry)	[21–23]
Filamentary (organic CBRAM)	Metallic filament	Limited (stochastic)	Few states	Low	[107, 108]

1.5.2 Organic Precedents for the Composite Strategy

The relevant precedents are not a chronological checklist but a set of partial solutions to the same materials problem: how to combine ion-mediated conductance control, analogue reversibility, and a simple device geometry. The earliest two-terminal organic memristive devices are usually taken to be the polyaniline (PANI) electrochemical junctions of Erokhin, Berzina, and co-workers [109–112], in which a PANI channel interfaced with a poly(ethylene oxide)-based electrolyte is doped electrochemically by ionic redistribution under bias. This work already paired a conjugated polymer with an ion-conducting polyether host—essentially the materials logic extended throughout this thesis—and framed the device from the start as an adaptive, analogue-tunable element supporting history-dependent conductance changes.

The LEC route is especially close to the platform used here. Zakhidov’s *light-emitting memristor* [97], in which an ionic $[\text{Ru}(\text{bpy})_3]^{2+}$ complex supplied both the emitter and the mobile ions, showed that redistribution in a polymer host yields memristive behaviour, while resistive switching in simple films such as poly(vinyl alcohol) [93] confirmed that the phenomenon is accessible in cheap matrices but typically binary and mechanistically opaque. The best analogue control came from the electrochemical-transistor route: Gkoupidenis’ PEDOT:PSS OECTs reproduced short-term plasticity, paired-pulse facilitation, and long-term potentiation and depression [105, 113], Xu’s core–sheath nanowire transistors [114] and the Van de Burgt–Fuller organic electrochemical neuromorphic device (ENODE) [21, 22] delivered linear, symmetric updates through ion exchange between electrolyte and channel [23], and halide-perovskite-gated OECTs extended the accessible timescales [115]. Each such synaptic element, however, requires a third terminal and a contacted

electrolyte volume, which makes a self-contained two-terminal device an attractive complement for the reservoir paradigms of section 1.3.5.

The two-terminal line developed in parallel. Liu’s biomimicking redox organic memristor [94] demonstrated coexisting short- and long-term memory in a passive two-terminal device, but also exposed the redox route’s partial irreversibility, with the long-term regime sometimes involving permanent chemical change rather than controlled state evolution [116]. Molecular memristors, especially Goswami’s metal-coordinated azo-aromatic layers [95, 96], achieved contrast, endurance, and multi-state operation through a demanding synthesis; stretchable and flexible devices [90, 91, 117, 118] demonstrated mechanical compliance; and perovskite–organic hybrids [77, 119] and organic electrochemical reservoir computing [35] broadened the landscape further, the latter showing that mixed ionic–electronic devices can serve as volatile physical-reservoir elements.

By the early 2020s the field had matured into a subfield consolidating around shared physical principles [23, 120], with the individually demanding features each demonstrated—ion-mediated operation in two- and three-terminal forms, analogue symmetric programming, short-to-long-term transitions, mechanical compliance, molecular precision—but not yet combined in one simple platform. The three-terminal route bought reversibility and analogue control at the cost of architectural complexity, the two-terminal redox route simplicity at the cost of full reversibility, and the molecular route precision at the cost of synthesis. In that 2022 publication context, the specific gap addressed by the proof-of-concept chapter was a two-terminal, reversible, multi-state, ion-mediated device with interpretable chemistry built from standard organic-electronic components. The route pursued here fills that gap with a composite of a conjugated semiconducting polymer, an ion-conducting polymer host, and a dissociated alkali-metal salt. These parallel developments are collected

in fig. 1.11.

That framing should not be read as a frozen 2026 state-of-the-art claim. Since the proof-of-concept publication, organic neuromorphic electronics has advanced quickly, especially through organic electrochemical neurons and circuits for perception and biointerfacing [121], organic electrochemical-device systems for multimodal robotic learning [122], and organic-iontronic memristors for artificial synapses [123]. Those developments strengthen the larger premise that mixed ionic–electronic organic devices are useful neuromorphic substrates. They do not, however, change the role of the present thesis, which is narrower: to use a self-contained two-terminal polymer-electrolyte composite as a chemically tunable fading-memory element for temporal computing.

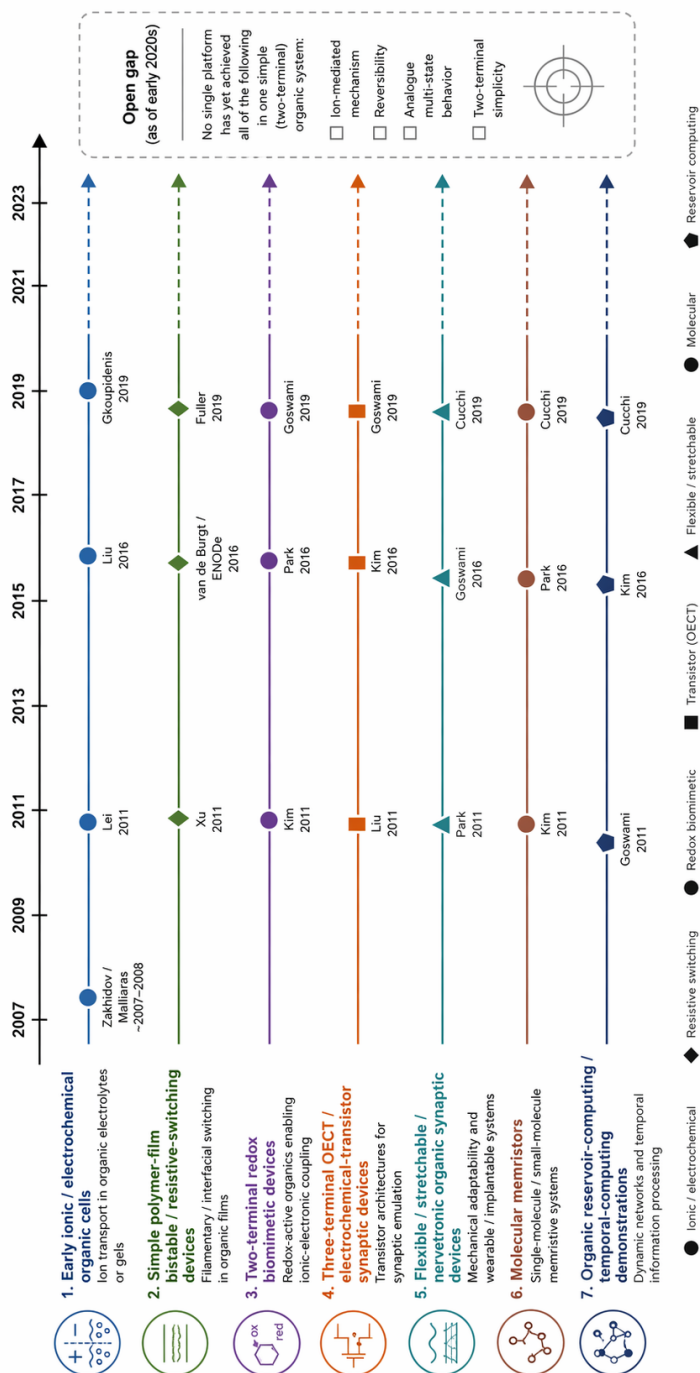


Figure 1.11: Historical map of organic memristive and organic synaptic devices, arranged by mechanism family and by year of representative publication. The map frames the literature context of the 2022 proof-of-concept publication: by the early 2020s, several demanding features of a neuromorphic organic device had each been demonstrated, but the specific combination targeted here was a self-contained two-terminal platform combining reversibility, multi-state analogue behaviour, and an interpretable ion-mediated mechanism.

1.6 Polymer Electrolyte Composites: The Enabling Strategy

The composite platform follows directly from the mixed ionic–electronic precedents above. It places a semiconducting conjugated polymer, an oxygen-rich ion-conducting polymer host, and a dissociated alkali-metal salt in one solution-processed active layer, then operates that layer in a standard two-terminal stack. Electronic transport, ion solvation, and mobile-charge supply are therefore assigned to separate components that can be changed independently.

1.6.1 Ion-Conducting Polymer Chemistry

An ion-conducting polymer is a macromolecular host in which dissolved or dispersed ions can move under an applied electric field without the help of a liquid solvent phase. Poly(ethylene oxide) (PEO) established the model system: alkali-metal salts were first shown to form solid complexes with PEO in 1973 [124], ionic conductivity was measured in those complexes in 1975 [125], and the resulting solid polymer electrolytes became practical ion conductors through the late 1970s and early 1980s [87, 88, 92]. Their relevance here is not battery performance but control of cation coordination, segmental motion, host crystallinity, and composition—the same variables that set memristive relaxation.

The polyether host has a high density of ether oxygens along the backbone or side chains. In PEO, those oxygens appear regularly along the $-\text{CH}_2-\text{CH}_2-\text{O}-$ repeat unit, and the chain conformation in the amorphous phase allows each cation to coordinate several ether oxygens from one or more polymer chains [92, 126]. Hybrane-type hyperbranched polyester-amide hosts arrange the oxygen density differently, through ether and ester groups attached as side chains on a branched polymer core [127], but the

coordination chemistry is of the same kind.

In both hosts, the oxygen atoms play the same chemical role. They act as hard Lewis bases that solvate alkali cations, stabilise salt dissociation, and define the local coordination environment of the cation. The anion, in contrast, is usually held more weakly (mainly through electrostatic compensation) and tends to be more mobile than the cation within the same host [128]. This is especially the case for soft, charge-delocalised anions such as triflate (CF_3SO_3^-) or bis(trifluoromethanesulfonyl)imide (TFSI, $\text{N}(\text{SO}_2\text{CF}_3)_2^-$).

Ion transport in these hosts is usually described through three related microscopic phenomena:

- From the *vehicular* side, a cation and its solvation shell move together through the polymer matrix, so long-range transport is coupled to polymer motion [129].
- From the *hopping* or site-to-site side, the cation instead transfers between coordination sites through a local rearrangement of oxygens, without macroscopic motion of the host. This last view is conceptually close to Grotthuss-type transport in proton conductors [130], even though the chemistry of the transfer itself is different.
- Finally, from the *segmental-motion-assisted* side, short polymer segments reorganise around the cation, creating and destroying coordination sites that allow effective diffusion [131, 132].

These three phenomena are not mutually exclusive, although the weight of each contribution during transport depends on host chemistry, cation identity, salt concentration, and temperature [87, 92].

Examining device dynamics more closely, the macroscopic ionic mobility of a polymer electrolyte generally correlates with the segmental mobility

of the host material. This segmental mobility is determined by the glass transition temperature T_g and by whether the host remains amorphous during operation. Below T_g , segmental motion is essentially frozen and ionic conduction drops. Above T_g , it is activated, and ionic transport becomes measurable, with the temperature dependence in this regime typically following the Vogel-Fulcher-Tammann form characteristic of fragile glass-formers [87, 133].

Even above T_g , however, not all of the host contributes equally. Crystalline regions contribute little to conduction and can even trap ions in ordered environments [126, 132]; the textbook picture is that conduction proceeds essentially through the amorphous fraction, although it has since been shown that specific crystalline stoichiometric phases can themselves support measurable ionic motion [134]. Low-crystallinity or fully amorphous hosts are therefore preferred for ion transport. A low T_g and a low crystalline fraction can be pursued through copolymerisation, branching, or plasticisation; hyperbranched architectures such as the Hybrane family are attractive here because branching itself suppresses crystallinity [135]. For room-temperature memristive operation the host should remain amorphous and mobile under ambient conditions, ideally without added plasticisers.

The strength of cation binding depends directly on the identity of the cation. For this matter, the concept of hard-soft acid-base (HSAB) reasoning [89] becomes important. Ether and ester oxygens in polyether and polyester-amide hosts are hard Lewis bases, meaning that their donor pairs are localised, their polarisability is low, and their coordination to alkali cations is primarily electrostatic. Among the alkali cations, Li^+ is the hardest Lewis acid (small, with a high charge density), and should therefore interact most strongly with hard oxygen donors. Na^+ and K^+ are progressively softer, with larger radii, lower charge densities, and weaker electrostatic coupling to the same oxygen environment.

This means that, in a common polyether host and at fixed composition, the residence time of the cation at a coordination site might be expected to follow the coordination-strength order $\text{Li}^+ > \text{Na}^+ > \text{K}^+$. That expectation provides a hypothesis for the devices' fading-memory timescale: harder cations should relax more slowly, softer cations more quickly. The experimental chapters test this hypothesis rather than assume it; in the archive used here, composition is the replicated quantitative variable, while cation order is not robust across host and anion. The coordination geometry, the three transport mechanisms, and the cation–oxygen binding hierarchy that motivate the hypothesis are summarised in fig. 1.12.

A second set of design parameters is provided through composition. A polymer electrolyte is defined not only by host and ion identity, but also by the salt-to-host ratio; in a composite device, the polymer-to-semiconductor ratio plays a role as well. The salt-to-host ratio sets the number and spatial density of occupied coordination sites, while the host-to-semiconductor ratio sets the volume available for ionic transport and the connectivity of the electronic phase. The two are therefore treated as independent composition axes in the later chapters.

These considerations motivate the host choices made in this thesis. Hybrane DEO750 8500 is a hyperbranched polyester-amide with a high density of ether and ester oxygens. It provides an amorphous, low-crystallinity ion-transport host and is used in chapter 2 as the host of the proof-of-concept Super Yellow (SY)/Hybrane/lithium triflate (LiOTf , LiCF_3SO_3) composite, where it supports the full synaptic functionality demonstrated in a single device.

PEO, in turn, acts as the central host for the broader comparative work. Its chemical role is similar to that of Hybrane: ether oxygens coordinate alkali-metal cations, segmental motion enables transport, and composition can be varied systematically while the electronic semiconductor is held fixed. The reason for choosing it as the anchor of the compar-

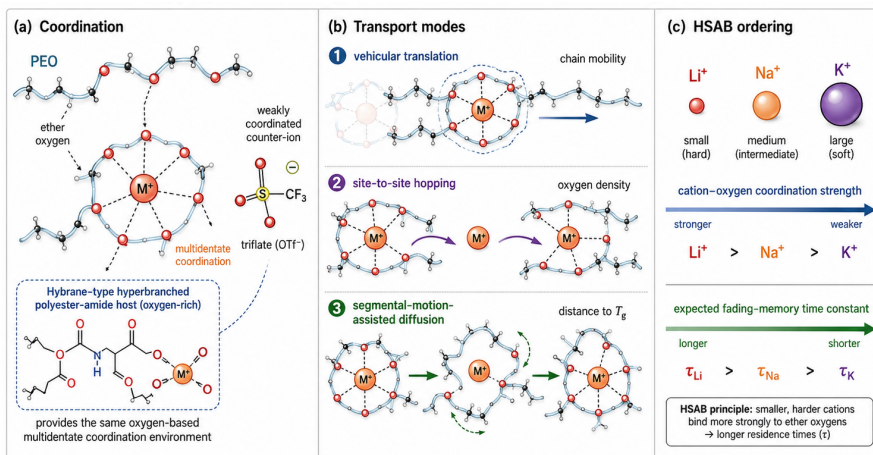


Figure 1.12: Molecular-level chemistry of ion transport in the polyether-hosted polymer-electrolyte composite. (a) A short poly(ethylene oxide) segment coordinates an alkali cation through four to six ether oxygens, with the triflate anion held nearby as a weakly coordinated charge-compensating counter-ion; the same coordination chemistry applies to the hyperbranched polyester-amide hosts used in chapter 2 [92, 126, 128]. (b) Long-range ionic transport in this host proceeds by three coupled contributions: vehicular translation of the cation together with its solvation shell, site-to-site hopping between coordination pockets through local oxygen-cage rearrangement, and segmental-motion-assisted diffusion in which short polymer segments reorganise to create and destroy coordination sites [129, 131, 132]. (c) Within a common polyether host, hard-soft acid-base reasoning ranks the alkali cations along the $\text{Li}^+ > \text{Na}^+ > \text{K}^+$ series in decreasing cation-oxygen coordination strength [89]. This ordering motivates the cation-timescale hypothesis tested later, but the thesis does not assume it as a host- and anion-independent law.

ative corpus, however, is not just chemical analogy. PEO is the most thoroughly characterised solid polymer electrolyte in the literature, with its phase diagrams, coordination geometries, conduction mechanisms, and salt-dependent dynamics having been mapped across decades of work [87, 92, 125, 126, 132]. Anchoring the replicated composition series in PEO therefore places the measured dynamics on a host whose structural and transport behaviour is already independently constrained.

A hyperbranched polyether, trimethylolpropane ethoxylate (TMPE), is used alongside PEO as a contrasting host in that comparison, so that the linear and branched polyether architectures can be set against each other directly. Host, anion, and cation substitutions are surveyed at smaller sample sizes rather than treated as material laws.

1.6.2 LEC-Derived Composite Operation

The coupling between ionic redistribution in a polymer host and electronic transport in a conjugated semiconductor can be both stable and reversible, and polymer light-emitting electrochemical cells (LECs) are the clearest precedent for it [136]. An LEC is a simple two-terminal sandwich [98]: a conjugated light-emitting polymer is mixed from a common solution with an ion-conducting polymer and a dissociated salt. Under bias, cations migrate toward the cathode and anions toward the anode, forming doped regions that inject carriers into the undoped bulk, and the resulting dynamic p-i-n profile produces electroluminescence in the intrinsic region between the doped edges [101].

Two models were historically used to describe this operation. In the electrochemical-doping model [98, 137], the dominant effect is bulk doping of the conjugated polymer; in the electrodynamic model [99, 100], it is the lowering of charge-injection barriers by ion accumulation at the contacts. Later experiments [101, 103] and a unifying framework [102] showed these to be limiting cases of the same ion redistribution under different injection regimes, not contradictory mechanisms. The two models, their reconciliation, and the bias-off configuration that the memristive composite reuses are compared on a common cathode–anode position axis in fig. 1.13; studies of electrochemical side reactions [106] additionally identify the degradation pathways that must be suppressed.

For the present purpose, the lessons are twofold. First, this coexistence

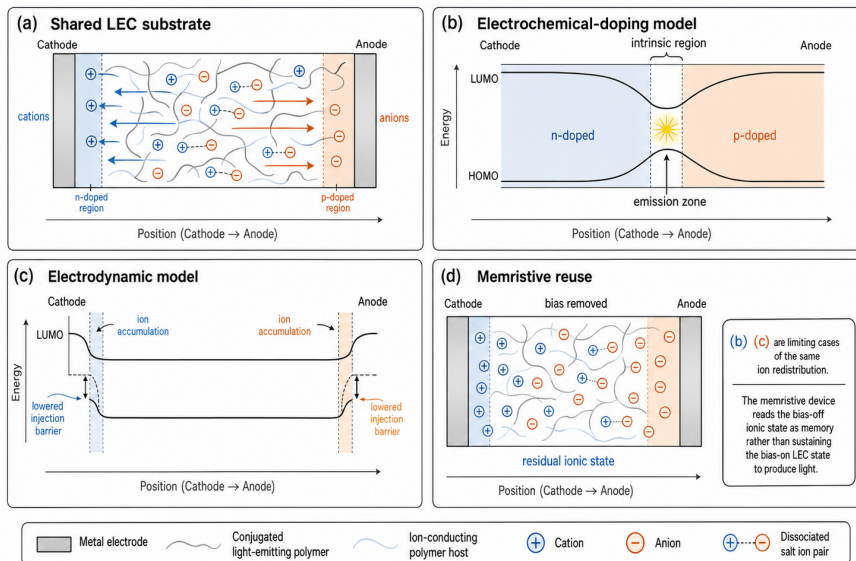


Figure 1.13: Light-emitting electrochemical cell operating models and their memristive reuse, drawn on a common cathode–anode position axis. (a) Shared substrate: under sustained bias, cations drift toward the cathode and anions toward the anode, forming doped regions at the two contacts. (b) Electrochemical-doping model: the dominant effect of the ionic redistribution is *bulk doping* of the conjugated polymer, producing a spatially wide p-i-n profile with emission localised at a narrow intrinsic region. (c) Electrodynamic model: the dominant effect of the same redistribution is *injection-barrier lowering* at the metal/polymer interfaces, producing thin ion-accumulation layers at the contacts and an essentially flat-band bulk. The two pictures were later reconciled as limiting cases of the same redistribution under different injection regimes. (d) Memristive reuse: with the bias removed, the residual ionic distribution left behind by the previous bias is the *stored internal state* of the device.

is durable: LECs operate over useful windows with lifetimes of hundreds to thousands of hours [104, 138], without destructive side reactions within a few cycles, provided the contacts are engineered—through metal choice and operating protocol—for repeated charge injection and withdrawal [139–141]. Second, the electrolyte plays the same role in both devices, supplying mobile ions whose redistribution modulates the conju-

gated polymer rather than conducting electrons itself; only its *use* differs. An LEC holds the bias on so the p-i-n profile sustains emission, whereas a memristive device writes an ionic distribution and reads the residual distribution left after the bias is removed as its stored state. That same redistribution can therefore plausibly serve as the internal state variable of a two-terminal memory device.

The most direct precedent is the Hybrane-based composite that anchors the proof-of-concept device of chapter 2. Mardegan and co-workers [104] demonstrated stable LECs based on Hybrane hyperbranched polyester-amide hosts blended with a conjugated light-emitting polymer, reporting long operational lifetimes under continuous bias. Their work validates the semiconducting-polymer / ion-conducting-host / salt architecture, the solvent system, and the processing protocol that chapter 2 then redeploys under pulsed bias to operate the same composite as a memristive device.

1.6.3 Rationale for the Composite Approach

The composite assigns the three roles—electronic transport, ion solvation and transport, and mobile-charge supply—to separate components rather than asking a single material to perform all three [98]: a conjugated polymer, an oxygen-rich host, and a dissociated salt, so that salt, cation, or blend ratio can be changed independently of the rest of the stack [23]. The salt is the clearest control variable: a small mass fraction that fixes the identity and density of the mobile ions writing the internal state. By the HSAB argument of section 1.6.1, the harder alkali cations bind ether oxygens more strongly and should be slower to displace, whereas charge-delocalised anions such as triflate interact only weakly and redistribute more easily [128]. This mobility asymmetry is the proposed origin of the device’s memory: under bias both species move but relax at different rates, so a fast component (mobile anions, near the contacts) and a slower

component (more tightly bound cations) let a single device retain traces of past stimulation over multiple timescales [109, 111, 112], in analogy with the short- and long-term plasticity of section 1.2.2. The cation is therefore a natural hypothesis for the slow response, but not a conclusion that can be assumed independently of host, anion, electrode, and drive protocol.

Composition adds a second, more robust level of control. The salt fraction sets the number of mobile ions and hence the achievable size of the conductance change [87, 88], while the host fraction sets the volume and connectivity of the segmentally mobile phase in which they rearrange; in principle these vary the amplitude and the rate of the response separately, though the comparative chapter treats their independence as an experimental question. The architecture is also experimentally convenient, since the three components share a common solvent: a new formulation is just a change of solution ratios deposited on the same electrode geometry, with no redesign of the stack. This makes a replicated composition grid realistic, while host, anion, and cation substitutions are surveyed as lower-power chemical contrasts.

Devices with the same polymers but different blend ratios should exhibit a spread of ionic relaxation times, with sample-limited chemistry and ordinary device-to-device variation adding further diversity. In a common circuit this family acts as a bank of temporal kernels with distinct decay constants [5, 80]—a liability for storing a precisely trained weight, but a resource for covering a range of fading-memory timescales [32–35]. The application chapter develops this idea quantitatively from the measured device behaviour.

The experimental chapters test this rationale first with the Hybrane/LiOTf proof-of-concept device of chapter 2, and then with the broader PEO-based corpus in which composition is varied as the replicated axis and chemistry is surveyed as illustrative side evidence. Figure 1.14 summarises

this concept used throughout the thesis, linking the two-terminal stack to the qualitative timescale ladder originally expected from cation and composition chemistry.

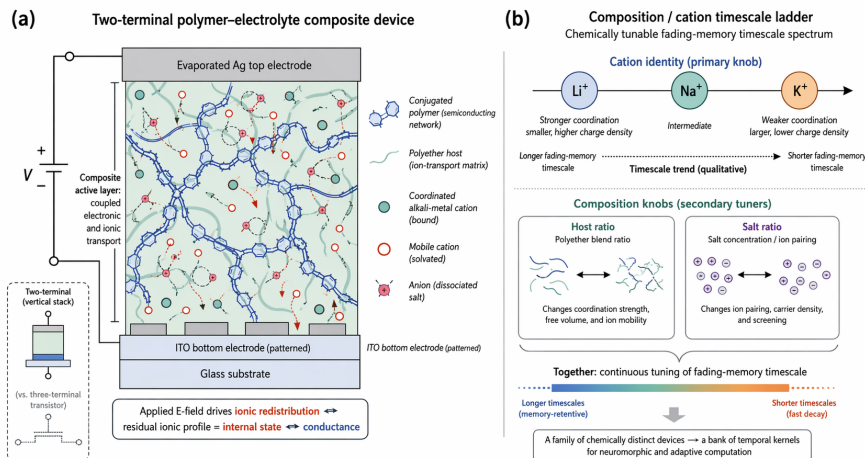


Figure 1.14: The thesis-specific polymer-electrolyte composite device and its chemically tunable dynamical timescale. (a) Vertical two-terminal stack: a solution-processed composite active layer, combining a conjugated semiconducting polymer network with an oxygen-rich ion-conducting host, coordinated alkali-metal cations, and a dissociated salt, is sandwiched between a patterned ITO bottom electrode and an evaporated silver top electrode; the internal state of the device is set by the residual ionic distribution that remains after a driving bias is removed. (b) Qualitative timescale ladder on which cation identity and the host/salt composition ratios were expected to place the fading-memory time constant of the device. Chapter 4 tests this expectation and shows that composition is the replicated quantitative variable, while the cation ordering is not robust across the available host/anion families.

From Background to Device

The resulting device target is specific: a solution-processed composite of a conjugated semiconducting polymer, an oxygen-rich ion-conducting host, and a dissociated alkali-metal salt, operated as a two-terminal device

whose residual ionic distribution is its internal state. The thesis-level motivation, scientific questions, objectives, and chapter structure are set out in the front-matter roadmap rather than repeated here. The next chapter takes the first step in that programme: the proof-of-concept SY/Hybrane/LiOTf device of chapter 2, its memristive and synaptic functionality, and the mechanistic interpretation that paves the way for the comparative and computational work that follows.

Chapter 2

Proof of Concept: Polymer-Based Composite Organic Memristive Devices

2.1 Introduction and Motivation

This chapter reports the proof-of-concept polymer-electrolyte memristive device announced at the end of chapter 1. The composite combines a semiconducting poly(phenylene vinylene) (PPV) derivative (*Super Yellow*, SY), an oxygen-rich hyperbranched polyester-amide ion-conducting host (*Hybrane*[®] DEO750 8500), and a dissociated lithium salt (lithium triflate, LiOTf, LiCF₃SO₃), all processed from a common solvent into a standard two-terminal (2-T) sandwich on indium tin oxide (ITO) and silver. The chapter has two purposes within the thesis. First, it tests whether the materials logic of section 1.6 translates into a working device that supports the full set of synaptic functions in a single platform. Second, it isolates the mechanistic ingredient (the cation–host coordination chemistry) that the comparative work of Chapter 4 then tests across cations and compositions.

At the start of this work the literature already contained several partial demonstrations of these functions in different architectures (reviewed in section 1.5.2): ion-mediated analogue switching in multi-layer polyaniline junctions [109, 111, 112], bistable ionic redistribution in a light-emitting memristor [97], tunable short-to-long-term transitions in a partially-irreversible redox device [94, 116], and reversible analogue control in three-terminal organic electrochemical transistors [21, 22, 105, 114]. Each covered part of the specification developed in section 1.2.3 (analogue, reversible, multi-state, low-energy conductance changes set by electrical pulses), but their combination in a single two-terminal organic device with an interpretable ion-mediated mechanism remained, at that point, an open problem.

The active-layer logic adopted here is the composite logic of section 1.6.3. The three functional roles — electronic transport, ionic transport, and mobile-charge supply — are assigned to three different components, so that each role can be tuned chemically without redesigning the rest of the stack. *Super Yellow* supplies the electronic-conductance pathway. *Hybrane* supplies an amorphous, low-crystallinity ion-transport host with a high density of hard oxygen donors. LiOTf supplies a hard alkali cation paired with a soft, charge-delocalised anion, so that the cation–host and anion–host interactions are asymmetric in the hard–soft acid–base sense of Pearson [89]. The mechanistic account developed in section 2.5 attributes the two memory regimes to this asymmetry.

The chapter shows that the composite supports, in a single 2-T platform, a full set of synaptic functions that fall into two families. The first family comprises three dynamical measurements: current–voltage hysteresis characteristic of memristive switching, potentiation as a function of the number of applied voltage pulses, and depotentiation as a function of the delay before read-out. These three measurements recur, on the same 2-T device geometry, throughout the comparative work of Chapters 4

and 5, and are introduced here as an explicitly named set. The second family comprises extended synaptic protocols that build on the first and are specific to this chapter: excitatory post-synaptic current (EPSC) responses, demonstrated here with four readily distinguishable readout states; the resolution of the depotentiation measurement into a short-lived and a longer-lived memory regime, denoted short-term memory (STM) and long-term memory (LTM) following the convention of the artificial-synapse literature [23, 90]; and spike-timing dependent plasticity (STDP) with a time constant in the biologically realistic range [55, 58]. The interpretation tying these observations together is developed in section 2.5: the fast component is associated with displacement of the weakly bound triflate anion, while the slow component is associated with displacement of the lithium cation from its hard oxygen coordination environment. The same interpretation yields the cation-substitution prediction tested in Chapter 4.

Within the arc of the thesis the present chapter is narrow in role: it is the only chapter in which the full synaptic suite is treated as the main evidence, while only the three dynamical measurements named above are carried forward to the comparative corpus. The two specific handoffs onto the rest of the thesis are taken up in the closing section.

2.2 Materials Design and Composite Formulation

2.2.1 Design Rationale

The operating principle of the device is the same one developed in section 1.6.3: ionic migration within a solid-state polymer matrix under an applied electric field redistributes mobile ions near the electrodes,

which in turn modifies the energy barriers for charge injection and the doping level of the semiconducting polymer, and therefore the electronic conductance of the device. The composite active layer is built to satisfy three functional requirements simultaneously:

1. **Electronic-conductance pathway:** a semiconducting polymer provides the medium through which the electronic current flows, and whose conductivity is susceptible to modulation by ionic redistribution and electrochemical doping.
2. **Ion-transport medium:** a polymer electrolyte provides a structural environment in which dissolved ions remain mobile under an applied field, yet are retarded enough by intermolecular interactions that, once the field is removed, the conductance change persists for seconds — long enough to be read out — before fading back toward equilibrium.
3. **Ion source:** a dissociated salt provides the mobile charge carriers (a cation and an anion) whose differential response to the field is proposed to underlie the distinct memory regimes reported in section 2.4.

The composition adopted here is built around a deliberate asymmetry in Lewis acid–base character between the two ions of the salt. LiOTf is chosen because Li^+ is a hard Lewis acid with a high charge-to-radius ratio ($\lambda = z/r \approx 1.3 \text{ \AA}^{-1}$ – $1.7 \text{ \AA}^{-1}$ across the four- to six-fold oxygen coordination that Li^+ adopts in polyether hosts, with r_{Li^+} in the range $0.59 \text{ \AA}$ – $0.76 \text{ \AA}$ for the corresponding Shannon effective radii [142]), whereas the triflate anion CF_3SO_3^- is a soft Lewis base with a diffuse, low-charge-density anionic centre. The ion-transport polymer, *Hybrane*, contains a high density of ether and ester oxygen atoms that act as hard Lewis bases. Under the hard–soft acid–base (HSAB) reasoning of Pearson [89]

already used in section 1.6.1, hard acids associate preferentially with hard bases. The Li^+ cations are therefore tightly coordinated by the oxygen atoms of *Hybrane*, while the CF_3SO_3^- anions interact only weakly with the host. Under bias, the weakly coordinated triflate anions are displaced at low voltage, while the strongly coordinated lithium cations require a substantially higher field to break free of their coordination environment. This differential mobility is proposed as the molecular origin of the two memory regimes (short-term memory and long-term memory) reported below.

2.2.2 Super Yellow: The Semiconducting Polymer

Super Yellow (commercial PDY-132, Merck) is the semiconducting PPV-family copolymer in the blend. Its role in the proof-of-concept device is to provide the electronic-conductance path whose injection barriers and local doping state can be modulated by mobile ions. The relevant material facts for the main argument are its solution processability in cyclohexanone, its established use in organic light-emitting devices [86], and frontier levels of approximately $E_{\text{HOMO}} = -5.4\text{ eV}$ and $E_{\text{LUMO}} = -3.0\text{ eV}$ [143]. The full copolymer name and component-level notes are collected in section A.1.

2.2.3 Hybrane DEO750 8500: The Ion-Transport Polymer

Hybrane is an oxygen-rich hyperbranched polyester-amide [127]. In this chapter it is used as the ion-transport host: ether and ester oxygen atoms provide hard Lewis-base coordination sites for alkali cations, while the hyperbranched architecture suppresses crystallisation and keeps the blend amorphous. Ion motion is therefore treated as segmental-motion-assisted hopping between local coordination environments, in the polymer-

electrolyte picture introduced in section 1.6.1. The relevant migration activation scale is of order 50 kJ mol^{-1} for polymer electrolytes near room temperature [87]; the longer discussion of host structure and transport is in section A.1.

2.2.4 Lithium Triflate: The Ionic Salt

LiOTf (lithium trifluoromethanesulfonate, LiCF_3SO_3) supplies the mobile ions. It is compatible with the cyclohexanone-processed *Super Yellow/Hybrane* blend and is widely used in polymer-electrolyte and LEC systems [98, 103]. Mechanistically, its value is the contrast between Li^+ , a small hard Lewis acid, and CF_3SO_3^- , a charge-delocalised soft anion. The chapter uses that asymmetry to interpret the two relaxation regimes, while the detailed ion-pairing rationale is kept in section A.1.

2.2.5 Composite Formulation and Preparation

The active layer uses a mass ratio of *Super Yellow:Hybrane:LiOTf* = 1:0.30:0.09, processed from cyclohexanone under dry nitrogen. After mixing, the final solution concentrations are 8.72 mg mL^{-1} (*Super Yellow*), 2.62 mg mL^{-1} (*Hybrane*), and 0.78 mg mL^{-1} (LiOTf). The ratio is the optimised formulation reported in the published proof-of-concept work [6]: higher salt contents degraded film homogeneity, whereas lower salt contents reduced the available mobile-carrier density. Detailed solution preparation and optimisation notes are given in section A.1. The same formulation had also been validated in related *Super Yellow/Hybrane*/lithium-salt LECs with long electroluminescent operation [104].

2.3 Device Architecture and Fabrication

2.3.1 Device Geometry: The Two-Terminal Vertical Structure

The device adopts a vertical sandwich geometry, in which the active layer is placed between a bottom transparent electrode and a top metallic electrode (see fig. 2.1). This 2-T architecture is the simplest memristive configuration available: it contains no gate electrode and requires no transistor to address the device, in contrast to the *three-terminal* (3-T) organic synaptic transistors discussed in section 1.5.2 [105, 113, 114].

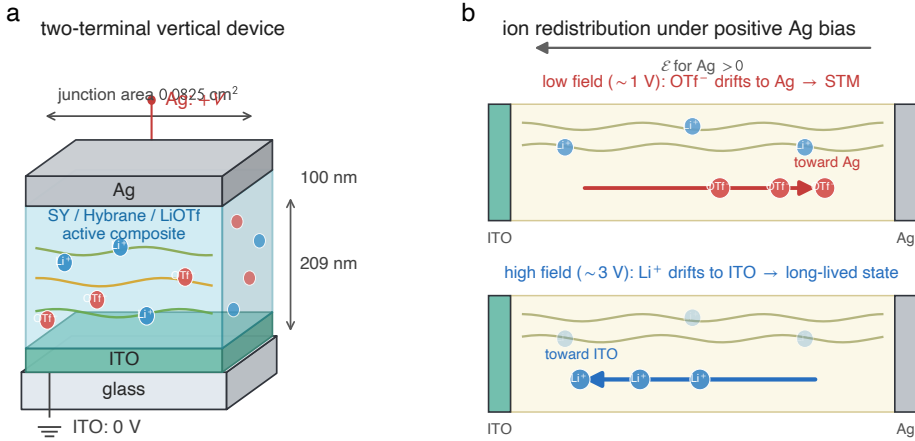


Figure 2.1: Schematic of the two-terminal vertical synaptic device and active-layer mechanism. The device stack is ITO-glass / *Super Yellow/Hybrane*/LiOTf composite / Ag; the composite separates the electronic pathway, ion-transport host, and mobile-ion supply. The proposed mechanism assigns the faster STM response to weakly coordinated triflate redistribution and the longer-lived state to displacement of strongly coordinated Li^+ . Schematic generated from the device geometry and mechanism reported in Ref. [6].

The 2-T choice matters for two reasons. First, it reduces fabrication

complexity, given that the device is made by a sequence of only three depositions (ITO substrate, spin-coated active layer, evaporated metal), which makes the same stack compatible with the comparative corpus of Chapter 4. Second, a 2-T device that shows coexisting short-term memory and long-term memory without an independent gating signal shows that the memory physics is intrinsic to the active material, rather than being supplied externally by a gate field acting on a separate channel.

The bottom electrode is a pre-patterned layer of ITO deposited on borosilicate glass. ITO combines optical transparency with adequate electronic conductance ($\sigma \approx 10^4 \text{ S cm}^{-1}$) [144], allowing optical inspection of the active layer and characterisation of the device stack by techniques such as UV-Vis spectroscopy and electroluminescence imaging. The work function of ITO ($\Phi_{\text{ITO}} \approx 4.7 \text{ eV}$) is well matched to the ionisation potential of PPV-family semiconductors [145], facilitating hole injection into the active layer.

The top electrode is silver (Ag), deposited by thermal evaporation. Ag was selected over gold for reasons of cost and practical feasibility; its work function ($\Phi_{\text{Ag}} \approx 4.3 \text{ eV}$) creates a modest electron injection barrier with the *Super Yellow* conduction band, consistent with asymmetric injection conditions often observed in LEC-type structures [99]. The junction active area defined by the shadow mask geometry is 0.0825 cm^2 .

2.3.2 Fabrication and Active-Layer Morphology

ITO-patterned glass substrates were cleaned by sequential ultrasonication, dried under nitrogen, and transferred into a nitrogen glovebox. The composite solution was spin-coated at 2000 rpm for 60 s and annealed at 75°C for 3 h. Contact profilometry gave an active-layer thickness of 209 nm at this spin speed, and tapping-mode AFM showed smooth films with RMS roughness in the range 1 nm–3 nm across the investigated

processing window [6]. The spin-speed optimisation, profilometry method, AFM parameters, and supporting morphology evidence are reported in section A.3.

The top electrode was 100 nm of Ag deposited by thermal evaporation through a shadow mask. The mask defines a junction area of 0.0825 cm^2 . Detailed cleaning and evaporation parameters are collected in section A.1; the main point for the chapter is that the complete stack is fabricated by solution processing plus one metal evaporation step.

2.4 Electrical Characterisation

Electrical measurements used a Keithley 2450 SourceMeter in a custom probe station. Unless otherwise stated, measurements were made under nitrogen with voltage applied to the Ag top electrode relative to grounded ITO. Protocol-level details are given in section A.2.

The characterisation follows the two-family distinction of section 2.1. Sections 2.4.1 to 2.4.3 report the three dynamical measurements carried forward to Chapters 4 and 5; sections 2.4.4 and 2.4.5 then report the device-specific extended protocols, the EPSC multi-state readout and spike-timing dependent plasticity. The short- and long-term memory regimes (short-term memory/long-term memory) are read off the variable-delay depotentiation measurement and so appear with it in section 2.4.3.

2.4.1 Current-Voltage Hysteresis

The fundamental electrical fingerprint of a memristive device is a current-voltage (I - V) hysteresis loop with a characteristic pinched appearance at zero voltage [59–61]. This shape arises because, at $V = 0$, the net current is zero regardless of the history of the device (a memristor cannot

spontaneously drive current), yet the *slope* of the I – V curve at zero voltage (i.e., the instantaneous conductance) depends on the history of applied voltage.

Figure 2.2 shows the I – V curves obtained by applying successive triangular voltage sweeps to the device at a sweep rate of 0.25 V s^{-1} , with the voltage ramped from 0 V to a maximum of 1.2 V and back to 0 V (single polarity, positive). A waiting period of 10 s was observed between successive sweeps to allow partial ionic relaxation. Ten sequential sweeps are shown.

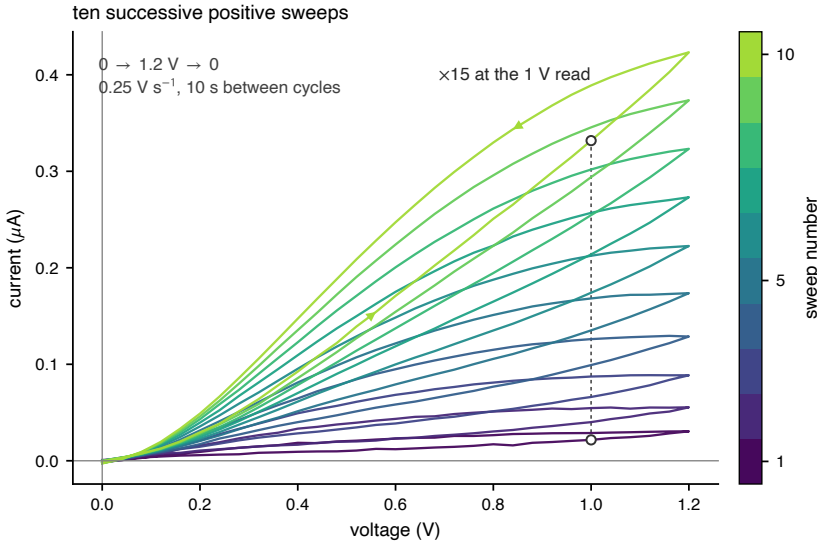


Figure 2.2: Current–voltage hysteresis of the composite memristive device. Ten successive positive triangular sweeps (0 V–1.2 V, 0.25 V s^{-1} , 10 s between cycles) fan out in near-uniform increments, the forward current at 1 V growing roughly 15-fold from the first to the tenth cycle. The clockwise loop closes at the origin (pinched hysteresis) without abrupt switching. Representative SY/Hybrane/LiOTf device (NM_v025) of the published 1:0.30:0.09 formulation, reproducing the hysteresis of Ref. [6].

The curves show four features:

Progressive, well-resolved increase of conductance. The current at any fixed voltage *increases* with each successive sweep, in well-separated increments that persist across the whole sequence rather than saturating after the first few cycles. At 1 V the forward current grows from 0.022 μA on the first sweep to 0.33 μA on the tenth — an on/off ratio of approximately 15 across ten cycles — with the cycle-to-cycle step staying substantial throughout (+85 % from cycle 1 to cycle 2, and still +13 % from cycle 9 to cycle 10). This continuous, finely graded modulation, spread over many distinguishable levels, is the hallmark of analogue rather than binary operation. It reflects the progressive redistribution of ions toward the electrodes with each sweep: each successive cycle leaves the ions in a slightly more asymmetric configuration than the previous one, producing a slightly lower resistance state.

Hysteresis without current rectification. In all cycles, the return sweep (decreasing voltage) produces lower current than the forward sweep (increasing voltage) at the same voltage, generating the characteristic clockwise hysteresis loop for positive voltage sweeps. This is distinct from simple rectification (diode behaviour): a diode would produce the same forward and reverse I – V curves; the hysteresis here reflects a time-dependent internal state variable (the ionic configuration) that lags the applied voltage.

Absence of abrupt switching. Unlike digital resistance switching (SET/RESET transitions), in which the resistance changes abruptly by orders of magnitude at a threshold voltage, the conductance here evolves continuously and smoothly with each cycle. This *analogue* gradation matters for the synaptic role outlined in section 1.2.3, since it allows the device state to be tuned to one of many distinguishable intermediate levels rather than being confined to two (or a few) binary states.

Behaviour under negative polarity. Application of negative voltage sweeps (not shown individually in the main figure) produces a qualitatively different hysteresis: the current decreases rather than increases with successive negative cycles. This has been attributed to electrochemical side-reactions in the vicinity of the ITO electrode under large negative bias, which generate a thin insulating layer at the ITO/active layer interface analogous to the behaviour reported in MEH-PPV-based LECs [103, 106]. This effect is not detrimental to memristive operation when devices are used in the single-polarity positive regime.

2.4.2 Variable-Number-of-Pulses Potentiation

Potentiation by a train of write pulses is the second of the three dynamical measurements introduced at the start of this section. Two complementary experiments are reported: a reversible potentiation/depression cycle that establishes the analogue, multi-state character of the conductance, and a measurement of the conductance change as a function of the number of applied pulses.

Reversible Potentiation and Depression

The ability to set the device conductance to a specific value by applying a controlled number of voltage pulses — and to subsequently decrease it back toward the initial value by pulses of opposite polarity — is demonstrated by the potentiation/depression experiment shown in fig. 2.3a.

A pulsed voltage protocol was applied: 50 consecutive write pulses of 1 V amplitude and 0.1 s duration, followed by 50 write pulses of -2 V. The conductance was monitored at the peak of each pulse (i.e., at the 1 V and -2 V levels, respectively). This protocol is analogous to the repeated pre-synaptic stimulation of a biological synapse and its subsequent inhibition.

The results demonstrate that the conductance can be continuously potentiated from its initial value by more than 200 % during the 1 V pulse train, and subsequently depotentiated back toward the initial value during the -2 V train. The potentiation and depression curves are smooth and monotonic, without any abrupt transitions. The full reversibility of the process — the conductance returns to a value near the initial level after the complete depression protocol — confirms that the underlying ionic redistribution is non-destructive and does not permanently alter the chemical composition or morphology of the active layer.

The energy cost of each synaptic event (one write pulse of 1 V, 0.1 s) can be estimated from the average current during the pulse. The resulting energy per event is approximately 50 nJ, or approximately $6 \text{ fJ } \mu\text{m}^{-2}$ of device area (the 0.0825 cm^2 junction). This value is consistent with those reported for organic synaptic devices operating in the picojoule-to-nanojoule range [77], and comparable to the picojoule-to-nanojoule switching energies reported for inorganic memristive devices [68].

Quantitative Pulsed Read–Write Protocol

To quantify the potentiation as a function of pulse number, and to separate the act of writing from the act of reading, a non-perturbing read protocol was adopted; the same protocol is reused for the depotentiation measurement of section 2.4.3. The read–write–wait sequence, detailed in section A.2.1, comprises three steps:

1. **Read.** A reading pulse ($V_{\text{read}} = 0.5 \text{ V}$) is applied to establish the initial conductance G_0 . This reading voltage is intentionally low: the activation threshold for ionic redistribution in this system was determined by independent experiments to lie above 0.7 V [6], so $V_{\text{read}} = 0.5 \text{ V}$ measures the conductance state without perturbing it.
2. **Write.** A train of N_{pulses} write pulses (V_{write} , duration 0.1 s each)

is applied.

3. **Wait and read.** After a waiting time t_{wait} under open-circuit (high-impedance) conditions, a second reading pulse is applied to measure the final conductance G_f .

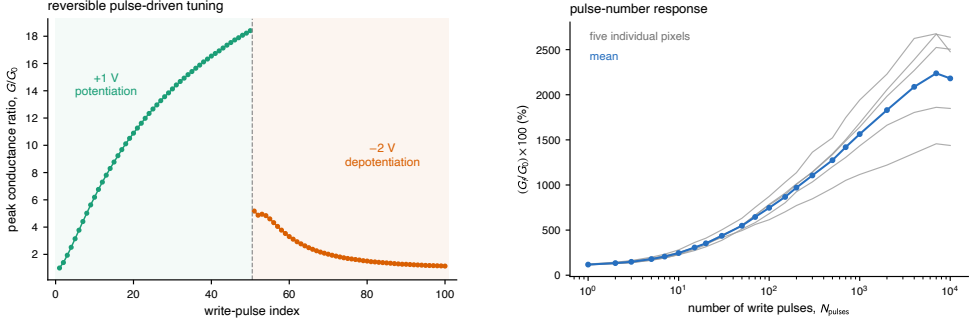
The conductance change is reported as the percentage ratio $\Delta G = (G_f/G_0) \times 100$.

Between experiments, the device is fully reset by connecting the junction to a short circuit for 300 s, which discharges the ionic double layers and restores the homogeneous equilibrium ion distribution. The suitability of this reset protocol was verified by confirming that G_0 returns to its original value within the measurement uncertainty before each new experiment.

Potential as a Function of Pulse Number

Figure 2.3b shows the dependence of ΔG on N_{pulses} for $V_{\text{write}} = 1$ V and $t_{\text{wait}} = 150$ ms. The conductance ratio increases monotonically with N_{pulses} , reaching values in excess of 10³% for $N_{\text{pulses}} > 10^3$. The relationship is approximately linear in log-log space for $N_{\text{pulses}} < 100$ and saturates at large pulse numbers, consistent with a scenario in which the degree of ionic asymmetry that can be built up under a given write voltage is bounded by the total number of mobile ionic carriers.

Within this archived pulse-train dataset, the low-pulse-count response is reproducible to approximately the 10 % level: for the five independent pixels in fig. 2.3b, the median relative standard deviation is 8.9 % and the largest value is 11.8 % over the points with $N_{\text{pulses}} < 100$. The dispersion increases modestly at large pulse numbers, reflecting the probabilistic nature of the ionic migration dynamics at large driving forces.



(a) Reversible potentiation/depression.

(b) Pulse-number response.

Figure 2.3: Pulse-driven conductance integration in the composite memristive device. (a) Fifty consecutive 1 V write pulses continuously potentiate the conductance; subsequent -2 V pulses depotentiate it toward the initial state. (b) Dependence of the conductance ratio $(G_f/G_0) \times 100\%$ on the number of write pulses for $V_{\text{write}} = 1$ V and $t_{\text{wait}} = 150$ ms. Thin grey lines are the five independent crossbar pixels on the NM_v055 substrate; the bold line is their mean. Data regenerated from the local archive associated with Ref. [6].

2.4.3 Variable-Delay Depotentiation: Short- and Long-Term Retention

The third dynamical measurement carried through the thesis probes how a written conductance state fades once the drive is removed: the depotentiation of the device measured as a function of the delay before read-out. This is one of the most important characterisation protocols for neuromorphic devices, because it determines whether the device can emulate short-lived and longer-lived synaptic states on experimentally accessible timescales. In the artificial-synapse literature these regimes are commonly labelled short-term memory and long-term memory; in the present system, the longer-lived regime remains on the sub-minute scale and should therefore be understood operationally rather than as a claim of indefinitely persistent storage.

The measurement reuses the pulsed read–write protocol of section 2.4.2,

with the waiting time t_{wait} now the swept variable and the pulse number held fixed. Figure 2.4 shows the dependence of ΔG on t_{wait} for three excitation conditions: (i) $V_{\text{write}} = 1 \text{ V}$, $N_{\text{pulses}} = 10$ (STM, blue); (ii) $V_{\text{write}} = 1 \text{ V}$, $N_{\text{pulses}} = 50$ (STM, red); and (iii) $V_{\text{write}} = 3 \text{ V}$, $N_{\text{pulses}} = 10$ (LTM, black).

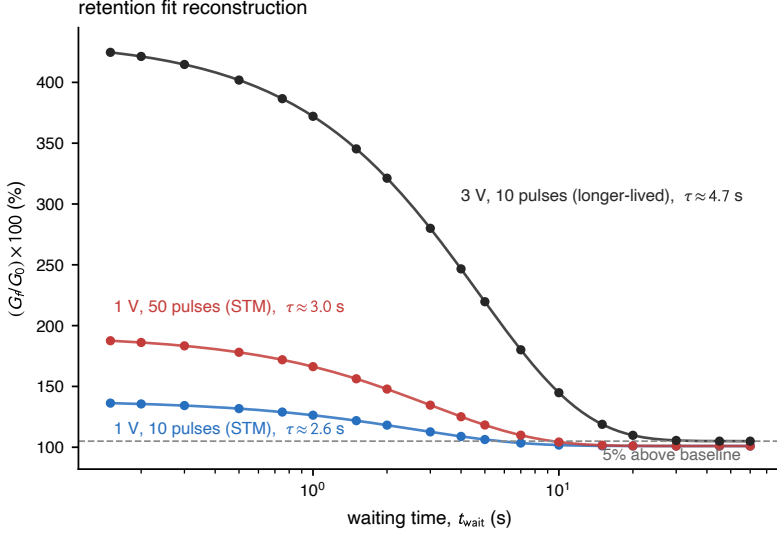


Figure 2.4: Memory retention kinetics. Conductance ratio $(G_f/G_0) \times 100\%$ versus waiting time t_{wait} for STM conditions (1 V; 10 and 50 pulses, blue/red) and the longer-lived regime denoted LTM (3 V, 10 pulses, black). Solid lines are standard-exponential reconstructions of the form in eq. (2.1); the fit parameters are summarised in section A.2.3. Data and fit parameters from Ref. [6].

For short waiting times ($t_{\text{wait}} < 500 \text{ ms}$ for STM conditions, $t_{\text{wait}} < 2 \text{ s}$ for LTM conditions), the conductance first *increases* slightly relative to the value immediately after the write pulse train. This counterintuitive behaviour is *not* ascribed to ionic inertia: ion transport in the polymer matrix is strongly overdamped, so any momentum imparted by the field relaxes on a timescale orders of magnitude shorter than the observed enhancement. It is instead attributed to the continued redistribution of ions after the drive is removed. At the end of the pulse train the

ionic configuration and its self-consistent space-charge field have not yet reached their quasi-steady profile; the residual internal field, together with the slow advance of the electrochemically doped region near the electrode, continues to raise the conductance for a brief interval before the diffusion-driven relaxation sets in. An analogous transient enhancement following a stimulus train has been described as “post-tetanic potentiation” in biological synaptic preparations [42, 43].

For longer waiting times, the expected exponential relaxation of conductance is observed as the ionic configuration relaxes back toward equilibrium by thermal diffusion. The fit record summarised in section A.2.3 reports least-squares fits in IGOR Pro 8 to a standard exponential:

$$\frac{G_f(t_{\text{wait}})}{G_0} = R_\infty + A \exp\left(-\frac{t_{\text{wait}}}{\tau}\right) \quad (2.1)$$

where τ is the characteristic relaxation time, A is the decaying amplitude, and R_∞ is the long-waiting-time conductance ratio. Equivalently, the published proof-of-concept fit corresponds to a stretched-exponential model with the stretch exponent fixed at $\beta = 1$. No free β parameter, confidence interval, or goodness-of-fit statistic was reported for these three traces, so the chapter 2 evidence is used here only to establish the two experimentally resolved relaxation times, not to claim a fitted distribution of relaxation barriers. The later comparative analysis in chapter 4 uses a free-exponent Kohlrausch descriptor only where the DELAYTIME archive is refitted consistently across the comparative corpus.

The fitted characteristic times are $\tau_S = 2.57$ s for the 1 V, 10-pulse STM trace, $\tau_S = 3.01$ s for the 1 V, 50-pulse STM trace, and $\tau_L = 4.73$ s for the 3 V, 10-pulse LTM trace. The corresponding offsets and amplitudes are tabulated in section A.2.3. The retention time — defined operationally as the waiting time at which G_f/G_0 returns to within 5 % of the baseline (i.e., $G_f/G_0 < 1.05$) — is 10 s–15 s for STM and exceeds 45 s for the

longer-lived regime labelled LTM. These values are directly relevant to the timescales of spike-based computation: the 10 s–15 s STM window is sufficient to maintain working memory over a series of sequential spike inputs, while the > 45 s LTM window shows that the device can preserve a write history over many subsequent events without approaching the retention regime of a conventional non-volatile memory.

The molecular mechanism underlying the STM/LTM distinction is discussed in detail in section 2.5.

2.4.4 Excitatory Post-Synaptic Current

The EPSC measurement is a standard protocol in the field of artificial synaptic devices, designed to quantify the number and stability of distinguishable conductance states accessible to the device [23, 90]. The protocol employed here closely mimics the biological experiment in which a patch-clamped neuron is subjected to a defined sequence of pre-synaptic inputs and the post-synaptic current is monitored.

The measurement protocol is as follows (fig. 2.5a):

1. **Initialisation.** The device is initialised to its ground state S_0 by applying a fixed read-out bias $V_0 = 1$ V for 1 s. This bias differs in purpose from the sub-threshold probe ($V_{\text{read}} = 0.5$ V) used for the non-perturbing read-out of sections 2.4.2 and 2.4.3: the EPSC states are defined operationally by the current response at a single common read-out bias against which all states S_0 – S_4 are referenced, not by a passive probe required to leave the state untouched.
2. **Excitation.** A short positive write pulse of 2 V for 50 ms is applied; the device is subsequently held at 1 V for 1 s, during which the conductance spontaneously relaxes to a new excited steady state S_1 .

The same sequence is repeated to produce a second excited state S_2 .

3. **Inhibition.** Two negative write pulses of -2.5 V for 50 ms are applied, producing inhibitory states S_3 and S_4 .

The four excited states are readily distinguishable from each other and from the ground state. Defining the EPSC readout ratio $R_n = I_{S_n}/I_{S_0}$ as the signed current response of state S_n normalised to that of the ground state S_0 under the fixed read-out bias used in the experiment, the values obtained are: $R_1 = 7.33$, $R_2 = 15.55$, $R_3 = -3.72$, $R_4 = -16.97$. (The ratio is formed from signed currents rather than conductances, so the inhibitory states carry a negative sign; conductance itself remains non-negative throughout.) The large absolute ratios (up to ~ 16 -fold increase or ~ 17 -fold decrease relative to S_0) provide a wide margin for reliable state identification in a practical device, even in the presence of noise. The four-state sequence used here is a representative demonstration rather than an upper bound: given the continuous, analogue conductance tunability established in section 2.4.2, the number of distinguishable readout states is not intrinsically limited to four, and many more intermediate levels could in principle be resolved by the same protocol within the available margin.

The states S_1 and S_2 are reached by *excitatory* (positive polarity) stimulation, while S_3 and S_4 are reached by *inhibitory* (negative polarity) stimulation, in direct analogy with long-term potentiation (LTP) and long-term depression (LTD) at biological synapses. The coexistence of both excitatory and inhibitory states in the same 2-T device, without an independent gate terminal, distinguishes the present platform from the 3-T organic implementations discussed in section 1.5.2.

2.4.5 Spike-Timing Dependent Plasticity

Spike-timing dependent plasticity is a timing-dependent Hebbian learning rule: it instantiates Hebb’s local correlation principle [41] in a form that was later observed experimentally in biological preparations [54, 55]. Here the timing delay is defined as $\Delta t = t_{\text{post}} - t_{\text{pre}}$. With this convention, if the presynaptic neuron fires shortly *before* the postsynaptic neuron (causal ordering, $\Delta t < 0$), the synapse is potentiated (LTP); if the presynaptic neuron fires shortly *after* the postsynaptic neuron (anti-causal ordering, $\Delta t > 0$), the synapse is depressed (LTD). The magnitude of the change decays exponentially with $|\Delta t|$. This temporal asymmetry is essential for the synaptic encoding of causal relationships in neural circuits and underlies the ability of STDP-based learning rules to implement unsupervised feature extraction and pattern recognition [78].

Experimental Protocol

The STDP experiment requires the emulation of pre- and postsynaptic spike waveforms by voltage pulses applied at the two electrodes of the device. In a 2-T device, both pulses must be applied at the same physical terminal (the Ag electrode), so the experimentally measured total voltage $V(t)$ is the algebraic sum of the presynaptic and postsynaptic voltage waveforms separated by the delay Δt :

$$V(t) = V_{\text{pre}}(t) + V_{\text{post}}(t + \Delta t) \quad (2.2)$$

Each spike waveform consists of a unipolar triangular pulse with maximum amplitude V_{sp} and duration 1.2 s. The delay Δt was varied from -600 ms to 600 ms. The amplitude $V_{sp} = 1.55$ V was chosen to be large enough to activate ionic migration (above the threshold of approximately 0.7 V) while remaining below the onset of electrochemical degradation of the

active layer (conservatively estimated at 2 V for the pristine device).

For $\Delta t < 0$ (pre fires before post), the two spike waveforms partially overlap in time with additive polarity, producing a composite pulse whose peak voltage exceeds V_{sp} and which is directed in the positive sense. The ions then redistribute toward the ITO electrode, the conductance increases, and the device records a net synaptic potentiation.

For $\Delta t > 0$ (post fires before pre), the overlap produces an anti-causal summation. Depending on the exact waveforms, the effective net voltage can be of reduced amplitude or of reversed polarity relative to the causal case, so the ionic redistribution is either weaker or reversed and the conductance decreases, in the form of a net synaptic depression.

STDP Curve and Fitting

Figure 2.5b shows the measured STDP function: the percentage conductance change ΔG plotted against the timing delay Δt . Under the sign convention above, the measured kernel is asymmetric Hebbian: negative Δt (causal) produces positive ΔG (potentiation), and positive Δt (anti-causal) produces negative ΔG (depression). Each branch (potentiation for $\Delta t < 0$, depression for $\Delta t > 0$) is fitted separately, as a function of the absolute delay $|\Delta t|$, by a decaying exponential:

$$\Delta G(\Delta t) = A_{\pm} \exp\left(-\frac{|\Delta t|}{\tau_{\pm}}\right) + G_0 \quad (2.3)$$

where $A_{\pm}$ and $\tau_{\pm}$ are the branch-specific amplitude and time constant (+ for the potentiation branch, – for the depression branch) and G_0 is a constant offset. The best-fit values of $\tau_{\pm}$ for both the potentiation and depression branches lie in the range 85 ms–90 ms. This is in close quantitative agreement with the characteristic timescale of biological STDP, which is approximately 100 ms [55, 58]. The device, therefore,

not only qualitatively replicates the STDP function but does so with biologically realistic time constants.

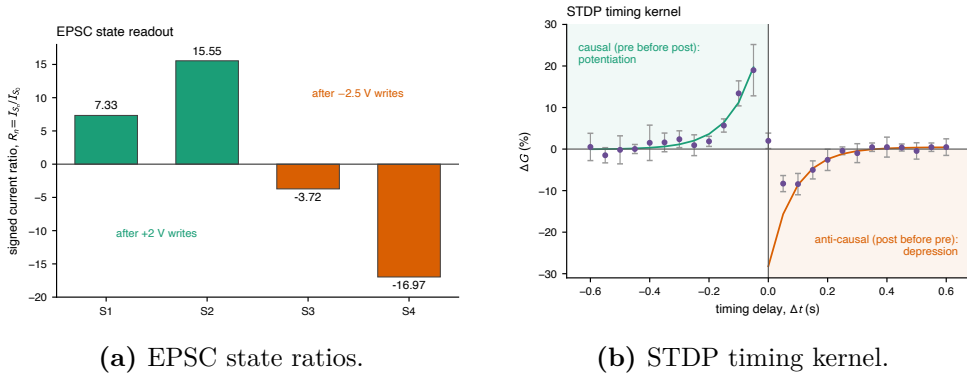


Figure 2.5: Compact summary of the extended synaptic measurements. (a) Signed EPSC readout ratios $R_n = I_{S_n}/I_{S_0}$ for the four demonstrated states. (b) Spike-timing dependent plasticity: percentage conductance change ΔG as a function of the pre/post spike-timing delay Δt , with each branch fitted to the decaying exponential of eq. (2.3). Data and reported state ratios from Ref. [6]; STDP curve regenerated from the local NM_v055 Day19 archive.

The result that the obtained learning rule corresponds to *asymmetric Hebbian* STDP reflects the specific voltage waveform design, the stated definition of Δt , and the polarity conventions of the electrode configuration. Different waveform designs or an opposite sign convention would be expected to produce differently labelled STDP kernels, offering an additional degree of tunability in the design of hardware neural network architectures built from these devices.

2.5 Theoretical Framework: Ion Transport and the Origin of Memristance

The injection and bulk-transport physics of the composite are described by two limiting-case models inherited from the polymer light-emitting

electrochemical cell (LEC) literature: an *electrodynamic* (injection-limited) model [99, 100] and an *electrochemical-doping* (ohmic) model [98, 101, 137], which van Reenen *et al.* [102] reconciled as two regimes of the same field-driven ion redistribution. The equilibrium ion distribution, the thermionic-injection description, and the two models in full are set out in section A.4; the focus here is on what they imply for the two observed memory regimes.

For the present SY/Hybrane/LiOTf system, the *Super Yellow* polymer has moderate hole mobility and a moderate injection barrier from ITO, so both mechanisms may contribute simultaneously. The interpretation adopted here associates the ED mechanism with the rapid conductance changes at short timescales (milliseconds to seconds), driven primarily by the fast-moving triflate anions, and the ECD mechanism with the larger, slower changes at longer timescales (seconds to tens of seconds), driven by the slower-moving Li^+ cations; on this reading the hierarchy of timescales is the physical origin of the STM/LTM dichotomy. The device-level measurements of section 2.4 do not by themselves separate the two contributions: a clean discrimination would require their temperature, film-thickness, or sweep-rate dependence, which is not pursued here. The assignment is therefore a working interpretation rather than an established fact. Its central falsifiable consequence — that the slow timescale should track the cation–host coordination strength — is examined directly in Chapter 4 by varying the cation at fixed host chemistry.

2.5.1 Lewis Acid-Base Chemistry and the Two Memory Regimes

The chemical mechanism proposed here for the two memory regimes is the same hard–soft acid–base reasoning developed in sections 1.6.1 and 1.6.3, applied to the specific salt and host of this device.

Within the HSAB framework, the strength of the interaction between a Lewis acid and a Lewis base in a coordination environment is set by the matching of their hardness or softness [89]. The hard–hard interaction between Li^+ (a small, hard Lewis acid with charge density $z/r \approx 1.3 \text{ \AA}^{-1}$ – $1.7 \text{ \AA}^{-1}$) and the ether oxygen atoms of *Hybrane* (hard Lewis bases) is governed predominantly by electrostatic forces, and is therefore thermodynamically strong and geometrically specific. In polyether environments Li^+ is coordinated by four to six oxygen donors — five in the crystalline complex $\text{PEO}_3\text{:LiCF}_3\text{SO}_3$ (three ether oxygens and two triflate oxygens) — with oxygen–lithium distances of $1.9 \text{ \AA}$ – $2.1 \text{ \AA}$ [126]. This local site *binding* should not be confused with the *migration* activation barrier that governs hopping: the latter is the saddle-point barrier between adjacent sites — of order 50 kJ mol^{-1} ($\approx 20 k_{\text{B}}T$; section 2.2.3) — and is therefore smaller than the full coordination well, while still being substantially larger than $k_{\text{B}}T$.

The interaction between the CF_3SO_3^- triflate anion and the host oxygens is, by contrast, much weaker. The diffuse, low-charge-density triflate anion does not engage in strong electrostatic interactions with the hard-base oxygen atoms, and the residual interaction is governed mainly by weak van der Waals forces [128, 146]. The energy barrier for displacement of a triflate anion from one interstitial site to the next is therefore expected to be substantially lower than the barrier for a Li^+ cation.

This differential activation barrier translates directly into a differential ionic mobility under an applied field, and hence into different voltage thresholds for significant ionic redistribution:

- A modest voltage of 1 V applied across a 209 nm film generates a field of approximately 4.8 MV m^{-1} . In the interpretation adopted here, this is sufficient to drive the more weakly bound triflate-anion redistribution observed experimentally [6]. The fast anionic redistribution produces a rapid conductance change, which then

relaxes on the timescale of triflate diffusion through the host. This is the short-term memory regime, with retention 10 s–15 s, driven primarily by triflate relaxation.

- A voltage of 3 V across the same film generates approximately 14.4 MV m^{-1} . This higher field is assigned to the onset of a slower cation contribution, consistent with the stronger Li^+ coordination expected in oxygen-rich polyether hosts [126, 128]. The cation contribution sets in, produces a delayed and stronger ionic redistribution, and relaxes on the much longer timescale of cation re-coordination. This is the long-term memory regime, with retention exceeding 45 s, driven primarily by cation redistribution.

This account yields a specific, experimentally testable prediction. Replacing Li^+ with a monovalent cation of larger ionic radius (Na^+ or K^+) should reduce the charge density, and therefore the hardness, of the Lewis acid. The cation–oxygen coordination with the polyether host should weaken accordingly, and the slow fading-memory timescale of the device should shorten as the cation is moved along the $\text{Li}^+ \rightarrow \text{Na}^+ \rightarrow \text{K}^+$ series. The prediction is tested in Chapter 4 on the main comparative corpus of the thesis: a PEO-based polymer-electrolyte series loaded with lithium, sodium and potassium triflate salts (PEO/LiOTf, PEO/NaOTf, PEO/KOTf) at fixed polymer and salt mass fractions, complemented by a dedicated PEO/LiOTf concentration sub-study. The prediction is addressed there as a statement about the shift of the measured variable-delay depotentiation timescale when the cation is varied at fixed host chemistry, not as a claim about the full synaptic suite reported in the present chapter.

2.6 Device Stability and Reproducibility

2.6.1 Ambient Shelf Stability

Two distinct notions of device longevity must be kept separate. *Cycling endurance* — the number of write/erase cycles a device sustains before its performance parameters drift outside specified tolerances — is the metric that ultimately governs suitability for the millions-to-billions of weight updates of on-chip training; it is not characterised for the present devices and is left for future work. *Shelf (calendar) stability* — the retention of performance during storage and intermittent operation under ambient conditions — is what is reported here. It is the more immediately relevant metric at the proof-of-concept stage, where the question is simply whether the unencapsulated composite survives realistic handling at all.

The devices reported here were characterised at regular intervals over a period of two weeks (~ 300 h) after fabrication, without encapsulation, while stored in ambient air at room temperature. The key metrics monitored were the baseline conductance G_0 , the maximum achievable conductance change $\Delta G_{\max}$, and the STM retention time τ_S ; the supporting stability record is collected in section A.3.

The results show that all three metrics remain within 20% of their initial values over the full two-week measurement period [6]. This stability compares favourably with other organic memristive devices reported in the literature, which typically exhibit significant degradation within hours to days under ambient conditions [119]. The enhanced stability is attributed to the solid-state nature of the composite: unlike devices based on liquid electrolytes or gel polymers, the Hybrane matrix provides a mechanically cohesive environment that protects the ions from evaporation and the polymer films from delamination, consistent with the stability of related Hybrane-based LEC blends [104].

2.6.2 Device-to-Device Reproducibility

Reproducibility is a persistent challenge for memristive devices, particularly those based on filamentary switching in inorganic materials, where the stochastic nucleation of conducting filaments produces large device-to-device and cycle-to-cycle variability [71]. The present system, which operates by uniform bulk ionic redistribution rather than localised filament formation, is intrinsically less susceptible to this type of variability.

The strongest reproducibility statement available from the archived proof-of-concept data is the one attached to the pulse-number experiment in fig. 2.3b. That dataset contains five independent crossbar pixels on the NM_v055 substrate, measured under the same 1 V write and 150 ms wait protocol. The shaded band is one standard deviation across those five pixels. For $N_{\text{pulses}} < 100$, the median RSD is 8.9% and the largest RSD is 11.8%; the appropriate claim is therefore reproducibility at about the 10% level in the low-pulse-count regime, not an exact upper bound below 10%.

The EPSC readout and STDP measurements remain important protocol demonstrations, but the recovered archive does not contain a normalized EPSC population table comparable to the PULSES and DELAYTIME tables used later in the thesis. They are therefore not used to support an independent EPSC RSD claim here. Likewise, the published proof-of-concept data establish same-substrate/pixel reproducibility and two-week shelf stability; they should not be read as a powered batch-to-batch reproducibility study. Campaign-scale reproducibility — a separate metric on a separate axis — and its resolution by a change of ion-transport host are taken up in chapter 3.

2.7 Comparison at the Time of Publication

The historical comparison table reproduced in table A.2 places this device alongside representative organic and hybrid memristive devices available at the time of the 2022 publication. Two combinations stand out as unrepresented in that set. The first is the coexistence of short-term memory and long-term memory in a single *fully reversible* 2-T device: the closest prior report, Liu *et al.* [94], achieved short-to-long-term transitions but through irreversible redox chemistry [116], whereas here the conductance returns to G_0 after the reset protocol. The second is STDP in a 2-T architecture, where the timing rule emerges from the superposition of pre- and postsynaptic pulses at the single Ag terminal rather than from the gate of a 3-T device [90, 105, 115]. On the operating-regime axes the device is competitive in unencapsulated shelf lifetime (~ 300 h, roughly an order of magnitude beyond comparable organic devices [119]) but not in energy per event (~ 50 nJ, above CMOS and inorganic-memristor budgets [27, 68]); the full table and a per-axis discussion are given in section A.5.

The comparison above retains the device-level metrics conventional in the organic memristive literature of the early 2020s: reversibility, retention window, energy per event, and operational lifetime. Subsequent work has broadened the field substantially, particularly through organic electrochemical neurons and neuromorphic circuits [121, 122] and through the wider organic-iontronic memristor landscape [123]. The table is therefore used only to locate the 2022 proof-of-concept result historically. Within the thesis-wide argument, those metrics play a narrower role: they establish that the polymer-electrolyte platform introduced here is reversible, dynamically addressable, and stable enough to support the comparative study of Chapter 4 (a PEO-based triflate composition and cation sweep, characterised through the same three common dynamical measurements)

and the temporal-computing reinterpretation of Chapter 5 built on those measurements. They are not invoked to claim that this device should be judged by the standards of a high-density non-volatile memory technology, nor to suggest that the full synaptic suite reported here is available across the full comparative corpus. Those measurements remain specific to the present SY/Hybrane/LiOTf device.

2.8 Summary

This chapter has reported the design, fabrication, and characterisation of a fully reversible 2-T organic memristive device that supports, within a single polymer-composite active layer, a full set of synaptic functions organised in two families: the three dynamical measurements carried through the rest of the thesis (I–V hysteresis, variable-number-of-pulses potentiation, and variable-delay depotentiation), and the extended synaptic protocols specific to this device (EPSC, the separated short-term memory/long-term memory reading of the retention data, and STDP).

A ternary *Super Yellow/Hybrane/LiOTf* composite satisfies the three functional requirements of section 2.2.1 (electronic pathway, ion-transport medium, ion source), fabricated as a 2-T ITO/composite/Ag sandwich by solution processing and one evaporation step, with a 209 nm active layer below 3 nm RMS roughness. On this single platform the three dynamical measurements carried forward to the comparative corpus were established: analogue I–V hysteresis (forward current at 1 V growing ~ 15 -fold over ten cycles in near-uniform increments); fully reversible pulse potentiation/depression exceeding 200 % at ~ 50 nJ per event, growing past 10³% for $N_{\text{pulses}} > 10^3$; and variable-delay depotentiation resolving two regimes (short-term memory, $\tau_S = 2.57$ s–3.01 s, retention 10 s–15 s; long-term memory, $\tau_L = 4.73$ s, retention > 45 s, from $\beta = 1$ fits). The extended single-device protocols add a four-state EPSC read-out (signed

ratios -17 to $+16$) and an asymmetric-Hebbian STDP window ($\tau = 85\text{ ms}$ – 90 ms , biologically realistic [55, 58]). The short-term memory/long-term memory dichotomy is attributed to the Lewis acid–base mobility asymmetry between the weakly coordinated triflate anion and the strongly coordinated Li^+ cation in the oxygen-rich host, with the injection and bulk-transport physics described by the electrodynamic and electrochemical-doping models in their reconciled form [102] (section A.4). Devices were stable over a $\sim 300\text{ h}$ unencapsulated ambient shelf (cycling endurance not characterised), with $\sim 10\%$ low-pulse-count variability across five pixels on NM_v055 (median RSD 8.9%); campaign-scale reproducibility is taken up in chapter 3.

Taken together, these results establish the SY/Hybrane/LiOTf composite as a thoroughly characterised exemplar of a polymer-electrolyte organic memristive device, and identify cation–polyether coordination chemistry as the physical hypothesis for memory-timescale control.

This hypothesis motivates the comparative study of Chapter 4, which replaces *Hybrane* with poly(ethylene oxide) and varies the PEO/LiOTf composition as the replicated quantitative axis, surveying host, anion, and cation substitutions as sample-limited side evidence; there the hypothesis is tested through the three common measurements rather than the full synaptic suite, and the outcome is cautious, with no robust host- and anion-independent Li/Na/K law claimed. The richer single-device metrics reported here (EPSC, separated short-term memory/long-term memory retention, and STDP) serve the later chapters only as lithium-device priors. Beyond that handoff, the volatile two-timescale behaviour established here anchors the temporal-computing reinterpretation of Chapter 5, in which behavioural models fitted to the comparative corpus drive the reservoir-computing simulations; there the device’s fading memory drives the computation rather than limiting it.

Chapter 3

Reproducibility, Degradation, and the Device-Provenance Infrastructure

3.1 The Reproducibility Problem

Chapter 2 established, on a single fully characterised *Super Yellow*, *Hybrane*, and LiOTf device, that a three-component polymer-electrolyte composite supports the complete set of memristive functions in one two-terminal platform, and that the proof-of-concept device was stable on a two-week ambient shelf scale. Chapter 4 will treat the volatile dynamics of these composites as something to be engineered across a replicated composition grid. Between those two results sits a problem that has to be confronted before the comparative study can be read on its own terms, and that is the subject of this short chapter: *as the campaign scaled from a single validated device to a fabrication programme, the memristive hysteresis became progressively harder to obtain*. Successive batches of nominally identical *Super Yellow*/*Hybrane*/LiOTf devices drifted away

from the low-current, high-area, multi-level behaviour of the proof-of-concept device toward a high-conductivity, low-window response with no usable switching — and therefore no usable potentiation.

Two notions of stability must be kept apart here, and chapter 2 validated only the first. *Calendar (shelf) stability of an individual device* — whether a finished device retains its characteristics over days to weeks — was demonstrated for the proof-of-concept device and, as shown below, holds quite generally for *Hybrane* devices. *Batch-scale reproducibility of the platform over the fabrication campaign* — whether a device built this month behaves like one built last spring — is a different metric on a different axis, and it is this second metric that failed. The failure is therefore a degradation *over calendar time of the fabrication line*, not an ageing of devices after they are made; the distinction is load-bearing for everything that follows and is established quantitatively in section 3.3.

The fabrication and characterisation notes place a sharp behavioural inflection at devices NM_v026–031 (2021-04-22), recorded at the time as “the first devices that changed electrical behaviour”, with the contemporaneous hypotheses left explicitly open — “could be evaporation rate ... the metal used for evaporation, and possible chemical degradation of substances” — and still unresolved in a note dated 2022-01-31, documenting some nine months of active investigation. Anchoring the analysis on that documented inflection rather than on a smooth trend is both more faithful to the campaign and statistically more powerful. The rest of the chapter follows that diagnostic chain: tested causes are eliminated, the degradation is quantified under four mandatory controls, a candidate chemistry is proposed, the PEO resolution is documented, and the resulting methodology is carried forward.

3.2 Eliminating Candidate Causes

Faced with a drifting fabrication line and a list of open hypotheses, the team crossed the candidate causes one at a time, each as a deliberate batch with its variables pinned. Table 3.1 summarises the campaign as a method of elimination: one row per tested cause, the batches that tested it, and the verdict. This is also where the provenance library that the rest of the thesis relies on was motivated — one cannot run an elimination of this kind without a record that pins every process variable per device.

Two entries deserve emphasis. First, device NM_v067 (2021-10-21) records that a second experimenter, working independently, “has the same problems with devices”: the failure crosses people, which removes individual technique as the cause and points to a shared material or systemic origin. (The same note also guessed at the ITO; that guess was not borne out by the substrate study.) Second, the *Super Yellow* semiconductor was an explicit co-suspect — an old bottle was noticeably “more viscous . . . maybe because of bonds degradation in the old one” — but the old-versus-new *Super Yellow* comparison produced no significant change, and the group kept buying fresh *Super Yellow* throughout the period in which devices continued to collapse. With illumination, atmosphere, transport, substrate, solvent, evaporation, and the semiconductor each crossed and excluded, the residual suspect is the remaining consumable that aged on the shelf between batches: the hyperbranched *Hybrane* host stock. The remainder of the chapter tests that attribution quantitatively and offers a candidate chemistry for it.

Table 3.1: Methods-of-elimination summary for the degradation investigation. Candidate causes were crossed as deliberate batches with variables pinned in the provenance record; the exclusions leave the hyperbranched *Hybrane* host stock as the residual suspect, pursued quantitatively in section 3.3.

Candidate cause	Crossed as (batches)	Verdict
Operator / technique	cross-person check (v067): “Lorenzo has the same problems with devices”	Eliminated. The failure appears across people, so the cause is systemic, not one operator’s technique.
Illumination (light vs dark)	v015–019, vt001–006, v041–044, v053–056	Excluded — no systematic recovery of the window.
Storage / measurement atmosphere (glovebox / air / vacuum)	v015–019, v088–099	Excluded.
Baby-chamber transport (full 2 × 2 on Au)	v083–087, v038–040	Excluded.
Substrate (ITO) lot / colour	v079–082; substrate-difference study	Not borne out — a one-off ITO guess by a colleague did not survive the substrate comparison.
Solvent (cyclohexanone) lot	v079, v091	Excluded.
Evaporation rate / silver holder	v045–048, v106–113	Recovery <i>attempted</i> by reverting to the “year-ago” holder; not reproducibly restored.
Annealing regime	late hot-anneal batches	<i>Partial</i> window recovery (section 3.3); consistent with the chemical hypothesis, not a process fix.
<i>Super Yellow</i> stock (old vs new)	v045–048 (“more viscous ... maybe bonds degradation in the old one”)	Eliminated as primary — old vs new <i>Super Yellow</i> gave no significant change, and fresh <i>Super Yellow</i> continued to be purchased while the collapse persisted.
<i>Hybrane</i> host stock	residual after the above	Retained suspect — pursued quantitatively and mechanistically below.

3.3 The Quantitative Degradation Signal

The campaign record is qualitative; the claim that the platform degraded must be made quantitative against the device archive, and made in a way that survives the confounds peculiar to this dataset. The analysis is reproduced in full by `scripts/bridge_hybrane_peo_reproducibility.py` and mirrors the slicing of the group’s interactive feature-explorer tool.

Four controls, each shown to matter. The analysis rests on four controls, each verified to change the conclusion if dropped: *per-device weighting* (the pixels-per-device count falls with date, Spearman $\rho = -0.80$, so every feature is reduced to one value per device before any test); a *standard-protocol corpus* ($n \approx 69$ devices, lead mass ratios, excluding the invasive recovery experiments); *sweep-amplitude matching* (the window features are read in the $\sim 3\text{ V}$ stratum where they are expressed, the conductivity features in a tightly matched $\sim 1.2\text{ V}$ band); and a *pre/post-inflection step test* anchored at the documented NM_v026 inflection, with every conductivity trend reported with and without the deliberately-reverted recovery batches. The four controls are stated in full, with the supporting correlations, in section B.1.

Principal result: the switching window collapses at $\sim 3\text{ V}$. On the standard corpus, per device, the switching window measured in the $\sim 3\text{ V}$ stratum collapses across the inflection (fig. 3.1, table 3.2). The median on–off ratio falls from 2.44 (early, $n = 10$) to 1.21 (later, $n = 42$; Mann–Whitney $p = 0.0019$); the median normalized hysteresis area falls from 0.235 to 0.027, an order of magnitude ($p = 0.0005$); and the potentiation, measured as the per-cent change in the maximum-voltage current over a sweep, flips from $+17.9\%$ to -2.1% ($p = 0.013$). These three are the *same functional event*: a device with no on–off window has

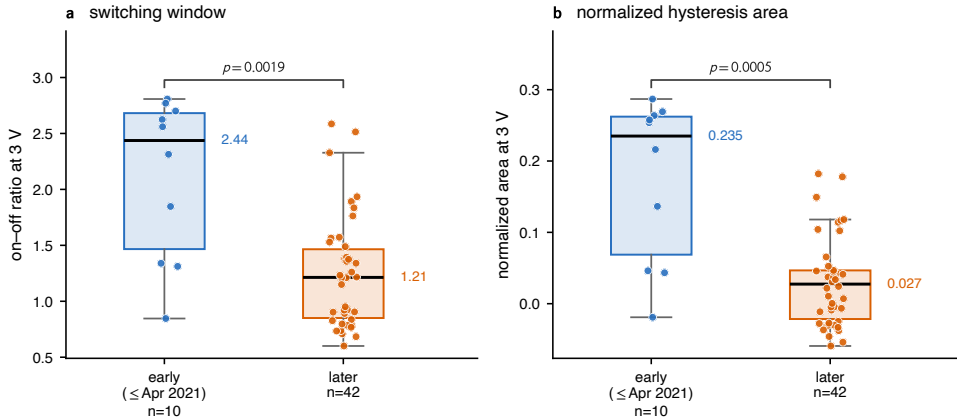


Figure 3.1: Switching-window collapse across the campaign on the standard *Super Yellow/Hybrane/LiOTf/Ag* corpus, read at the ~ 3 V stratum where the window is expressed, one value per device. (a) On-off ratio and (b) normalized hysteresis area, both for devices fabricated before the NM_v026 inflection ($\leq$ Apr 2021) versus later. Boxes are quartiles with the median annotated beside each box; points are individual devices; the bracket gives the two-sided Mann–Whitney p . The window collapses by roughly a factor of two in on-off ratio and an order of magnitude in area. Reproduced by `scripts/bridge_figures.py` (F1).

no room to potentiate, so the loss of hysteresis area and the sign change of the potentiation are two readouts of one failure. The magnitude matches the team’s recollection of an early on-off near 2.5 falling to the 1–1.7 range later, now anchored to the documented inflection.

Mechanism: ohmic drift at matched ~ 1.2 V. The complementary signal lives in the conductivity, read in the tightly matched ~ 1.2 V band so that the sweep-amplitude confound is held fixed. There, the absolute current rises with fabrication date: per device, the current at maximum voltage correlates with date at $\rho = +0.45$ ($p = 0.0002$), the raw hysteresis area at $\rho = +0.57$, and the current difference at the on-off point at $\rho = +0.51$ (all $n = 64$; fig. 3.2a, table 3.2). These correlations *strengthen* when the deliberately-reverted recovery batches ($\geq 2022-03$) are dropped

3.3. THE QUANTITATIVE DEGRADATION SIGNAL

Table 3.2: The degradation as a multi-feature signal on the standard corpus, per device. *Window collapse* is read at ~ 3 V as an early-vs-later step at the NM_v026 inflection (Mann–Whitney). *Ohmic drift* is read at matched ~ 1.2 V as a per-device Spearman correlation with fabrication date, reported with all dates and with the recovery tail ($\geq 2022-03$) dropped; the correlation *strengthens* when the reverted batches are removed, as expected if they are partial recoveries. Health flags are shown for completeness and are discussed in the text.

Window collapse @ ~ 3 V (early $\leq$ Apr 2021, $n = 10$ vs later, $n = 42$)			
Feature	early	later	Mann–Whitney
on–off ratio	2.44	1.21	$p = 0.0019$
normalized area	0.235	0.027	$p = 0.0005$
% change in max-V current	+17.9 %	−2.1 %	$p = 0.013$
Ohmic drift @ matched ~ 1.2 V (per-device Spearman vs date, $n = 64$)			
Feature	all dates	drop recovery tail	
current at max V	$\rho = +0.45$ ($p = 0.0002$)	$\rho = +0.55$ ($p < 10^{-4}$)	
hysteresis area ($V \cdot \mu A$)	$\rho = +0.57$ ($p < 10^{-4}$)	$\rho = +0.69$ ($p < 10^{-4}$)	
current difference at on–off	$\rho = +0.51$ ($p < 10^{-4}$)	$\rho = +0.63$ ($p < 10^{-4}$)	
Health flags (per-device, ~ 1.2 V, $n = 64$)			
is_broken decreases ($\rho = -0.40$, $p = 0.001$) — a handling metric, <i>not</i> degradation;			
is_saturated (fails to potentiate) rises ($\rho = +0.27$, $p = 0.03$).			

— to $\rho = +0.55$, $+0.69$, and $+0.63$ respectively — as expected if those late batches are partial recoveries pulling against the underlying trend. Because the amplitude is held within a narrow band, this is a drift of the material toward more ohmic conduction, not an artefact of the sweep range. The window collapse (relative hysteresis $\rightarrow 0$) and the ohmic drift (absolute current $\uparrow$) are two readouts of the same degradation. The magnitudes are reported as correlations and medians rather than as a single multiplicative “conductivity rise”, because the absolute current depends on the sweep amplitude being controlled for.

What the health flags do and do not say. A naive degradation metric would be the fraction of broken devices; it points the wrong

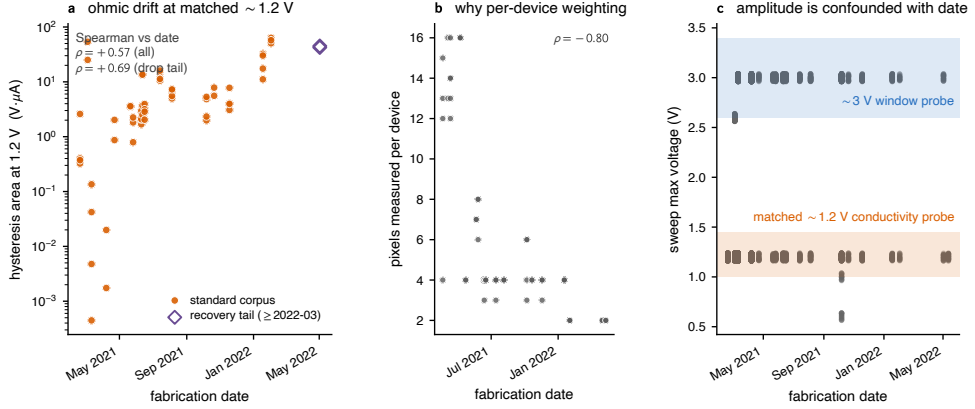


Figure 3.2: Mechanism and methodology. (a) Per-device hysteresis area read at the matched ~ 1.2 V band rises with fabrication date (Spearman $\rho = +0.57$ over all dates, strengthening to $+0.69$ when the recovery-tail batches, open diamonds, $\geq 2022-03$, are dropped; the statistics use all $n = 64$ devices, while the log axis cannot display the few non-positive areas). (b) The number of pixels measured per device falls over the campaign ($\rho = -0.80$), which is why every statistic is computed per device rather than per pixel. (c) Sweep amplitude is itself confounded with date: the matched ~ 1.2 V conductivity probe and the ~ 3 V window probe (shaded) are the two strata used throughout, never pooled. Reproduced by `scripts/bridge_figures.py` (F2).

way. The curve-level `is_broken` flag — erratic or open-circuit behaviour, a contact and handling failure — in fact *decreases* over the campaign ($\rho = -0.40$, $p = 0.001$), tracking the handling and contacting problems the team progressively fixed (the notes are full of them). It is therefore *not* a degradation metric. The flag that does track the failure is `is_saturated` — a curve whose maximum current fails to exceed the previous one, i.e. a device that no longer potentiates — which rises with date ($\rho = +0.27$, $p = 0.03$). The two flags mean different things and are reported separately so that the improvement in handling is not mistaken for the worsening of the device physics.

3.3.1 A Candidate Chemistry: Moisture-Driven Hydrolytic Aging of the *Hybrane* Stock

The rest of this thesis is mechanistic, so attributing the failure to the stock without identifying a mechanism would leave a gap. A definitive answer, however, is not supportable from the available record: no electrochemical impedance spectra were ever recorded on any *Hybrane* device, no gel-permeation chromatography, nuclear-magnetic-resonance or Karl-Fischer analysis was run on the reagent stock, and the aged stock no longer exists in its degraded state. What follows is therefore presented strictly as a hypothesis together with the experiments that would confirm it.

The hypothesis. *Hybrane* is a hyperbranched polyester-amide [127, 135]: ester- and amide-rich, and strongly hygroscopic. The leading hypothesis is that, over roughly a year in the bottle, the stock takes up atmospheric water and slowly hydrolyses (chain scission), and that this process could account for the measured electrical signatures without requiring a separate finished-device ageing mechanism.

- **Ohmic drift** (current $\uparrow$ at matched 1.2 V): absorbed water together with polar scission products raises the baseline ionic and leakage conductivity and plasticises the matrix.
- **Window collapse** (on-off $\rightarrow \sim 1.2$, area $\rightarrow \sim 0.03$, $\% \Delta I \rightarrow -2\%$): memristance needs a matrix stiff enough to *hold* a field-displaced ionic profile (the metastable-polarisation picture of chapter 2). A plasticised, lower-molecular-weight, water-laden host relaxes and leaks, so the loop closes and successive sweeps stop building up conductance.
- (*secondary*) acidic hydrolysis products could partially dope the PPV-type *Super Yellow*, adding a fixed, irreversible conductance — a

second route to the “high-conductivity, undesirable” state recorded in the notes.

Internal support. Three lines of evidence already in hand are consistent with the hypothesis, though none closes it. *First*, an annealing-recovery test (fig. 3.3): among late, degraded-era devices, aggressive annealing ($\geq 150^\circ\text{C}$) restores the $\sim 3\text{ V}$ window to a median on-off of 1.95 ($n = 6$) against 1.22 for the standard- 75°C late devices ($n = 44$; Mann–Whitney $p = 0.024$, one-sided), back to the early-standard reference level (1.74). Driving off absorbed water and re-densifying the film is the natural reading; a handling-only or inert-stock story does not predict a thermal recovery. *This test is suggestive, not decisive*: $n = 6$ elevated-temperature devices, with co-varying process tweaks, and two of them had measurement-pin issues. *Second*, the full electrical fingerprint — ohmic drift plus window and potentiation collapse — is consistent with a water-plasticised, ion-leaky film. *Third*, the material class is, through its ester and amide linkages, intrinsically susceptible to hydrolytic chain scission, while its polar, hyperbranched architecture is hygroscopic by construction [127, 135].

Confirming experiments (future work). The hypothesis would be tested directly by Fourier-transform infrared spectroscopy (loss of the ester $\text{C}=\text{O}$ band, growth of $-\text{OH}$) and gel-permeation chromatography (a molecular-weight drop) on *retained aged versus fresh* stock; by Karl-Fischer titration or thermogravimetry for water content; and by impedance spectroscopy to show that the ionic resistance fell. None were run on the *Hybrane* corpus, so the hypothesis stays open — and may never fully close, because it concerns a consumable that no longer exists in its aged state.

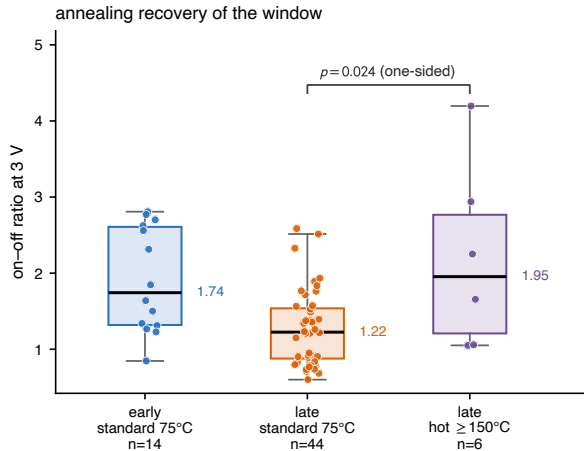


Figure 3.3: Annealing-recovery test of the hydrolysis hypothesis (*Hybrane*/Ag, on-off ratio at ~ 3 V, per device). Among late, degraded-era devices, an aggressive $\geq 150^\circ\text{C}$ anneal restores the window (median $1.22 \rightarrow 1.95$) toward the early-standard reference (1.74); one-sided Mann-Whitney $p = 0.024$. Suggestive only ($n = 6$ hot devices, co-varying tweaks, two with pin issues). Reproduced by `scripts/bridge_figures.py` (F4).

3.4 Resolution: Switching to a PEO Host

The crisis was resolved not by recovering the *Hybrane* line but by changing the ion-transport host. Replacing the hyperbranched *Hybrane* with linear poly(ethylene oxide) (PEO) — the host that carries the comparative study of chapter 4 — delivered two distinct advantages. Laboratory notes first record PEO in use by 2021-12 (NM_v083), placing the host transition at the height of the investigation.

A wider, stronger window. At a matched ~ 3 V sweep, comparing valid (non-broken, non-saturated) devices per device, the PEO host opens a markedly larger switching window than the contemporaneous *Hybrane* devices: median normalized area 0.26 (*Hybrane*) $\rightarrow$ 0.42 (PEO), Mann-Whitney $p = 0.018$; median on-off ratio $2.45 \rightarrow 4.91$ (fig. 3.4a,b).

Sustained temporal reproducibility. Across its full life, the PEO/Ag platform *sustained* a usable switching window over roughly two and a half years (2022–2024, $n = 46$ devices): the per-device on–off ratio at ~ 3 V held at a median of 2.61, with yearly medians of 3.3/2.6/1.9 and no significant downward trend over time. The *Hybrane* platform, by contrast, sat at a median on–off of 1.35 after its stock-driven collapse (fig. 3.4c). PEO did not exhibit the supply-degradation failure mode, and that absence, sustained over years, is the resolution of the crisis.

What “reproducible” means here. The claim is precise: it is *temporal* / *batch* reproducibility — no stock-driven collapse over years — and not a claim of superior within-batch, device-to-device uniformity. At matched ~ 3 V the device-to-device coefficients of variation are only modestly different (PEO ≈ 0.38 vs *Hybrane* ≈ 0.47), and the chapter does not lean on that gap. The case for PEO rests on the two advantages above: a larger window and, above all, the disappearance of the supply-degradation failure over the multi-year span. The attribution of the original failure to the *Hybrane* host stock (with *Super Yellow* eliminated; section 3.2) is consistent with this resolution: changing the suspected reagent removed the failure.

3.5 Methodological Legacy

The diagnosis of the previous sections required instrumentation that did not exist at the start of the campaign. Building it — provenance capture, a normalized characterization procedure, an automated feature pipeline, and trend-discovery tooling — is itself a contribution, and it is the same instrumentation that produces every quantitative result in the comparative and temporal-computing chapters that follow. Five durable outputs came out of it.

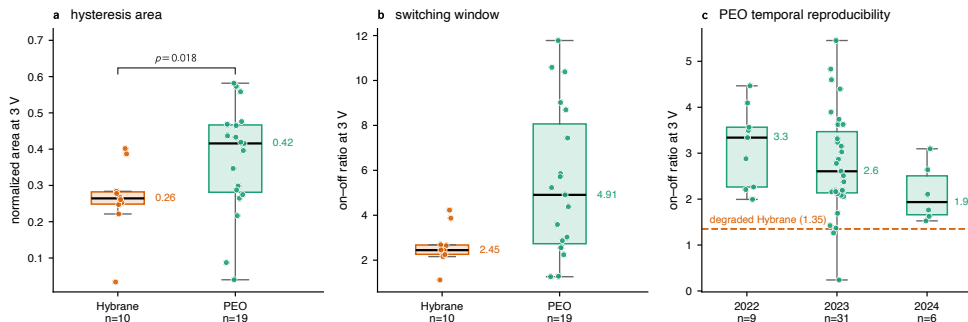


Figure 3.4: Resolution of the reproducibility crisis by switching to a PEO host (silver electrode, ~ 3 V, per device). (a) Normalized hysteresis area and (b) on-off ratio are larger for PEO than for contemporaneous *Hybrane* valid devices ($p = 0.018$ for area). (c) PEO temporal reproducibility: the per-device on-off ratio holds at a usable level (median ≈ 2.6) across 2022–2024 with no collapse, well above the degraded-*Hybrane* level (dashed line, 1.35). Reproduced by `scripts/bridge_figures.py` (F3).

- **A fabrication-provenance standard.** The per-device library schema (storage atmosphere, illumination, transport, reagent and solvent lots, operator, evaporation profile, and free-text fabrication and characterisation notes) exists *because* the crisis forced the pinning of every variable. It made the elimination of section 3.2 possible and is now the permanent record for every device in the archive.
- **A normalized electrical-characterization procedure.** A hard lesson of section 3.3 is that the protocol sets the response: sweep amplitude alone co-determines the apparent conductivity and whether a window is even expressed. The campaign converged on fixed, documented drive and read protocols so that devices and batches are finally comparable — the precondition for the comparative chapter.
- **An automated feature pipeline.** The raw instrument traces are reduced to per-device, per-pixel, and per-curve feature tables by a dedicated pipeline; every feature used in this chapter’s panel, and in

chapter 4 and the temporal-computing chapter, is computed there.

- **Trend-discovery and curation tooling.** An interactive feature explorer (with which the present trends were originally found, and whose slicing the four controls of section 3.3 reproduce) and a curation tool that produces the screened device set turn the archive into discoverable trendlines.
- **Figure generation.** A thesis-figure pipeline renders the quantitative figures from the same feature tables, so that every plotted number is traceable to the archive.

The *Hybrane* failure is what made the rest of the thesis measurable. The same apparatus is what the comparative study of chapter 4 and the data-driven temporal-computing study build upon.

3.6 Reconciliation and Bridge

This chapter must be squared with chapter 2, which attributed enhanced shelf stability to the *Hybrane* matrix. The resolution is a partition, not a retraction, and it turns on the two-metric distinction of section 3.1.

- The *published proof-of-concept device* was genuinely stable on the validated $\sim$ two-week ambient shelf scale [6]; that claim stands. Indeed, *Hybrane devices* hold their characteristics over weeks to months — a separate, deliberately long multi-day stability study confirms it, and is reported with the proof-of-concept device — which is precisely why the failure here is identified as a degradation of the *supply*, not of finished devices.
- The *batch-scale reproducibility of the Hybrane platform over the fabrication campaign* is a different metric, and it is the one that

collapsed over the following year, in the manner quantified in section 3.3.

The two statements are about different axes (one device over two weeks; the fabrication line over a year) and are both true. The Chapter 2 stability result is therefore preserved intact, and this chapter adds the campaign-scale reproducibility axis that the proof-of-concept study did not probe.

With the reproducibility crisis diagnosed and resolved by the move to a PEO host, the platform is finally in a state where the volatile dynamics can be varied deliberately and compared across many nominally identical devices. That is the task of chapter 4, which treats the *Super Yellow*/PEO/LiOTf system on the silver electrode as a replicated composition grid — using the very provenance standard, normalized protocols, and feature pipeline that this chapter’s crisis brought into being — and asks how the fading-memory dynamics can be engineered through composition and chemistry.

Chapter 4

Compositional and Chemical Control of Volatile Polymer-Electrolyte Memristive Dynamics

4.1 Introduction and Scope

Chapter 2 established, on a single fully characterised *Super Yellow*, *Hybrane*, and LiOTf device, that a three-component polymer-electrolyte composite supports the full set of memristive functions in one two-terminal platform. The natural next question — and the subject of this chapter — is how the *dynamics* of these devices respond to deliberate changes in the composite, so that the volatile, fading-memory behaviour can be treated as something to be engineered rather than merely observed.

Three families of measurement recur on the same 2-T geometry throughout this chapter and the next: (i) current–voltage hysteresis, which reports the steady-state switching window; (ii) potentiation as a function of the

number of applied write pulses, which reports how the conductance is built up; and (iii) depotentiation as a function of the delay before read-out, which reports how a written state fades. Introduced as a named set in chapter 2, they are the shared, abundant basis of the comparative work. The richer single-device synaptic protocols of chapter 2 — excitatory post-synaptic current, separated short- and long-term memory, and spike-timing dependent plasticity — are *not* repeated here; where invoked, they serve only as prior evidence from the chapter 2 device.

Evidence tiers. **Composition** — variation of the ion-transport polymer and salt mass fractions in the *Super Yellow*/PEO/LiOTf system — is supported by a replicated design (typically two to four nominally identical devices per composition cell) and is presented as a *quantitative* result. **Chemistry** — variation of the ion-transport host, the salt anion, and the alkali cation — is supported, after data-quality screening, by at most one or two devices per matched cell; it is therefore presented as an *illustrative* landscape, with sample sizes stated explicitly and no claim of statistical power. The **electrode** comparison is also illustrative, available only at the lead composition (section 4.8). A separate **methodological** result establishes that the *drive protocol* itself sets the apparent fading-memory timescale, which both explains why the chemistry comparisons must be read cautiously and provides a design rule for future comparative work.

4.2 Materials, Devices, and Measurement Protocol

The devices retain the architecture of chapter 2: a vertical two-terminal sandwich of ITO (bottom, transparent), a solution-processed composite active layer, and a thermally evaporated silver top electrode, with a junction

area of $8.25 \times 10^{-2} \text{ cm}^2$. *Super Yellow* is retained as the semiconducting component in every device of this chapter, so that the electronic-transport role is held fixed and the variation is confined to the ion-transport phase.

Two design axes are varied. The *composition* axis varies the mass fraction of the ion-transport polymer poly(ethylene oxide) (PEO) and of the lithium triflate salt (LiOTf) relative to the semiconductor. The *chemistry* axis substitutes the ion-transport host, the salt anion (triflate or bis(trifluoromethanesulfonyl)imide, TFSI), and the alkali cation (Li^+ , Na^+ , K^+ , as the triflate or TFSI salt). The comparative work replaces the hyperbranched polyester-amide *Hybrane* host of the proof-of-concept device (chapter 2) with two hosts used here at the same level: linear PEO and the hyperbranched polyether trimethylolpropane ethoxylate (TMPE), so that the host axis contrasts a linear against a branched polyether at matched chemistry. Every device reported *quantitatively* uses the silver electrode. A separate corpus of gold-electrode devices also exists, entirely at the lead composition; rather than discard it, the electrode is treated as a third, illustrative axis in its own right (section 4.8), where it both contrasts an active against an inert electrode and provides an independent-electrode check on the chemistry comparisons.

Film thickness is a controlled covariate, not a confound. Because adding more ion-transport polymer thickens the solution-processed film, the spin-coating speed was, in most batches, raised for the higher-PEO cells specifically to partially equalise thickness across the grid (spin speeds span 1500 rpm–3500 rpm); this compensation was deliberate but not applied uniformly to every comparison batch. The natural question is therefore whether the composition trends below are in fact thickness effects. They are not, and the point is settled directly against the profilometry record (the silver composition series, $n = 30$; fig. 4.1). The compensation is real but incomplete, so a residual covariation survives —

median film thickness rises from ≈ 227 nm at PEO 0.3 to ≈ 298 nm at PEO 1.2 (Pearson $r \approx 0.68$ between PEO fraction and thickness). Four lines of evidence show that thickness is nonetheless not the operative variable: the reported metrics are dimensionless ratios and timescales in which a geometric thickness effect cancels; the activation voltage is flat (2.2 V–2.4 V, $r \approx -0.06$) across the full $2.6\times$ thickness range; a 38% within-cell thickness spread leaves the fading-memory time essentially unchanged; and, most directly, the partial correlation of thickness with $\log t_{1/2}$ controlling for PEO vanishes ($\approx +0.05$) while the composition effect survives controlling for thickness (≈ -0.42). Thickness covaries with PEO as a downstream consequence of polymer loading, but the dynamics track composition. The four arguments with their full statistics, and a column-by-column audit confirming that every other fabrication variable is held constant or crossed with composition rather than confounded with it, are given in section C.2.

Drive and read protocols. Because the comparisons below turn on them, the electrical protocols are stated explicitly — and they are *not* identical across the three measurements (table 4.1). Hysteresis uses a triangular voltage sweep. The potentiation (N -pulse build-up) was acquired with a 3 V write and 1.5 V read at a variable pulse count ($N \leq 1000$); the depotentiation (fading-memory decay) used a fixed train of 30 write pulses at a 4 V write and 2 V read. The inter-pulse interval is effectively common (0.104 s for potentiation, 0.103 s for depotentiation), but the *write amplitude differs* between the two (3 V vs 4 V) — a point that matters when the two are later composed (section 4.10, Chapter 5). The read voltage is above the ionic activation threshold established in chapter 2, a point returned to in section 4.6. An older 2022 device generation used a 6 V write with a 3 V read; the modern and old amplitudes are never pooled, and the same-device contrast between two amplitudes is itself the drive-protocol result (section 4.6).

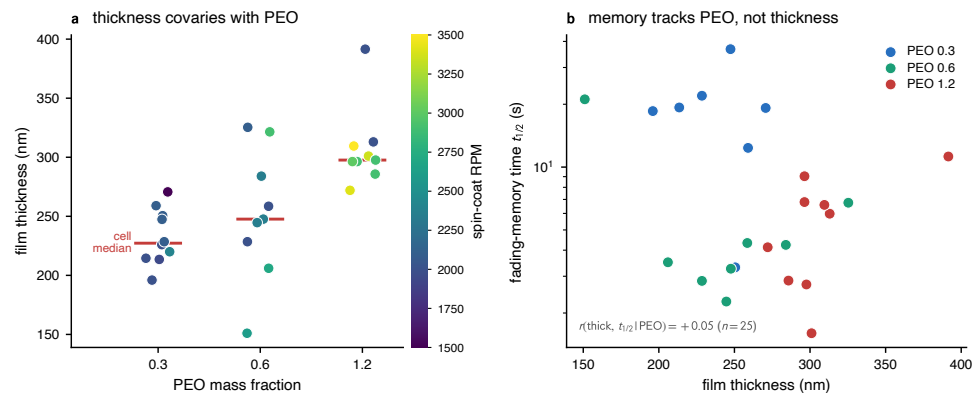


Figure 4.1: Film thickness covaries with composition but does not drive the dynamics (silver composition series, $n = 30$). (a) Film thickness rises with PEO fraction (red bars are cell medians); the spin-coating speed (colour) was raised for higher-PEO cells to partially equalise thickness, but the compensation is incomplete. (b) The fading-memory time $t_{1/2}$ is set by composition (PEO colour), not by thickness (horizontal axis): within each PEO band the timescale is flat against thickness, and the partial correlation of thickness with $t_{1/2}$ controlling for PEO is ≈ 0 .

Table 4.1: The three common measurements use distinct drive protocols (modern *Super Yellow*/PEO/LiOTf/Ag corpus). Potentiation and depotentiation differ in write amplitude, so the chapter never pools them and Chapter 5 composes them only as an explicit assumption.

Measurement	Write	Read	Pulse train
Hysteresis (HYST)	triangular sweep	—	—
Potentiation (PULSES)	3 V	1.5 V	variable $N \leq 1000$, 0.104 s
Depotentiation (DELAYTIME)	4 V	2 V	30 pulses, 0.103 s
Protocol overlay (v114)	3 V, 6 V	1.5 V, 3 V	30 pulses

4.3 Methods of Analysis

Hysteresis curves are summarised by the on–off conductance ratio and by a normalised loop area; potentiation curves by the per-decade growth

of the conductance ratio with pulse number, by the peak ratio, and by whether the response turns over (decreases) at the highest pulse counts; and depotentiation curves by a fading-memory time constant.

For the depotentiation (fading-memory) decay, the comparative archive is refitted with the more general Kohlrausch (stretched-exponential) function

$$\Delta G(t_{\text{wait}}) = \Delta G_0 \exp\left[-\left(\frac{t_{\text{wait}}}{\tau}\right)^\beta\right] + \Delta G_\infty, \quad (4.1)$$

with characteristic relaxation time τ , stretch exponent $0 < \beta \leq 1$, initial change ΔG_0 , and residual ΔG_∞ [147]. Two practical points follow from the data. First, the eight-to-thirteen-point decay curves often under-constrain β and τ jointly; the fitted τ is therefore reported only where the fit is well identified, and a robust, model-free half-enhancement time $t_{1/2}$ (the delay at which the conductance enhancement has fallen to half its initial value) is used as the primary timescale. Where both are available they agree. Both are *apparent* timescales: they depend not only on the material but on the amplitude used to write and read the state, a methodological point established in section 4.6 that disciplines every cross-protocol timescale comparison below — and the reason the composition series and the chemistry comparisons are each measured under a single fixed protocol. Second, fitted β values lie predominantly in the range 0.6–0.9, i.e. genuinely stretched relaxation, consistent with the heterogeneous distribution of activation barriers invoked in chapter 2.

Data-quality control and aggregation. Two screening layers are applied before any number is reported: erratic or non-physical measurements are flagged and excluded, and a per-device visual review stores per-curve verdicts (used as measured, cleaned, or discarded) in a curation registry applied automatically by the analysis scripts. All timescales are recomputed after this screening, because the instrument-side fit is

not robust to anomalous points. Summaries then follow a conservative curve→pixel→device→cell ladder of medians, and the representative curves of fig. 4.2 are drawn by script from devices near their cell medians rather than hand-picked. The full raw-to-summary provenance trail, the aggregation ladder, and the stretched-exponential identification thresholds are documented in section C.1.

4.4 Composition Control of the Dynamics

The quantitative core of this chapter is the *Super Yellow*/PEO/LiOTf composition grid, in which the PEO mass fraction is varied over 0.3, 0.6 and 1.2 and the salt mass fraction over 0.045, 0.09 and 0.18, with two to four nominally identical silver-electrode devices per cell. Each measurement was acquired under its own fixed protocol (table 4.1); the composition contrasts below are therefore always drawn *within* a single measurement, never by pooling the 3 V potentiation with the 4 V depotentiation. Before the grid is summarised, fig. 4.2 shows what the three underlying measurements actually look like on representative devices, so that the on–off ratio, the growth exponent, the turnover, and the fading-memory time can be read off real curves before they are reduced to per-cell numbers. Three trends, mutually consistent and each visible across the grid, define the result.

First, the steady-state switching window contracts as the PEO fraction rises. At fixed salt mass fraction 0.09, the median on–off ratio falls from ≈ 3.3 at PEO 0.3 to ≈ 2.4 at PEO 1.2, and the normalised loop area falls correspondingly from ≈ 0.33 to ≈ 0.18 ; the activation voltage is, by contrast, essentially flat across the whole grid (2.2 V–2.4 V), indicating that composition tunes how *much* the conductance moves rather than the voltage at which it begins to move. The widest window in the grid sits in the low-PEO row, with the largest loop area at the low-salt corner

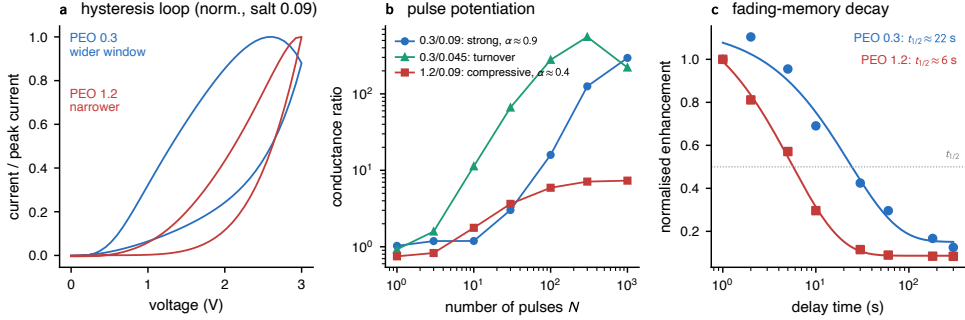


Figure 4.2: Representative raw curves of the three common measurements, on devices chosen near their cell medians (*Super Yellow*/PEO/LiOTf/Ag). (a) Hysteresis loops, each normalised to its own peak current so the comparison is the loop *openness* (the switching window) rather than absolute conductance: the low-PEO device (PEO 0.3, blue) opens a wider window than the high-PEO one (PEO 1.2, red), which carries more current but switches over a narrower range. (b) Pulse potentiation (conductance ratio vs pulse number, log–log): a strong near-linear accumulation ($\alpha \approx 0.9$), a low-salt cell that *turns over* and then declines, and a weak, compressive high-PEO response ($\alpha \approx 0.4$). (c) Fading-memory decay (enhancement vs delay, normalised to the first read) with stretched-exponential fits and the half-enhancement level dotted: the low-PEO device crosses $t_{1/2}$ far later (≈ 22 s) than the high-PEO one (≈ 6 s).

(PEO 0.3, salt 0.045: area ≈ 0.44); the per-sweep on–off ratio is strongly right-skewed there (cell median ≈ 3.4 but individual sweeps reaching ~ 40), so robust cell medians rather than means are reported throughout to avoid letting a few outlying sweeps define a cell.

Second, potentiation weakens, and changes shape, as the composition is varied (fig. 4.5). Two axes act distinctly. The PEO fraction controls the *strength* of pulse integration: the per-decade growth of the conductance ratio (the log–log slope α) falls with PEO content — most cleanly at the lowest salt, where it drops from $\alpha \approx 1.1$ (near-linear accumulation) at PEO 0.3 to $\alpha \approx 0.3$ (strongly compressive) at PEO 1.2 — and the peak conductance ratio, i.e. the usable dynamic range, is largest in the low-PEO, low-salt corner ($\approx 480\times$ at PEO 0.3/salt 0.045) and falls to a

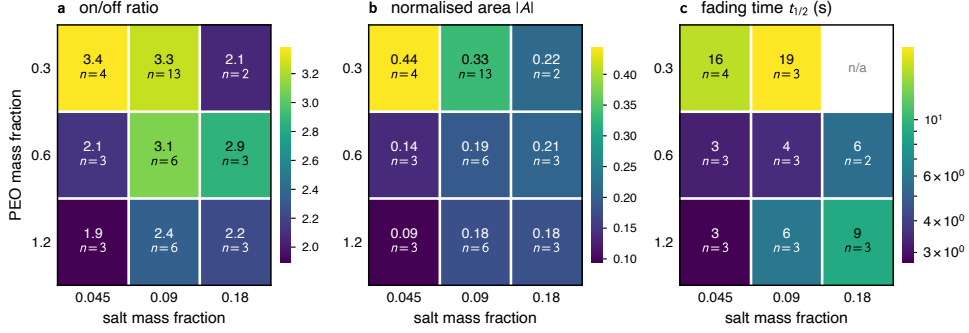


Figure 4.3: Composition control of the steady-state and dynamical response of *Super Yellow*/PEO/LiOTf devices. The switching window (a, b) and the fading-memory time (c) both decrease as the PEO mass fraction increases at fixed salt content.

few-fold at high PEO. The salt fraction, by contrast, controls *how far* the response can be driven before it saturates: at the lowest salt (0.045) every cell *turns over* — the conductance ratio peaks and then declines — by $N \sim 100\text{--}300$ pulses, whereas at the highest salt (0.18) the response grows monotonically through the largest counts tested ($N \approx 1000$), with salt 0.09 intermediate. Two caveats are kept in view: the α trend is noisier at the higher salts, where the sparsely sampled PEO 0.3/salt 0.18 cell ($n = 2$, curation-salvaged) departs from it; and the turnover sets a practical ceiling on the usable pulse count. Read constructively, these responses are a composition-tunable, predominantly compressive and (in the low-salt cells) non-monotone integration of the input pulse train — the nonlinear input encoding that Chapter 5 builds on.

Third, and most important for the rest of the thesis, the fading-memory timescale is composition-tunable over roughly 2s–20s. The longest retention occurs in the low-PEO row (at PEO 0.3, $t_{1/2} \approx 19$ s at salt 0.09, with an identified $\tau \approx 25$ s); increasing the PEO fraction to 0.6 or 1.2 shortens the timescale to a few seconds. A single curation-salvaged device at the still-lower PEO 0.15 level ($t_{1/2} \approx 22$ s) is consistent with

this trend, though it lies outside the replicated grid and is not counted toward it. The same ordering is reproduced by the instrument-side single-exponential time constant computed independently, which provides an internal cross-check on the trend. The device-to-device scatter within each cell is substantial — the fading-memory time constant varies by $\sigma(\ln \tau) \approx 0.85$ between nominally identical devices at fixed composition, a spread not explained by film thickness, spin speed, anneal, or device age (section C.2; quantified in section D.3) — and is not treated here as mere noise (fig. 4.4): the spread of timescales *across* the grid, together with this within-cell variability, constitutes the heterogeneous bank of fading-memory elements that Chapter 5 exploits; the realised palette across the whole device library, beyond the quantitative core, is mapped in section C.4. The same composition axis is visible in the potentiation strength: the growth exponent α falls monotonically with PEO fraction (cell medians $\approx 0.9 \rightarrow 0.7 \rightarrow 0.35$), again with real within-cell spread rather than a single value per composition.

Taken together, these observations support a single physical reading: increasing the proportion of the ion-transport phase produces a weaker, faster-relaxing memristive response, while the salt content governs how far that response can be pushed before it saturates. This is the quantitative core of the chapter, and the one result that rests on replicated devices; the per-cell decay and potentiation values behind the heatmaps are tabulated in section C.3. The PEO gradient is statistically secure for the fading-memory time (per-device Spearman $\rho = -0.52$, permutation $p = 0.009$) and falls in the same direction but only marginally ($p \approx 0.06$ – 0.07) for the window and potentiation descriptors, so the timescale trend is reported as established and the others as suggestive; a two-way decomposition confirms that PEO and salt act near-orthogonally, with no significant interaction term, and bootstrap confidence intervals on every cell median are given in section C.7. The mechanism it implies — that a more ion-

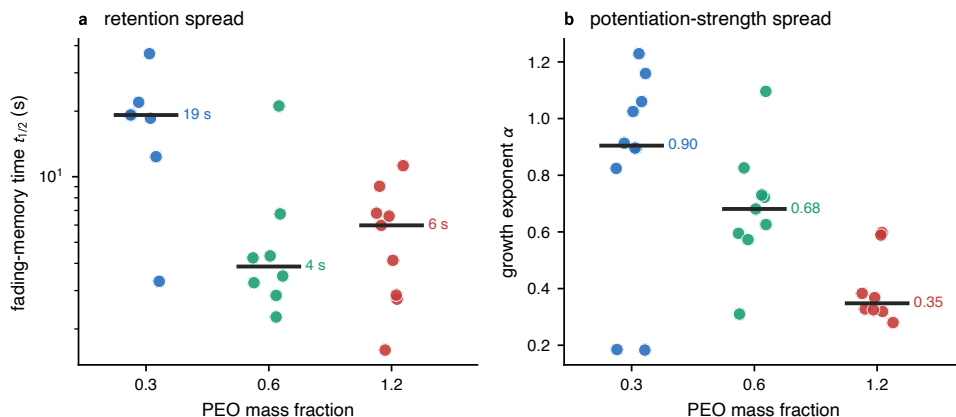


Figure 4.4: The cell medians of fig. 4.3 are summaries of genuinely spread distributions, not tight clusters (one point per device, *Super Yellow*/PEO/LiOTf/Ag; black bar is the cell median). (a) Fading-memory time $t_{1/2}$ and (b) potentiation growth exponent α , each against PEO fraction. The PEO fraction sets the broad level, but within every composition cell the device-to-device variation is comparable to the steps between cells — the heterogeneity that Chapter 5 treats as a resource rather than averaging away.

rich matrix relaxes a displaced ionic profile more readily — makes an independent, *equilibrium* prediction about the ionic resistance of these films; that prediction is tested directly by impedance spectroscopy in section 4.7.

4.5 Chemical Tuning of the Dynamics

Beyond composition, the identity of the electrolyte components shifts the dynamics further. These comparisons are reported as an illustrative survey: each matched cell (fixed composition, electrode, and protocol) contains at most one or two screened devices, so the magnitudes below are indicative and the sample size is given in every case. They are complemented here by structural and optical probes on *separate* sample sets

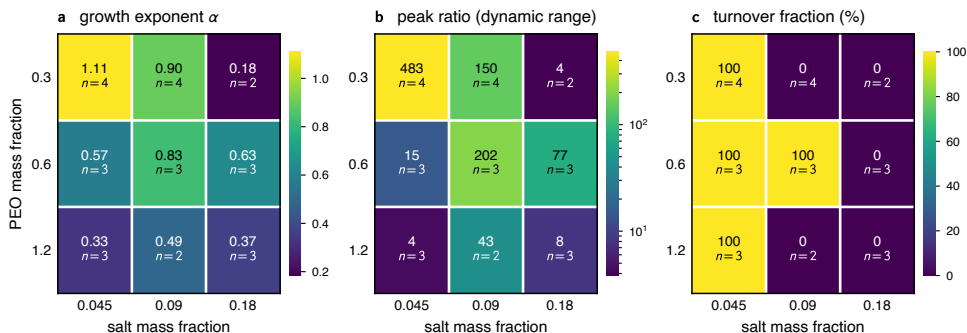


Figure 4.5: Potentiation (pulse-integration) response across the *Super Yellow*/PEO/LiOTf composition grid: (a) the log–log growth exponent α , (b) the peak conductance ratio (usable dynamic range, log scale), and (c) the fraction of devices whose response turns over within the measured range ($N \leq 1000$). Increasing the PEO fraction lowers α and the dynamic range; increasing the salt fraction suppresses turnover, sustaining growth to higher pulse counts. Per-cell median, with the device count n annotated in each cell.

— blend films and scratched powders at the lead composition (0.3/0.09)
 — examined by ATR-FTIR, X-ray diffraction, and UV-Vis absorption, which probe ion association, host backbone order, crystallinity, and the semiconductor’s electronic gap directly, independently of how a device is driven (figs. 4.6 and 4.8). These too are single specimens per chemistry and are read illustratively.

Host. Substituting the linear polyether PEO by the hyperbranched polyether TMPE, at fixed lithium-triflate chemistry and composition, shortens the fading-memory time by roughly a factor of six (PEO/LiOTf: $t_{1/2} \approx 20$ s–25 s, three screened silver devices; TMPE/LiOTf: ≈ 4 s, one screened silver device with a clear decay). This is the best-replicated of the chemistry comparisons and the clearest single chemistry effect, although the TMPE side rests on a single clean device. A caveat of provenance applies: the cleanest test would have been the lithium pair from a single designed host $\times$ cation batch, but the PEO/LiOTf corner of that batch

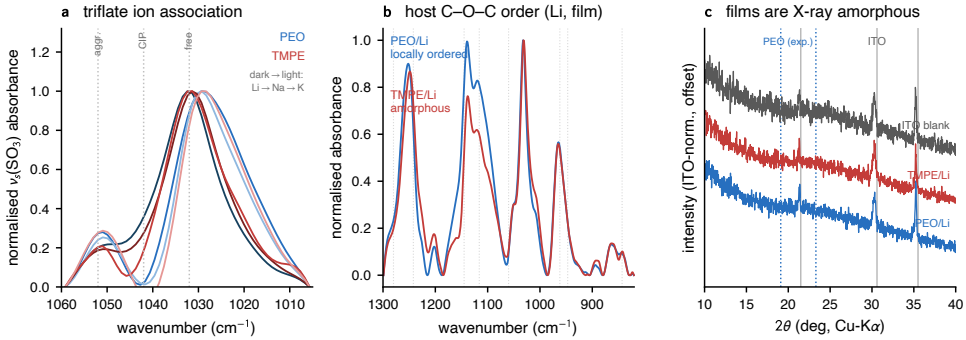


Figure 4.6: Microscopic corroboration of the chemistry axis (electrode-free *Super Yellow*-blend films/powders, 0.3/0.09, one specimen per chemistry). (a) ATR-FTIR triflate ion-association band $\nu_s(\text{SO}_3)$, whose position reports the ion-association state (spectroscopically free near 1032 cm^{-1} , contact ion pair near 1042 cm^{-1} , aggregate above; reference positions dotted [146]; PEO hosts in blues, TMPE hosts in reds, darker to lighter for Li/Na/K): it sits at 1029 cm^{-1} – 1032 cm^{-1} in *every* host $\times$ cation sample with no cation-resolved shift — microscopically consistent with the cation not being a clean retention variable. (b) ATR-FTIR C–O–C envelope: the linear PEO host (blue) shows a resolved multiplet, i.e. retained *local* (conformational) ether-backbone order, whereas the hyperbranched TMPE host (red) collapses it into a single broad, amorphous band [146]. (c) X-ray diffraction of PEO/Li and TMPE/Li with a bare-ITO blank: all patterns show only the crystalline ITO substrate reflections (grey lines; 21.5° , 30.6° and 35.5°), with no sharp PEO crystalline doublet (blue dotted; 19.1° and 23.3°) and no crystalline salt phase — the composite films are X-ray amorphous, so PEO’s order in (b) is short-range, not bulk crystallinity [126, 146]. ATR spectra are normalised after a local/rubber-band baseline and XRD is normalised to the ITO peak; intensities are not comparable across specimens (mixed film/powder sampling and thin films), so only band *positions* and *shapes* are read.

was a broken device and had to be discarded, so the comparison instead pairs the TMPE device against the nearest-matched silver PEO/LiOTf devices (same operator, fresh semiconductor, glovebox), which differ in fabrication date.

An independent microscopic correlate of the host substitution is visible

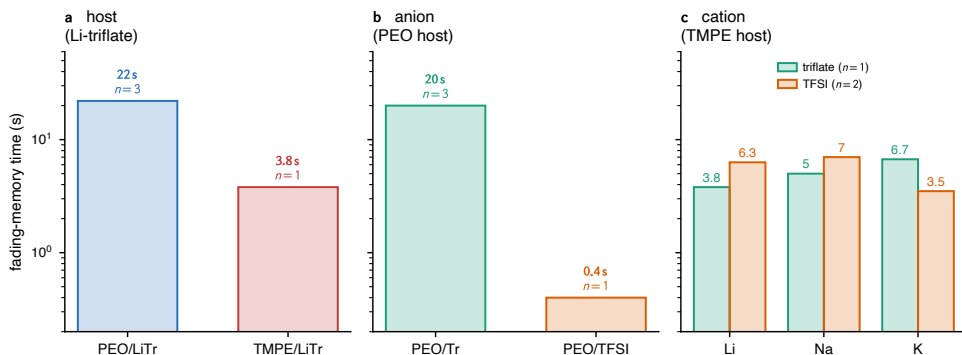


Figure 4.7: Illustrative effect of electrolyte chemistry on the fading-memory time constant (*Super Yellow*, 0.3/0.09, silver, matched 4 V/2 V protocol). Host (PEO vs TMPE, Li-triflate) and anion (triflate vs TFSI, PEO host) produce the largest shifts. The cation panel shows the same three cations in the TMPE host under each anion: the order *inverts* with the anion (potassium longest in triflate, shortest in TFSI), so the cation produces no consistent, host- and anion-independent ordering. Sample sizes are small (n as labelled); see text.

in the polyether backbone itself, and at two length scales. In ATR-FTIR the linear PEO film shows a resolved C–O–C stretching multiplet, with the weak order-sensitive marker bands near 843 cm^{-1} , 962 cm^{-1} and 1242 cm^{-1} , indicating retained *local* (conformational) order of the ether backbone, whereas the hyperbranched TMPE film collapses these into the single broad band of an amorphous polyether (fig. 4.6b) [146]. At the longer length scale of crystallites, however, X-ray diffraction on the same sample set finds *neither* host crystalline in the composite: every pattern is dominated by the ITO substrate (confirmed against a bare-substrate blank), with no sharp PEO crystalline reflections and no segregated crystalline salt phase (fig. 4.6c), so the films are X-ray amorphous and PEO retains short-range order without forming long-range crystallites [126, 146]. A host that is locally more ordered, even while globally amorphous, set against a fully amorphous TMPE, is consistent with the lower segmental mobility and slower ionic relaxation that the

longer PEO fading-memory time would require: in polyether electrolytes ion transport is carried by segmental motion of the host above its glass transition, so the relaxation of a displaced ionic profile is rate-limited by that segmental mobility [88, 131, 132]. With one specimen per host this is offered as a structural correlate rather than a demonstrated cause. The host ordering is moreover not specific to the silver electrode: on the independent gold-electrode corpus at the same composition, PEO again retains far longer than TMPE and supports a markedly wider switching window and stronger potentiation (section 4.8), so the host effect survives a change of electrode. That the salt forms no crystalline phase in any chemistry additionally supports reading these blends as homogeneous, salt-dissolved ion-conducting media.

Anion. Replacing triflate by the more charge-delocalised TFSI anion, in the PEO host, shortens the fading-memory time dramatically — from ≈ 20 s (triflate) to of order 1 s (TFSI). The direction is the one the electrolyte literature would anticipate: the delocalised TFSI charge weakens ion pairing and lowers the salt lattice energy, so that TFSI salts dissolve and dissociate more freely and plasticise the polyether host more strongly than triflate, raising ionic mobility [128, 148] — all of which should speed the relaxation of a driven ionic profile. The TFSI decays additionally tend to a sharper, more abrupt (compressed) shape rather than the stretched form of the triflate devices, suggesting a qualitatively different relaxation-time distribution. This is the most weakly controlled of the chemistry comparisons: the single PEO/TFSI device belongs to a later fabrication generation than the triflate devices and differs from them not only in the anion but also in spin speed and in whether the salt was filtered before use. The effect is large enough, and the read-out metrics dimensionless enough (section 4.2), that these accompanying differences are unlikely to account for a twenty-fold change, but the comparison is reported as illustrative for this reason as much as for its sample size.

Cation. The original motivating hypothesis — that the alkali cation would order the fading-memory timescale as $\text{Li}^+ > \text{Na}^+ > \text{K}^+$, by analogy with the hard–soft acid–base argument of chapter 2 [89] — is *not* supported by the data. The most controlled evidence available here is a pair of cation series measured in the *same* host (TMPE), at the same composition, electrode, and protocol, differing only in the salt anion (the cation panel of fig. 4.7). With triflate, the timescale *increases* along the series — $\text{Li}^+ \approx 3.8\text{ s}$, $\text{Na}^+ \approx 5.0\text{ s}$, $\text{K}^+ \approx 6.7\text{ s}$ (one screened device per cation, all $R^2 \geq 0.93$) — so potassium retains *longest*, the reverse of the hypothesised order. With TFSI, the same three cations invert: potassium is now *shortest* ($\approx 3.5\text{ s}$) and lithium and sodium are together and longer ($\approx 6\text{ s}$ – 7 s , two devices per cation). A single change of anion thus flips which cation persists longest, while in each series the within-cation device-to-device variation is comparable to the between-cation differences. The conclusion is that, in this archive, cation identity does not produce a robust, host- and anion-independent retention ordering: any apparent ordering (such as potassium being shortest in the TFSI family, which is consistent with the weakest cation–oxygen coordination) is contingent on the rest of the chemistry. A properly powered test of the cation hypothesis would require a dedicated series at fixed host, anion, protocol, and electrode with at least three replicates per cell, and is identified as future work in section 4.10.

This cation null is not an artefact of the driven measurement: three independent probes reach it from three directions. The first is ion association. The triflate symmetric SO_3 stretch $\nu_s(\text{SO}_3)$ sits at 1029 cm^{-1} – 1032 cm^{-1} in *all six* host $\times$ cation films and powders, with a spread at or below the instrumental resolution and no ordering by cation (fig. 4.6a); the band lies at the spectroscopically-free / contact-ion-pair boundary in every case. At the level of ion pairing, that is, the cation leaves no host-independent fingerprint — the same conclusion the driven retention

reaches, by an independent probe. The second is a change of electrode: on the gold-electrode TMPE/triflate cation series at the same composition, the steady-state switching window is flat across Li^+ , Na^+ , and K^+ (on-off ratio ≈ 1.3 – 1.5 with no monotone ordering; section 4.8), so a second electrode and a second observable return the same absence of a clean cation ordering. The third probe is the electronic structure: UV-Vis absorption of *Super Yellow*/TMPE blend films across the cation series (a separate, later Au-electrode generation) places the *Super Yellow* π – π^* absorption peak of this poly(phenylene-vinylene) derivative [86] at 446 nm with an onset of ≈ 2.50 eV *identically* for Li^+ , Na^+ , and K^+ (a salt-free control matches at the band edge), with no new sub-gap feature (fig. 4.8). The cation thus perturbs neither the ion-association state nor the semiconductor’s optical gap; consistent with the design premise of the chapter, the electrolyte chemistry tunes the ionic dynamics while leaving the *Super Yellow* electronic backbone fixed.

The hierarchy that emerges from these comparisons is therefore that the host and the anion shift the fading-memory time more strongly, and more legibly, than the cation does.

4.6 The Drive Protocol Sets the Timescale

A methodological result underlies, and disciplines, every comparison above. Because the read voltage used to interrogate the state is above the ionic activation threshold, and because the write amplitude controls how far the ionic profile is driven, the *drive protocol* itself influences the apparent fading-memory time. The point is made cleanly by a single device measured under both protocols: its fitted relaxation time increases from ≈ 4.6 s under the 3 V-write/1.5 V-read protocol to ≈ 15.5 s under the 6 V-write/3 V-read protocol — a roughly three-fold change in the measured timescale with no change in material, only in drive (fig. 4.9).

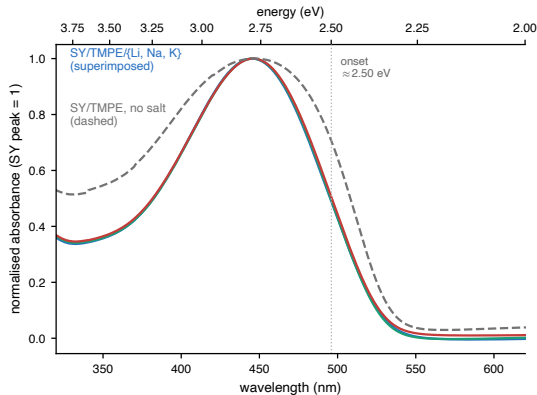


Figure 4.8: UV-Vis absorption of *Super Yellow*/TMPE blend films (lead composition 0.3/0.09; Au-electrode 2025 generation; one specimen per chemistry), normalised to the *Super Yellow* π - π^* peak. The peak (≈ 446 nm) and the absorption onset (≈ 2.50 eV) are superimposed for Li^+ , Na^+ , and K^+ ; the salt-free control (dashed) coincides at the band edge (its differing blue side and lower absolute absorbance are thickness/scattering, not electronic). The cation does not shift the semiconductor’s optical gap. Only band *positions* are read because absorbance magnitudes are thickness-confounded.

The mechanistic reading is that a larger write amplitude displaces more of the ionic charge (and, at the largest amplitudes, may drive electrode-level electrochemical processes), producing a deeper, longer-lived written state, while the supra-threshold read perturbs the state during interrogation. Two consequences follow. First, the composition result of section 4.4 is unaffected, because it was obtained entirely within a single, fixed protocol; only the chemistry comparisons, which unavoidably span device generations, are exposed to this confound, and they are reported accordingly. Second, the result is a positive design rule for the field: the volatile timescale of these devices is not a fixed material constant but is co-determined by how the device is driven, and any meaningful cross-device or cross-chemistry comparison of retention must hold the write/read protocol, the electrode, and the composition fixed.

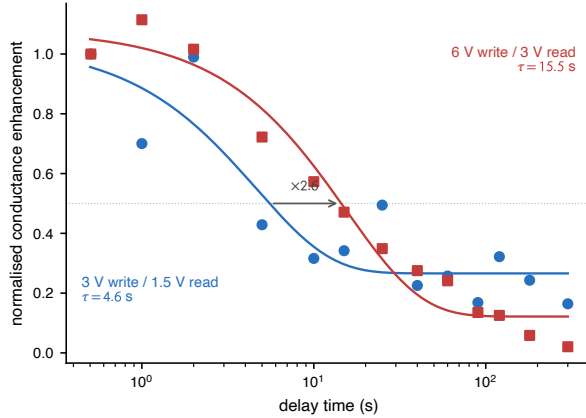


Figure 4.9: The apparent fading-memory time is set in part by the drive protocol. The same device, measured under a larger write/read amplitude, shows a markedly longer apparent relaxation time: the fitted single-exponential time constant rises from $\tau \approx 4.6$ s to ≈ 15.5 s, and the half-enhancement crossing (dotted, arrow) shifts by $\approx 2.8\times$.

4.7 Microscopic Corroboration: Impedance Spectroscopy

Every result so far rests on the *driven* response — pulses and decays. Small-signal impedance spectroscopy offers an independent test of the mechanism proposed for the composition trend: if increasing the PEO fraction shortens the fading-memory time because a more ion-rich matrix lets the displaced ionic profile relax more readily, then the *equilibrium* ionic resistance should fall along the same composition axis. The archive contains such a measurement, acquired on the same devices: 0 V-DC spectra (40 mV AC, 1 Hz– 1×10^6 Hz) on 26 silver-electrode *Super Yellow*/PEO/LiOTf devices spanning the composition grid.

Each spectrum is summarised in two separate ways. The first is *model-free*: the real part of the impedance at the apex of the main Nyquist semicircle, $Z_{\text{real}}^{\text{apex}}$, which fixes the resistance scale of the spectrum without

committing to any circuit. The second is *model-based*: the bulk ionic resistance R_{ion} of an equivalent circuit adapted from the polymer light-emitting electrochemical cell analysis of Munar *et al.* [149], reported only as corroboration of the model-free quantity because the per-spectrum circuit fits are genuinely ill-conditioned. The circuit topology, its two automated fits, and the per-cell values of both observables are detailed in section C.5.

Both observables fall steeply and monotonically with PEO fraction (fig. 4.10a). The model-free $Z_{\text{real}}^{\text{apex}}$ drops by more than three orders of magnitude across the grid — cell medians from $\sim 10^8 \Omega$ at PEO 0.15–0.3 to $\sim 10^5 \Omega$ at PEO 1.2 — with a rank correlation against PEO of -0.69 per device ($n = 26$) and -0.83 on cell medians; the fitted R_{ion} gives the same picture (-0.84 on cell medians). This is exactly the direction the mechanism requires: more ion-transport polymer, lower ionic resistance.

The EIS resistance scale and the *independently* measured fading-memory time fall along the *same* composition axis (fig. 4.10b): both decline together as the PEO fraction rises, and across cells $\log Z_{\text{real}}^{\text{apex}}$ covaries with $\log t_{1/2}$ at $r \approx +0.74$ ($+0.61$ per device, $n = 23$). Three qualifications bound this reading:

- The PEO fraction is the *common cause* of both quantities, so this correlation confirms that the same composition axis moves the equilibrium ionic resistance and the driven retention together — it is not evidence of a second, independent factor.
- It is the resistance *scale* that carries the composition signal, not the EIS relaxation *frequency*: the apex frequency corresponds to a sub-second RC time and does *not* track the seconds-long fading-memory time, consistent with impedance spectroscopy reporting the small-signal ionic-transport bottleneck rather than the slow relaxation that governs forgetting. The two timescales are nonetheless reconcilable

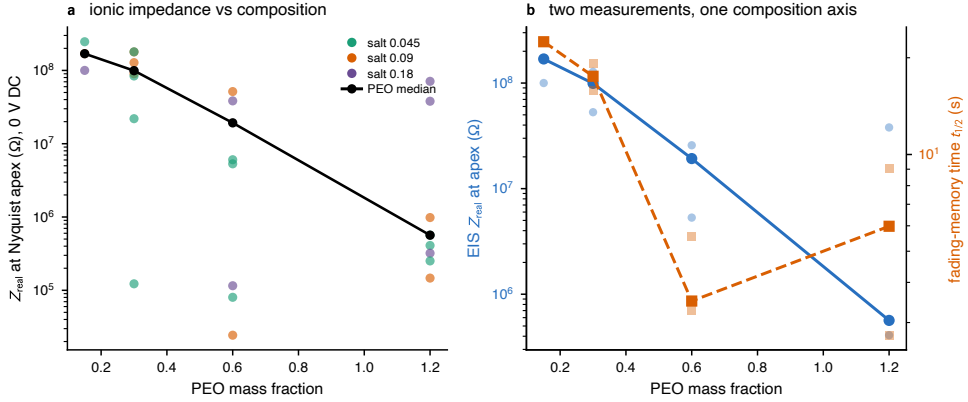


Figure 4.10: Equilibrium (0 V-DC) impedance spectroscopy corroborates the composition control of the dynamics (silver *Super Yellow*/PEO/LiOTf grid). (a) The impedance resistance scale — here the model-free real impedance at the Nyquist apex — falls by more than three orders of magnitude as the PEO fraction rises (points are devices, coloured by salt; the black line is the PEO median). (b) The equilibrium impedance scale (left axis) and the *independently* measured fading-memory time $t_{1/2}$ (right axis) decline along the *same* PEO composition axis (cell-level points with per-PEO median trends): the two are co-driven by composition, not independent variables.

in *magnitude*: the sub-second apex RC is the geometric-capacitance response, and scaling it by the chemical-to-geometric capacitance ratio measured independently from the low-frequency branch ($\gtrsim 33$) predicts a large-signal relaxation of several seconds, inside the observed $t_{1/2}$ range (section C.9).

- The effect is not a film-thickness artefact — $Z_{\text{real}}^{\text{apex}}$ spans $\sim 10^4\times$ while the film thickness spans only $2.6\times$, and the PEO dependence survives controlling for thickness (partial correlation -0.60 ; section 4.2).

Impedance spectroscopy thus supplies, for the ionic-resistance half of the mechanistic picture below, the independent microscopic support that the driven measurements alone cannot.

4.8 Electrode Dependence: Active Silver versus Inert Gold

Every result so far uses the silver top electrode. The archive also holds a gold-electrode corpus (50 devices), and rather than set it aside it is read here as a third, illustrative axis. One structural fact governs how it can be read: *every* gold device sits at the single lead composition (PEO or TMPE host, 0.3/0.09), so the gold corpus is a chemistry-axis set at fixed composition and carries *no* information about the composition series of section 4.4, which therefore remains a silver-only result. What the gold corpus can do is twofold: contrast an electrochemically active electrode (silver) against a noble, inert one (gold) at matched chemistry, and provide an independent-electrode check on the chemistry comparisons of section 4.5.

At matched chemistry and composition the electrode shifts all three behavioural families, and does so coherently (fig. 4.11). In the lead PEO/LiOTf cell, replacing silver by gold narrows the steady-state switching window (median on–off ratio $3.3 \rightarrow 2.0$, normalised loop area $0.33 \rightarrow 0.12$), weakens and flattens potentiation (peak conductance ratio $\approx 124\times \rightarrow 29\times$, log–log slope $\alpha \approx 0.9 \rightarrow 0.45$), and raises the activation voltage ($\approx 2.4\text{ V}–2.7\text{ V}$) — yet *lengthens* the fading-memory time, from $t_{1/2} \approx 19\text{ s}$ on silver (three devices) to $\approx 87\text{ s}$ on the two clean gold devices ($R^2 \approx 0.97–0.99$, $\beta \approx 0.6$). The same window-and-potentiation contraction holds in the TMPE/LiOTf cell (on–off $3.8 \rightarrow 1.3$; peak $54\times \rightarrow 1.9\times$). The per-cell metrics for every chemistry on both electrodes, including the host-free and salt-free negative controls, are tabulated in section C.6.

The natural reading is the textbook contrast between an active and an inert electrode. Silver is electrochemically active — mobile Ag^+ can be injected and electrodeposited, the basis of electrochemical-metallisation switching — and so contributes a large but *volatile* component to the response: a

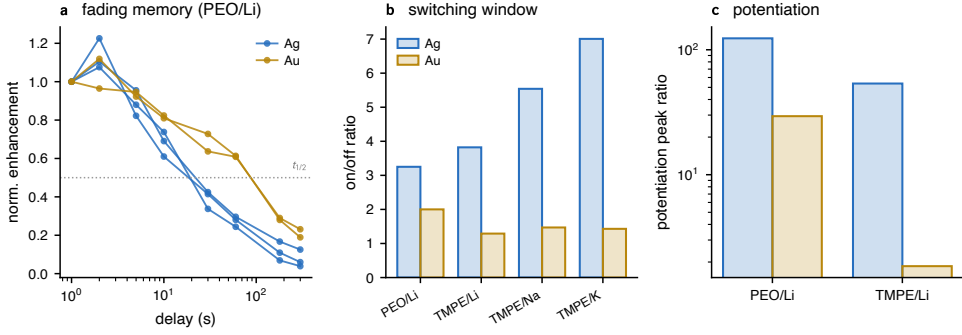


Figure 4.11: Illustrative electrode contrast at the lead composition (0.3/0.09). (a) Fading-memory decays of PEO/LiOTf devices, normalised to the first read: the gold curves (gold) cross half-enhancement later than the silver ones (blue), i.e. gold retains *longer*. (b) The steady-state switching window (on-off ratio) is narrower on gold for every chemistry, and is flat across the TMPE cation series on gold (no Li/Na/K ordering). (c) Pulse-potential dynamic range (peak ratio, log scale) is far smaller on gold. The inert electrode thus gives a weaker, higher-threshold, but more retentive response. Small n per cell, no human curve curation on the gold side, and a generation offset between the corpora bound the comparison (see text).

wider window, a stronger and steeper potentiation, a lower threshold, and a fast-relaxing written state [69, 70]. Gold is noble and inert, so the switching is carried by the polymer-electrolyte ionic mechanism alone: smaller in amplitude, higher in threshold, but more persistent. That the higher activation voltage appears on gold despite its films being *thinner* (spun faster) argues against a film-thickness origin and for an interfacial, electrode one [106].

Two of the chemistry conclusions of section 4.5 recur on this second electrode, which is the more useful outcome for a set of comparisons that is otherwise singly sampled. The host ordering persists — PEO retains far longer than TMPE on gold as on silver — and the cation null persists: the gold TMPE/triflate series shows a flat switching window across Li^+ , Na^+ , and K^+ (on-off ≈ 1.3 – 1.5 , no monotone trend), the same

absence of ordering the silver retention reported, now seen through a different observable. The gold corpus additionally supplies clean negative controls that the silver set lacked: host-free (semiconductor-plus-salt) and salt-free (semiconductor-plus-host) films show neither a switching window (on-off ≈ 1.0) nor potentiation (peak ≈ 1), confirming that both the ion-transport polymer and the dissolved salt are required for the memristive response.

These statements are illustrative, and for the familiar reasons. The gold retention rests on two clean devices; the gold curves were not subjected to the per-device visual curation applied to the silver chemistry set (section 4.3); and the corpora differ in fabrication generation (the gold devices are all 2024–2025, the silver chemistry and lead-cell decays 2022–2023), so electrode and generation are partly confounded — the one silver batch fabricated alongside the gold batch, by the same operator, was not measured under the decay protocol. The electrode is therefore reported here as a coherent, mechanistically sensible effect and as cross-electrode corroboration of the chemistry axis, not as a quantitative law; isolating it cleanly would require a single matched-generation batch split between the two electrodes.

4.9 Discussion

The picture that emerges is of a device whose volatile dynamics are controlled by a hierarchy of variables. The strongest and most reliable is the *composition* of the ion-transport phase, which tunes the switching window, the potentiation strength, and the fading-memory time together and in the same direction: more ion-transport polymer yields a weaker, faster-relaxing device. Next come the *host* and *anion* of the electrolyte, which shift the fading-memory time over a wider range still but are documented here only illustratively. The *cation*, which the proof-of-concept

chapter’s mechanism had suggested would be the controlling variable, is in fact the weakest and least legible variable in this dataset, and its effect is not separable from host, anion, and drive protocol at the available sample sizes. Cutting across all of these is the electrode itself: an inert gold contact, set against the active silver one, weakens the switching and potentiation but lengthens retention (section 4.8), a coarse, mechanistically distinct effect that acts on the electrode interface rather than on the electrolyte chemistry. The hard–soft acid–base framework of chapter 2 is retained here as a qualitative organising idea — it correctly anticipates, for instance, that the most weakly coordinating cation relaxes fastest within one family — but it is explicitly not advanced as a quantitative, transferable law on the strength of these data.

A working mechanistic picture. The composition trends admit a cautious interpretation, offered here as a working hypothesis rather than a demonstrated microscopic mechanism. A natural first guess — that increasing the PEO fraction simply dilutes and interrupts the *Super Yellow* electronic-percolation pathway — is in fact ruled out by the data: the *absolute* on-state conductance *rises* steeply with PEO (more than three orders of magnitude across the grid; section C.8), the opposite of what network dilution would produce. The window contraction is therefore better read as a *rising conductance floor*: the ion-rich phase adds a parallel, weakly switchable conduction path, so the off-state conductance climbs even faster than the on-state and the on–off ratio closes from below. The same ion-richer matrix should also let the displaced ionic profile relax more readily, shortening the fading-memory time — so that window, potentiation strength, and retention move together, as observed. The salt fraction, by contrast, plausibly sets the number of mobile, chargeable ionic states available before saturation, which would explain why a higher salt content sustains potentiation to larger pulse counts before turning over. The host and anion effects are most naturally read in

terms of cation–ether coordination strength, ion pairing, and segmental mobility [128, 132] — that is, the breadth and depth of the distribution of local activation barriers that underlies the stretched-exponential relaxation [147]. For the ionic-resistance step in this picture there is now direct, independent support: the equilibrium impedance of these films falls along the composition axis and tracks the fading-memory time (section 4.7). Vibrational spectroscopy and X-ray diffraction add further microscopic correlates on the chemistry axis: ATR ties the host’s effect to its *local* ether-backbone order (retained in PEO, absent in amorphous TMPE), while XRD shows the composite to be X-ray amorphous in every chemistry — so the order is short-range rather than bulk crystallinity, and the salt is dissolved rather than segregated — and ATR shows the cation leaving no ion-association fingerprint while UV-Vis shows it leaving the *Super Yellow* optical gap unshifted, consistent with the cation’s weak and inconsistent effect on retention and with the chapter’s premise that the electrolyte chemistry moves ion transport, not the *Super Yellow* electronic backbone (section 4.5). The remaining links — in particular the anion and barrier-distribution arguments — are not established microscopically here; temperature-dependent measurements and a chemistry-resolved impedance study (section 4.10) would be needed to test them, and the hard–soft acid–base idea is kept strictly qualitative.

The variability that runs through the chapter is itself a usable property. The spread of fading-memory times, both across compositions (roughly an order of magnitude) and between nominally identical devices, is precisely the kind of heterogeneity that a physical reservoir requires: a bank of elements with diverse but individually reproducible-enough time constants, whose fading memory of past input is the defining property a reservoir computes upon [5, 18, 28]. Both halves of that requirement can be put to number. The composition axis supplies, conservatively, between two and five resolvably distinct memory timescales (a two- σ criterion

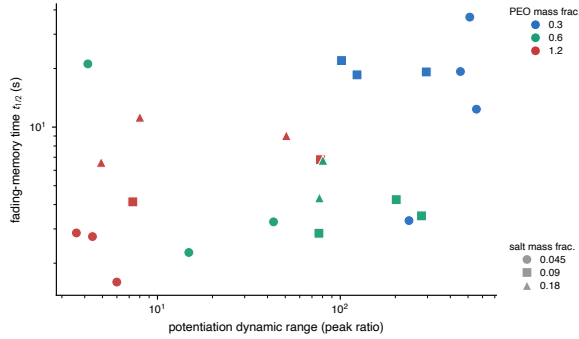


Figure 4.12: Per-device design space of the *Super Yellow*/PEO/LiOTf/Ag composition grid: fading-memory time $t_{1/2}$ against potentiation dynamic range (peak ratio), coloured by PEO fraction and marked by salt fraction (one point per device, median over its pixels). The PEO fraction selects the broad region a device occupies — low PEO (blue) in the long-memory, high-dynamic-range corner, high PEO (red) in the opposite one, so memory and nonlinearity are not independently tunable but *paired*. The substantial within-composition scatter is the genuine device-to-device heterogeneity that Chapter 5 exploits.

on the cell-median span of $\Delta \ln t_{1/2} \approx 2.1$ against the within-cell scatter; section C.11) — a handful of cleanly separated bands, not a continuum — while each junction reproduces its switching window to $\approx 15\%$ sweep-to-sweep (section C.10), far tighter than the diversity the bank spans, so the elements are reproducible enough to be used as fixed nonlinear filters. This is made concrete in fig. 4.12, which places every screened silver device in a fading-memory ($t_{1/2}$) versus dynamic-range (peak ratio) plane: the PEO fraction sets the broad region a device occupies — low PEO in the long-memory, high-dynamic-range corner; high PEO in the opposite one — so a single composition selects a *paired* operating point, while the substantial scatter within each composition is the genuine device-to-device diversity. Chapter 5 takes up this point directly.

4.10 Limitations and Future Work

The principal limitation is one of statistical power off the composition axis. Only the *Super Yellow*/PEO/LiOTf composition grid is replicated; the host, anion, and cation comparisons rest on one or two screened devices per matched cell, and are therefore illustrative. Four specific consequences should be stated plainly:

1. The cation hypothesis is not tested here at adequate power, and the apparent absence of a $\text{Li}^+ > \text{Na}^+ > \text{K}^+$ ordering should be read as “not demonstrated in this archive” rather than as a positive refutation.
2. The supra-threshold read and the amplitude dependence of the apparent timescale (section 4.6) mean that absolute retention values are protocol-specific.
3. The electrode is now reported as an illustrative axis in its own right (section 4.8) rather than simply held out, but it remains partly confounded with fabrication generation, and device ageing is still an uncontrolled variable that was not modelled.
4. Most consequentially for Chapter 5, pulse integration and forgetting were characterised *separately and at different write amplitudes*: the potentiation curves were acquired at a 3 V write and the fading-memory decays at a 4 V write (table 4.1), at an effectively common inter-pulse interval, and the archive contains no experiment that varies the input timing to probe the interaction of integration and forgetting directly.
5. The fading-memory protocol measured a single junction per device, so repeat-decay reproducibility of $t_{1/2}$ is not characterised at the population level (only one device carries two junctions; section C.10).

Within-device reproducibility is established only for the steady-state switching window, from its repeated sweeps.

Because the drive amplitude itself co-sets the apparent timescale (section 4.6), composing the 3 V potentiation shape with the 4 V decay in Chapter 5 assumes the two regimes are compatible — an explicit modelling assumption rather than a measured fact. The clean test is a single matched-amplitude experiment that sweeps both pulse count and inter-pulse interval; it is named as future work.

The corresponding future-work programme is therefore well defined: a dedicated cation series in a single host and anion, at a single fixed protocol and electrode, with a sub-threshold read voltage (as in the proof-of-concept device of chapter 2) and at least three replicates per cell, would convert the illustrative chemistry survey into a powered comparison. The impedance corroboration of section 4.7 is likewise extensible: it is reported here only for the replicated PEO/LiOTf composition axis at 0 V DC, whereas the archive also holds bias-resolved spectra (up to 3 V DC) and chemistry-varied devices that were not modelled, so a chemistry- and bias-resolved impedance study would test the host, anion, and barrier-distribution arguments that the present 0 V analysis leaves open. A single matched-generation batch split between silver and gold top electrodes would likewise convert the illustrative electrode contrast of section 4.8 into a clean, generation-controlled comparison. Most decisive of all for the mechanism, however, would be a temperature-dependent measurement of the relaxation time: the entire host-, anion-, and barrier-distribution reading of this chapter rests on segmental mobility above the glass transition and on a distribution of activation barriers, yet none of it is currently falsifiable without a temperature axis. A $\tau(T)$ series yielding an Arrhenius or Vogel–Fulcher–Tammann activation energy would tie the host and anion differences directly to a segmental-relaxation barrier, predict the temperature dependence of the stretch exponent β , and is

the natural partner to the chemistry-resolved impedance study above; it is the single most informative experiment the present archive lacks. Encapsulation and a systematic ageing study would address the stability dimension.

4.11 Bridge to Temporal Computing

This chapter has shown that the volatile dynamics of *Super Yellow* polymer-electrolyte composites are controllable, and that the control is dominated by composition. Quantitatively, the fading-memory time of the *Super Yellow*/PEO/LiOTf system is tunable over roughly 2 s–20 s by composition alone, with the switching window and potentiation strength varying in concert; chemistry (host, anion) extends this range further, illustratively; and the cation, contrary to the initial hypothesis, is not a robust controlling variable at the available power. A methodological result — that the drive protocol co-sets the apparent timescale — frames how all of this must be measured and compared.

What Chapter 5 requires from the present chapter is precisely a set of behavioural ingredients: a composition-indexed family of fading-memory time constants spanning a useful range, a characterisation of how conductance is built up by pulses (including its saturation and turnover), and an account of device-to-device heterogeneity. Those ingredients are now in hand. They also define the shape of the design space the next chapter inherits: the fading-memory time and the input nonlinearity (α) are *coupled* through the PEO fraction (fig. 4.12) — a single composition selects a paired (memory, nonlinearity) operating point rather than two independent settings — while the salt fraction acts as a partially separate control on the usable dynamic range and turnover. A temporal-computing substrate must therefore be assembled from a *bank* of compositions to span the behaviour it needs, rather than tuned to a single optimum; the

heterogeneity catalogued here is what makes that bank possible. The next chapter builds its temporal-computing substrate directly from this heterogeneity.

Chapter 5

Data-Driven Temporal Computing with Polymer-Electrolyte Memristor Reservoirs

5.1 Introduction and Scope

Chapters 2 and 4 treated the slow, volatile response of these polymer-electrolyte composites first as a phenomenon to be demonstrated and then as a property to be controlled. This chapter treats it as a computational resource. The guiding idea is simple: a device whose conductance carries a fading record of its recent input is, in miniature, a temporal processor, and a collection of such devices is a *reservoir* — a physical substrate that projects a time-varying input into a high-dimensional state from which simple, linear read-outs can solve temporal tasks.

The chapter rests on a single thesis that connects it to chapter 4: composition and drive *set* the device timescale, and a task *selects* the timescale it

needs. The design problem is therefore one of *matching*, and the open question is what a *spread* of timescales — the low-cost, intrinsic heterogeneity of a solution-processed device family — adds on top of a single well-chosen one. The target application is low-power, wearable *affective computing* — the machine recognition of human emotional and physiological state — whose signals are slow, multi-timescale, and a natural match to these devices. The evidence moves from task-agnostic reservoir benchmarks, to real physiological streams, to WESAD affective classification, and finally to deployment-style tests under sensor noise, consumer wrist-band sensing, and energy and training-cost constraints.

The evidence base is intentionally constrained. Every result is *in silico*: each device node is a compact behavioural model whose parameters are taken directly from the chapter 4 fits (section 5.2), and every read-out is a single linear regression, so that no claimed computation can be hidden in a nonlinear decoder or a hand-tuned dynamical system. No new devices are fabricated. Two demonstrations run throughout: Demonstration A asks whether a *single* composition suffices for a single-timescale feature, and Demonstration B asks what a heterogeneous *bank* adds when the target is multi-timescale. The scoped answer is that the measured device models support temporal-computing demonstrations under constrained read-out conditions, that fading memory is the active ingredient, and that timescale heterogeneity is valuable for memory breadth and task generality but not a guaranteed improvement on every task.

5.2 Measured Behavioural Model

Chapter 4 established that the fading memory of these composites is an engineerable quantity, dominated by composition. To put that memory to work, this chapter reduces each composition cell to a compact behavioural model and treats it as the unit cell of a reservoir computer. The model is

deliberately minimal: it is fitted only to the three common measurements of chapter 4 — hysteresis, pulsed potentiation, and delayed depotentiation — so that every simulation in this chapter traces back to measured data rather than to a hand-tuned dynamical system.

Each composition cell is summarised by a *parameter card*: the write nonlinearity ϕ_c from the potentiation curves, the fading-memory law λ_c from the depotentiation curves, and an associated device-to-device spread. The fading memory is the stretched-exponential (Kohlrausch) retention identified in chapter 4,

$$\lambda_c(\Delta t) = \exp\left[-(\Delta t/\tau_c)^{\beta_c}\right],$$

where τ_c and β_c are the per-cell relaxation time and stretch exponent; for the few cells without an identified stretched fit, a single exponential reproducing the model-free half-life is used instead ($\tau_c = t_{1/2}/\ln 2$, $\beta_c = 1$). The write nonlinearity is the compressive, saturating build-up of conductance with pulse number measured in chapter 4, parametrised by its log-log growth exponent α_c and its peak enhancement.

Driven by a rate-coded input $u_n \in [0, 1]$ sampled at interval Δt , a single device node updates as

$$x_n^{(i)} = \lambda_i x_{n-1}^{(i)} + \left[\max(w_i u_n, 0)\right]^{\alpha_i}, \quad \lambda_i = \exp\left[-(\Delta t/\tau_i)^{\beta_i}\right],$$

so that the node integrates a compressive write of its (masked) input and then leaks by exactly the measured retention λ_i over the sample interval; w_i is the reservoir input mask. The internal state is read sub-threshold and is therefore assumed state-preserving, as in the proof-of-concept device of chapter 2. Every read-out in this chapter is a single *linear* ridge regression on the device states, so that any computation demonstrated is attributable to the devices and not to a nonlinear read-out.

Two constraints are carried explicitly from chapter 4 into every result

below. First, the write nonlinearity and the fading-memory law were measured at *different write amplitudes* (3 V for potentiation, 4 V for depotentiation; table 4.1); composing them in a single node is an explicit modelling assumption, not a measured fact, and it most likely underestimates the true coupling between drive and retention. Second, both were measured at a single, effectively common inter-pulse cadence, so the model is not entitled to claims about input timing finer than that cadence. Both caveats would be removed by the single bench experiment set out in section 5.11. The per-cell parameter cards inherited from chapter 4 and the full set of simulation and read-out settings are tabulated in section D.2.

5.3 Reservoir Computing and the Common Metric Language

The reservoir principle. Reservoir computing emerged from the observation that a fixed, randomly-parametrised dynamical system — the *reservoir* — can perform useful temporal computation provided only that a *linear* read-out is trained on its state, leaving the dynamics themselves untrained [17, 18, 30]. Two properties make a physical system a useful reservoir candidate: a *fading memory* of past inputs, so that the state reflects recent history but eventually forgets, and a *nonlinear* response, so that the state mixes inputs in ways a linear read-out can exploit. These are precisely the two properties characterised in chapter 4 — the stretched-exponential depotentiation supplies the fading memory, the compressive potentiation supplies the nonlinearity — which is what makes a bank of these devices a natural candidate substrate for reservoir computing. The same recognition has driven a wave of memristive and organic reservoir demonstrations [5, 32, 34, 35], of which the polymer-electrolyte devices studied here are a low-power, biocompatible instance.

Architecture. The reservoir used throughout this chapter is a spatial bank of N *independent* device nodes, each a single behavioural model (section 5.2) driven by a randomly-masked copy of the input. Independence is a deliberately conservative choice: with no inter-node recurrence, all of the reservoir’s richness must come from the spread of intrinsic timescales across compositions, the device-to-device scatter, and the input mask, so any benefit attributed to heterogeneity cannot have been inherited from engineered connectivity. Where a single composition is the subject (Demonstration A), the same node can equally be deployed as a time-multiplexed, delay-feedback reservoir whose virtual nodes are distributed along a delay line [150]; the two architectures are interchangeable for the present claims, and the spatial bank is used for uniformity.

Common metrics. So that the demonstrations remain comparable, every result is reported on a common set of metrics. The linear *memory capacity* MC_k and its total [151] quantify how much past input is linearly recoverable at each lag, and so visualise timescale coverage directly; NARMA-10 [152] is a standard nonlinear system-identification benchmark; the information-processing-capacity framework [80] separates linear from nonlinear contributions and is reported directly in fig. 5.3; and the affective tasks are scored by class-balanced macro-F1. Robustness is probed by injecting the measured device-to-device and cycle-to-cycle spread into the bank, reported in fig. 5.8.

5.4 Benchmark Tier: Capacity and System Identification

The first tier of evidence is deliberately task-agnostic. Two capacity measures — linear memory capacity and the information-processing

capacity that splits it into linear and nonlinear parts — test reservoir suitability directly, since together they measure the two properties a reservoir must have: fading memory and a usable nonlinearity. A third benchmark, the NARMA-10 system-identification task, is then used not to establish absolute performance but to *rank* the compositions against one another. All three use a random rate-coded input, a bank of N device nodes, and a linear ridge read-out, and all are reported on the same metric ruler so that the two demonstrations of this chapter remain directly comparable.

Memory capacity. The linear memory capacity of Jaeger measures how much of the past input a reservoir can linearly reconstruct: MC_k is the squared correlation between a ridge reconstruction of the input k steps ago and its true value, and the total memory capacity is the sum of MC_k over all lags, bounded above by the number of nodes N [151]. Figure 5.1 compares two banks of equal size ($N = 24$): a *homogeneous* bank of jittered copies of the lead cell (PEO 0.3/0.09, differing only by device-to-device scatter) and a *heterogeneous* bank spanning the composition grid (τ from ≈ 3 to ≈ 26 s). The heterogeneous bank broadens the memory profile across lags and raises the total memory capacity from 4.2 to 6.4 (means over ten random input/bank seeds), a factor of 1.54 ± 0.10 , with near-perfect single-step recall (MC_1 rising from 0.51 to 1.00) and the gain concentrated at short-to-intermediate lags. The benefit is positive in all ten seeds and clears a paired test (Wilcoxon signed-rank $p = 0.002$, its floor at this sample size), so the heterogeneity gain is reported with the same statistical discipline as the affective-label null of section 5.7. Because the nodes are independent — a bank with no inter-node recurrence — this is the hard case for a reservoir, and the heterogeneity benefit, which arises purely from the spread of timescales and the device scatter, is therefore a conservative estimate.

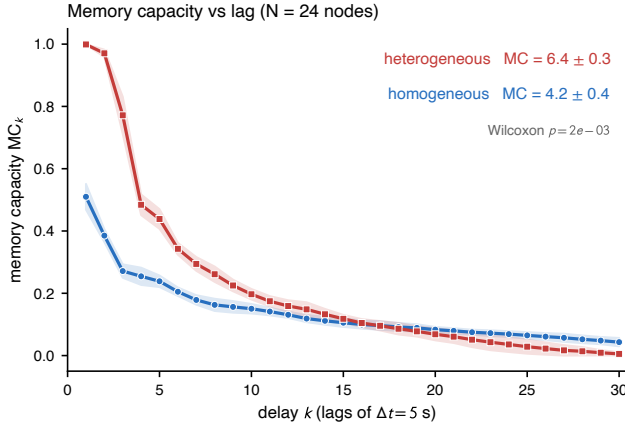


Figure 5.1: Linear memory capacity MC_k as a function of delay k for a homogeneous bank (all PEO 0.3/0.09 nodes) versus a heterogeneous composition bank ($\tau \approx 3\text{--}26\text{ s}$). The heterogeneous bank broadens recoverable memory across lags, raising total memory capacity from 4.2 to 6.4 (a $1.54\times$ increase; Wilcoxon $p = 0.002$ over ten seeds), with the gain concentrated at short-to-intermediate lags. Shaded bands are ± 1 SD over ten random input/bank seeds. $N = 24$ leaky nodes, random uniform input, linear (ridge) read-out; in-silico devices built from the Chapter 4 parameter cards.

Composition sweep and the single-node optimum. The same benchmarks, run on single-composition banks, rank the compositions against one another and so test the Demonstration-A claim that one cell suffices for a single-timescale task. Figure 5.2 shows that the lead cell PEO 0.3/0.09 attains the highest single-composition memory capacity (3.98 ± 0.19 , mean $\pm$ SD over ten seeds), while the neighbouring PEO 0.3/0.045 cell is marginally better on the NARMA-10 system-identification benchmark [152] (NRMSE 0.587 ± 0.019 against the lead’s 0.616 ± 0.031 , a close second). Both optima fall in the low-PEO row that chapter 4 identifies as carrying the longer fading-memory times, so the benchmark ranking is consistent with the composition result rather than an artefact of the simulation. This supports the Demonstration-A statement: for a single-timescale feature, the lead composition is the memory-capacity optimum and a heterogeneous bank is unnecessary. The

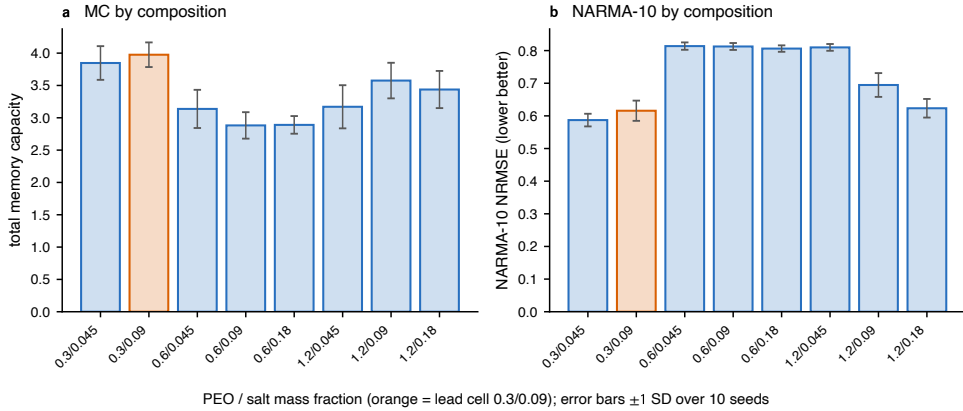


Figure 5.2: (a) Total linear memory capacity and (b) NARMA-10 reconstruction error (NRMSE, lower is better) for single-composition reservoirs across the PEO/LiOTf grid. The lead cell PEO 0.3/0.09 (orange) attains the highest memory capacity; the neighbouring 0.3/0.045 cell is marginally better on NARMA-10. Both optima fall in the low-PEO row anticipated by chapter 4. Only the *relative* composition ranking is claimed: absolute NARMA-10 NRMSE is high because a bank of independent, non-recurrent leaky nodes is a weak NARMA-10 reservoir.

NARMA-10 numbers carry only *relative* weight: the absolute error is high (≈ 0.6 – 0.8) because a bank of independent, non-recurrent nodes is a weak NARMA-10 reservoir, so the stronger evidence here comes from the capacity measures above (figs. 5.1 and 5.3) and NARMA-10 serves only to corroborate the composition ranking — a limitation a recurrent or delay-feedback architecture would relax.

Is the measured timescale distribution special? A natural control asks whether the *particular* timescales the compositions realise matter, or only that they are *spread*. Replacing the device bank with a generic echo-state network of the same size — leaky nodes whose relaxation times are drawn at random rather than measured — answers it. A random bank spanning the *same* range as the compositions ($\tau \approx 3$ – 26 s) matches the device bank’s memory capacity to within a twentieth (6.7 against the

5.4. BENCHMARK TIER: CAPACITY AND SYSTEM IDENTIFICATION

device bank’s 6.4; the small difference favours the random bank, $p = 0.05$), and a random bank spanning a *broader* range (0.5–60 s) exceeds it (7.6; $p = 0.002$). The reading is deliberately unflattering and clarifying: the device family’s specific τ values are not uniquely optimal, and the resource is the *spread* of timescales, which a solution-processed composition grid supplies *intrinsically* and at no design cost. The broader-range result also shows where headroom lies — extending the timescale window, the drive-amplitude variable of chapter 4, would raise capacity further — and motivates the tonic-band reach discussed in section 5.5.

Linear and nonlinear capacity. Memory capacity counts only what is *linearly* recoverable. The information-processing-capacity framework of Dambre et al. completes the picture by decomposing the total capacity into orthogonal polynomial degrees: degree-one (single-lag) terms give the linear capacity — essentially the memory capacity — while degree-two terms (per-lag squares and cross-lag products of the input) give the nonlinear capacity that only a nonlinear reservoir can supply, the whole bounded above by the node count N [80]. Figure 5.3 reports the split. Even the homogeneous bank carries a substantial nonlinear capacity (2.7 of a total 7.1), which quantitatively supports the interpretation that the compressive potentiation of chapter 4 supplies a usable nonlinear response — the second of the two reservoir prerequisites, alongside fading memory — rather than merely storing a linear echo. Heterogeneity raises the total capacity from 7.1 to 11.6 (paired Wilcoxon $p = 0.002$), lifting the linear and nonlinear parts together. The nonlinear capacity is almost entirely degree-two *self* terms (the per-node squaring of the write); cross-lag product terms are negligible, which is the signature of an *independent*, non-recurrent bank that does not multiply inputs across nodes — consistent with this architecture being, by design, the conservative hard case. That the total sits well below the N bound is likewise expected for leaky nonlinear nodes.

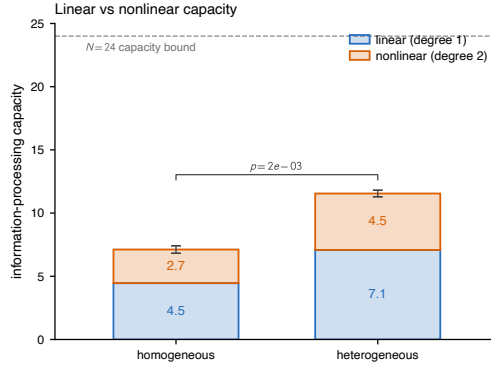


Figure 5.3: Information-processing capacity [80] split into linear (degree-one) and nonlinear (degree-two) parts, for the homogeneous versus the heterogeneous bank ($N = 24$, ten seeds). A nonzero nonlinear capacity in both banks indicates that the compressive write supplies a usable nonlinear response, not just linear memory; heterogeneity raises the total from 7.1 to 11.6 (paired Wilcoxon $p = 0.002$). The dashed line is the $N = 24$ upper bound; error bars are ± 1 SD on the total over seeds.

5.5 Affective Computing

Affective computing — the recognition of human emotional and physiological state by machines — was framed by Picard as a route to more natural human–computer interaction, classically from *peripheral* physiological signals that are slow, wearable-friendly, and unobtrusive to acquire [153]. It is the central application of this chapter, for two reasons that follow directly from chapter 4. First, the device timescale window — fading-memory times from a few seconds to tens of seconds — sits natively on top of the bands that carry affective information, so no aggressive time-rescaling is needed to bring signal and substrate into register (table 5.1). Second, affective information is *intrinsically multi-timescale*: a fast phasic electrodermal response, a respiration- and heart-rate-variability rhythm of a few seconds, and a tonic level that drifts over minutes all carry state at once [154–156]. That structure is the signal-driven motivation for a heterogeneous bank, and it is also the deployment case for an organic,

Table 5.1: Characteristic timescales of the peripheral affective signals used in this chapter, and the composition cells whose fading-memory window covers them [154–156]. The grid of chapter 4 spans the full range, with the lead PEO 0.3 cells serving the slow LF-HRV band.

Affective signal	Timescale	Device coverage
Phasic EDA (SCR)	$\sim 1\text{--}10\text{ s}$	low–mid PEO ($t_{1/2}$ 3–19 s)
Respiration	$\sim 3\text{--}5\text{ s}$	mid PEO
HRV, HF band	$\sim 2.5\text{--}7\text{ s}$	mid PEO
HRV, LF band	$\sim 7\text{--}25\text{ s}$	PEO 0.3 ($\tau \approx 19\text{--}26\text{ s}$)
Tonic SCL	minutes	slowest / drive-boosted

flexible, biocompatible, low-power substrate worn on the skin. Figure 5.4 makes the match concrete: it places each composition cell at its measured fading-memory time τ on the same axis as the affective bands, so that the coverage of the device family — and the one band it does not reach with composition alone — can be read off directly.

Closing the tonic gap. The minutes-scale tonic band that composition alone does not reach (the shaded gap in fig. 5.4) is exactly what the drive-amplitude variable of chapter 4 exists to fill, since raising the write amplitude lengthens the fading-memory time. Simulating that variable — extending the heterogeneous bank with drive-boosted, minutes-scale nodes — closes the gap as the design rule predicts: long-lag memory recall beyond 75 s nearly doubles (raising total memory capacity from 6.4 to 7.0), and on the real physiological streams the slowest-context reconstruction, tested now out to 120 s delays, improves by +0.019 in held-out R^2 (paired $p = 0.002$, all ten seeds; section D.5). This is a deliberately-labelled *modelled* extrapolation — no minutes-scale cell was measured — but it confirms that the timescale *window*, and not only the spread within it, is an actionable design variable, and it names the one experiment that would convert the tonic reach into a measured fact.

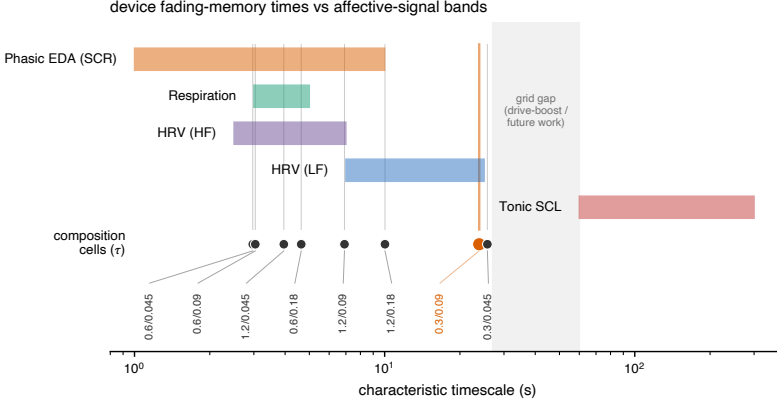


Figure 5.4: The bridge from Chapter 4 to this chapter, shown graphically. Each PEO/LiOTf composition cell is placed at its measured fading-memory time τ (markers, lower track) on the same logarithmic time axis as the characteristic timescale bands of the peripheral affective signals (coloured bars; table 5.1). The composition grid ($\tau \approx 3\text{--}26\text{ s}$) spans the phasic-EDA, respiration and HRV bands, with the lead cell PEO 0.3/0.09 (orange) sitting in the slow LF-HRV band. The tonic, minutes-scale band is not reached by composition alone (shaded gap) and would require the drive-amplitude control of chapter 4 or a slower future cell. This is the visual form of the design rule: an application selects a timescale, and performance follows from how well the device family covers it.

The experiments that follow use the Wearable Stress and Affect Detection corpus (WESAD) of Schmidt et al. [157]: fifteen subjects wearing a chest sensor that records electrodermal activity, respiration, temperature, and an electrocardiogram, with periods labelled baseline, stress, or amusement. WESAD is used in two complementary ways — as a source of real physiological streams whose multi-lag temporal context the reservoir must encode (section 5.6), and as a downstream affective-classification task (section 5.7).

5.6 Physiological Temporal-Context Reconstruction

The benchmarks of section 5.4 prove that a spread of device timescales broadens memory on a *random* input. This section puts that property to a more demanding test on the real, intrinsically multi-timescale affective signals introduced in section 5.5, isolating the multi-timescale *encoding* question before turning to final labels (section 5.7).

The reservoir is run continuously over each WESAD session at $\Delta t = 1$ s on four chest channels — electrodermal activity, respiration, temperature, and an electrocardiogram-derived heart rate — and a linear read-out reconstructs each channel at delays of 1, 3, 8, 20 and 45 s. These delays span fast phasic and beat-to-beat context, respiration- and HF-HRV-scale context, and tonic, LF-HRV-scale context [154, 155]. Evaluation is leave-one-subject-out over all fifteen subjects, with $N = 48$ nodes per bank.

Table 5.2 reports the held-out reconstruction quality. A memoryless bank barely improves on instantaneous input ($R^2 = 0.674$ against 0.673): with no fading memory, the past is simply not available. Memory of a single timescale lifts R^2 to 0.74, and the heterogeneous composition bank reaches the best single-bank score, 0.756 — a gain of +0.084 over instantaneous input and +0.012 over the best same-size homogeneous control. This last increment is small but unambiguous: scored per subject and paired against the best homogeneous bank across the fifteen subjects, the heterogeneous bank wins in *all fifteen* (Wilcoxon signed-rank $p < 10^{-4}$), so unlike the affective-label increment of section 5.7 it separates cleanly from zero. The mechanism is visible in the band-resolved columns: the fast homogeneous bank is best at the fastest context and the slow homogeneous bank at the slowest, but neither covers the whole vector, whereas the heterogeneous bank is the best *single* bank across the full fast-to-slow physiological

Table 5.2: Leave-one-subject-out held-out R^2 for multi-lag physiological context reconstruction on WESAD ($N = 48$ nodes; means over ten bank seeds where applicable). Bands are reconstruction delays: fast = 1–3 s, mid = 8 s, slow = 20–45 s. The heterogeneous bank gives the best single-bank score across the full context vector; each homogeneous bank wins only its own band. The heterogeneous advantage over the best homogeneous bank holds in all fifteen subjects (paired Wilcoxon $p < 10^{-4}$).

Bank	overall R^2	fast	mid	slow
Instantaneous	0.673	0.715	0.670	0.632
Memoryless	0.674	0.721	0.667	0.630
Homogeneous, fast	0.741	0.815	0.735	0.669
Homogeneous, slow	0.744	0.787	0.724	0.711
Heterogeneous	0.756	0.815	0.737	0.708

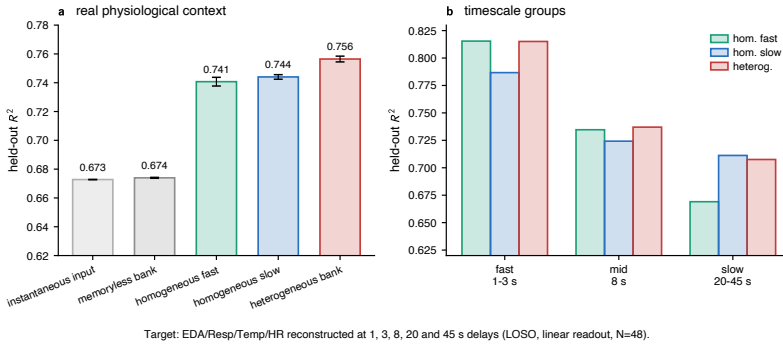


Figure 5.5: Multi-lag physiological context reconstruction on real WESAD streams (chest EDA, respiration, temperature, and ECG-derived heart rate). A linear read-out reconstructs each channel at delays of 1, 3, 8, 20 and 45 s from the reservoir state (leave-one-subject-out, $N = 48$). The heterogeneous composition bank gives the best single-bank held-out R^2 (0.756) across the full fast-to-slow context vector, ahead of the best same-size homogeneous control (0.744) and well ahead of instantaneous input (0.673). Fast and slow homogeneous banks each win only their own band.

context (fig. 5.5). This is the affect-aligned statement of “heterogeneity is a computational resource” on real signals, and it is exactly the result that the final-label classifier of section 5.7 cannot expose.

5.7 WESAD Affective Classification

The downstream task is affective-state classification from the WESAD physiological streams with the device reservoir and a leave-one-subject-out linear read-out. It is reported as a scoped application demonstration: the measured device models support affective classification under the constrained read-out used here, and the contribution of timescale heterogeneity to final labels is reported with error bars, including where it is null. For external calibration, the study that introduced WESAD reports a three-class chest accuracy of 76.5% at macro-F1 ≈ 0.72 with engineered window features and conventional classifiers under the same leave-one-subject-out protocol, and a binary stress-detection accuracy of $\approx 93\%$ [157]; the device reservoir’s scores below are of the same order. Placed against the wider WESAD literature (table 5.3), which spans engineered-feature classifiers and trained deep networks, the device reservoir’s leave-one-subject-out macro-F1 sits in the published performance band, but metric and protocol differences preclude ranking. The comparison shows that a bank of slow organic-memristor models read by a single linear regression can reach a meaningful WESAD performance level under this protocol.

Demonstration A: a single composition suffices for a single-timescale feature. Binary stress-versus-baseline classification from electrodermal activity, read out at the window level from the lead composition alone, reaches a leave-one-subject-out macro-F1 of 0.894 (fig. 5.6a). This supports the Demonstration-A claim that one well-matched composition can act as a useful affect-aligned node. It comes, however, with an instructive caveat: a static feature baseline (the mean, standard deviation, last value and slope of the same signal) matches it (0.899). The 60-second-window task is therefore quasi-static — the discriminative information sits in the instantaneous signal level, not in its temporal history — which is exactly why timescale heterogeneity has nothing to contribute

Table 5.3: The device reservoir against representative published WESAD systems. Figures are as reported by each study: classification *accuracy* for the engineered-feature and deep-network systems, the stricter class-balanced *macro-F1* for this work. Metrics and validation protocols differ across studies (Schmidt et al. and this work use leave-one-subject-out), so the table is context rather than a head-to-head ranking; its purpose is to show that slow organic-device states with a single linear read-out reach a meaningful WESAD performance level without a trained feature extractor. Binary = stress vs. non-stress; three-class = baseline/stress/amusement.

Study	Approach	Three-class	Binary
Schmidt et al. (2018) [157]	engineered features, AdaBoost/RF	76.5% (F1 0.72)	93.0%
Bobade & Vani (2020) [158]	engineered features, neural net	84.3%	95.2%
Gil-Martín et al. (2022) [159]	CNN on spectral features	85.1%	96.6%
This work	device-reservoir states, linear read-out	F1 0.758	F1 0.894

here, and why a memory-demanding reformulation is needed to test the heterogeneity claim fairly.

Demonstration B: streaming three-class tracking. The fair test is a streaming one: the reservoir runs continuously over each session and the affective class (baseline, stress, or amusement) is read out at *every* step, so that the task is designed to use fading memory rather than a static window summary. Table 5.4 decomposes the result into four matched-size banks. For reference, trivial baselines scored under the identical leave-one-subject-out protocol sit far below: a majority-class predictor reaches macro-F1 0.23 and a stratified-random predictor 0.33, so the reservoir’s 0.758 is well above chance and not an artefact of class imbalance. Significance of the bank comparisons is assessed across the fifteen *independent* held-out subjects, so the within-subject temporal autocorrelation of a per-step stream does not inflate the test. Relative to

Table 5.4: Leave-one-subject-out macro-F1 for streaming three-class WESAD tracking (means $\pm$ SD over ten bank seeds; matched input masks). Each increment is the gain over the row above: fading memory contributes a small, consistent gain, while the timescale-heterogeneity increment is within noise (*ns*).

Bank	macro-F1	increment
Instantaneous (no memory)	0.709	—
Memoryless (dimensionality only)	0.737	+0.028
Homogeneous (memory, single τ)	0.753 ± 0.009	+0.016
Heterogeneous (memory, τ spread)	0.758 ± 0.006	+0.005 (<i>ns</i>)

instantaneous input, adding dimensionality (a same-size memoryless bank) contributes +0.028, adding fading memory of a single timescale a further +0.016, and adding timescale heterogeneity a final +0.005. The measured fading-memory gain over a same-size memoryless bank is a modest but consistent +0.014 to +0.020. The timescale-heterogeneity increment, however, is $+0.005 \pm 0.011$ — within seed noise (positive in seven of ten seeds; across sampling intervals from 0.5 to 4 s the heterogeneous-minus-homogeneous gap stays in $[+0.002, +0.007]$ but never separates from zero). Per class, the best bank reaches F1 0.90 on baseline and 0.92 on stress but only 0.55 on the amusement class, which is the hard class for every model. The reason is an elicitation limit, not a memory-horizon one: amusement is the shortest WESAD condition (roughly 5.6 ks of labelled data against 17.6 ks of baseline and 10.0 ks of stress) and the lowest in autonomic arousal, so its physiological signature is both weak and under-represented. Consistently, no bank from memoryless to heterogeneous recovers it, which is why the class limits every model rather than discriminating between them.

Why heterogeneity does not pay off on the labels. The null is not a failure of the devices but a property of the task. Affect labels are slowly varying, sustained states; their discriminative memory demand is domi-

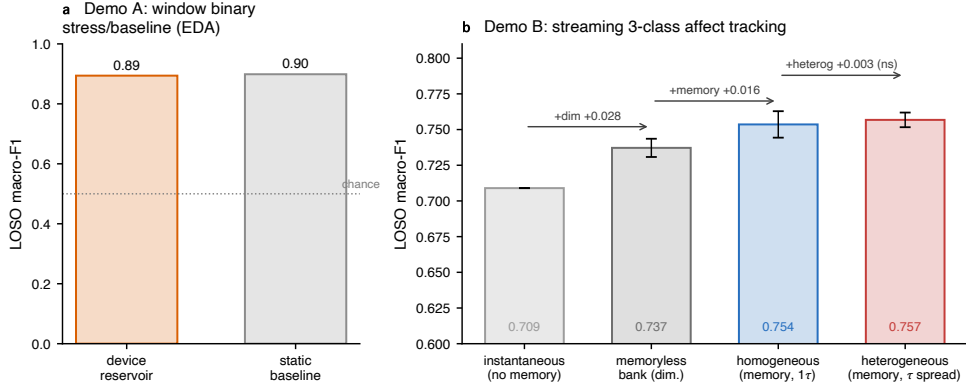


Figure 5.6: Affective classification with in-silico device reservoirs (leave-one-subject-out, linear read-out). (a) Demonstration A, window-level binary stress-versus-baseline from EDA: the single lead composition matches a static feature baseline, i.e. the 60 s-window task is quasi-static and requires no fading memory. (b) Demonstration B, streaming three-class affect tracking (baseline/stress/amusement) decomposed as instantaneous $\rightarrow$ +dimensionality $\rightarrow$ +fading memory $\rightarrow$ +timescale heterogeneity. Fading memory contributes a small but consistent gain; the timescale-heterogeneity increment ($+0.005 \pm 0.011$) is within seed noise. Error bars are ± 1 SD over 10 bank seeds.

nated by a single slow, tonic band that the lead composition’s relaxation time ($\tau \approx 19\text{--}26$ s) already serves. Timescale heterogeneity converts into task performance only when the target requires memory at many distinct lags *simultaneously* — which the memory-capacity benchmark (fig. 5.1) and the physiological-context reconstruction (table 5.2) both demand, and on which heterogeneity does pay, but which a single sustained label set does not. The reading carried into section 5.10 is that the measured device models support affective classification under this protocol, that fading memory helps, and that composition-timescale *matching* is the operative design variable — with heterogeneity a generality and robustness margin rather than a guaranteed improvement on any one label set.

5.8 Continuous Monitoring Under Deployment Conditions

The scores of section 5.7 are dominated by long stretches of sustained state, where the instantaneous signal level already carries the label — which is precisely why a static feature baseline ties the reservoir on the quasi-static window task. A wearable in the field is judged on what those averages hide. It must keep working while the wearer moves and the sensor is noisy, and motion artefact is the dominant failure mode of wrist-worn electrodermal and photoplethysmographic sensing. This is a temporal problem on which fading memory has a structural advantage: a leaky integrator is a temporal low-pass that averages transient artefact away, whereas an instantaneous read reacts to every spike.

Figure 5.7 makes the point on a continuous binary stress detector — stress versus baseline-plus-amusement, read out at every step, leave-one-subject-out, with the same single linear read-out used throughout. On clean streams every bank is competitive (macro-F1 $\approx$ 0.91–0.92, of the order of the $\approx$ 93% binary benchmark of the corpus [157]), the static read among them, exactly as the window task warned. As progressively stronger sensor noise and motion-artefact bursts are injected, the instantaneous and memoryless controls degrade steeply — macro-F1 from 0.914 to 0.840 by an injected noise of $\sigma = 0.4$ of the signal range — while the fading-memory banks barely move, holding $\approx$ 0.92 across the whole range. At $\sigma = 0.4$ the advantage of the heterogeneous memory bank over the instantaneous read is +0.088, positive in all fifteen subjects (paired Wilcoxon $p < 10^{-4}$); and even on clean streams, fading memory roughly halves the false-alarm rate ($0.046 \rightarrow 0.026$), the spurious-alert cost that most limits a deployed affect monitor. The active ingredient is memory itself, not its spread: the homogeneous and heterogeneous banks are equally robust, consistent with the design rule below. Two boundaries are

reported with the result rather than around it: the accuracy gain in the narrow window straddling label transitions is positive ($0.747 \rightarrow 0.775$) but not resolved across subjects, and the detection latency is comparable across banks, set by the prediction-smoothing horizon rather than by the device. The deployment-relevant statement is therefore specifically about robustness — under the corruption a real wearable faces, the temporal device sustains a stress detector that the static, instantaneous baseline, competitive only in laboratory-clean conditions, does not.

Real motion artefact, not a Gaussian proxy. The corruption above is synthetic, so the test was repeated with the noise *gated on the measured chest accelerometer*: at each step the injected amplitude tracks the wearer’s actual movement, concentrating artefact where the body moves rather than spreading it uniformly. The ordering replicates — the instantaneous read falls (macro-F1 $0.914 \rightarrow 0.898$ at the strongest gating) while the memory bank holds near 0.92, a $+0.032$ advantage that is positive across subjects (paired $p = 0.04$), and a classical moving average again matches the device ($+0.036$; section D.4). The effect is smaller than under the aggressive synthetic injection because the chest strap is mechanically stable and the protocol largely seated — the wrist, where motion artefact is severe, is the harder site examined in section 5.9 — but the robustness conclusion does not rest on the Gaussian noise model.

A classical-filter control: what the device does and does not buy. The mechanism just invoked — temporal low-pass rejection of transient artefact — is generic, so it must be asked honestly whether the robustness needs a device at all, or whether any smoother would do. The control is a classical one: take the same raw channels, apply an ordinary exponential moving average, and feed the smoothed signal to the same linear read-out, with no reservoir. It does. A single-time-constant moving

average matched to the lead cell’s relaxation recovers essentially all of the robustness — macro-F1 0.927 at $\sigma = 0.4$ against the instantaneous read’s 0.840 (+0.091, all fifteen subjects, $p < 10^{-4}$) — and indeed marginally exceeds the device bank itself (0.916). Spreading the moving average over the device timescale range adds little (+0.005, *ns*), and the device’s compressive *nonlinearity* adds nothing further over that linear filter bank on this binary task (−0.008, *ns*; table D.3). The honest reading sharpens rather than weakens the deployment argument. The noise robustness is not an exotic property of the reservoir; it is the property of *any* fading-memory filter, and the contribution of these devices is that they *are* that filter — the temporal averaging a digital pipeline would compute with an explicit, clocked filter is performed here passively by the material’s own relaxation, at the microwatt budget of section 5.9 and with no dynamics to train. Where the device’s nonlinearity and timescale spread do earn their place is the *multi-timescale* tasks of sections 5.4 and 5.6 — memory capacity and physiological-context reconstruction — on which a single moving average cannot compete and the heterogeneous bank is the resolved best; the single-timescale stress monitor simply does not demand them.

Cross-corpus replication. A single corpus cannot establish generality, so the robustness test was repeated on an independent dataset: the PhysioNet Non-EEG corpus of Birjandtalab et al. [160], twenty subjects wearing a *different* wrist sensor (an Affectiva Q recording electrodermal activity and skin temperature, with a pulse-oximeter heart rate) through relaxation and cognitive, emotional, and physical stress phases. Taking cognitive and emotional stress as the stress class against relaxation — the psychosocial analogue of the WESAD stressor, with the physical-exertion phase held out as a different mechanism whose racing heart rate is trivially separable from the instantaneous signal alone — and running the identical detector reproduces the WESAD result on this independent sensor and

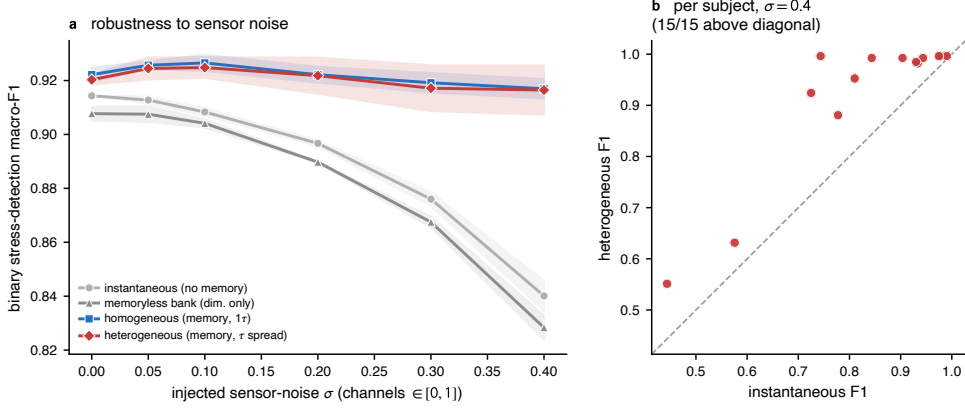


Figure 5.7: Continuous stress monitoring with device reservoirs (binary stress vs. baseline-plus-amusement, per-step, leave-one-subject-out, linear read-out). (a) Detection macro-F1 against the magnitude σ of injected sensor noise and motion-artefact bursts: the fading-memory banks (homogeneous, heterogeneous) stay flat at ≈ 0.92 , while the instantaneous and memoryless controls collapse, because temporal integration rejects transient artefact. Bands are ± 1 SD over seeds. (b) Per-subject scores at $\sigma = 0.4$: the heterogeneous memory bank beats the instantaneous read in all fifteen subjects (paired Wilcoxon $p < 10^{-4}$). Memory, not its heterogeneity, is the active ingredient; the homogeneous bank is equally robust.

population (table 5.5). The corpus is harder in absolute terms (clean macro-F1 ≈ 0.65 – 0.68 , against ≈ 0.91 on the cleaner WESAD chest signals) and, as on WESAD, the clean-stream memory advantage is small and not resolved. But under sensor noise the same divergence reappears, more sharply: the instantaneous detector falls from 0.651 to 0.516 while the fading-memory bank holds near 0.66 , a $+0.130$ advantage at $\sigma = 0.4$ that is positive in nineteen of the twenty subjects (paired Wilcoxon $p < 10^{-5}$). The noise-robustness of the temporal substrate is therefore not a property of one dataset; it transfers to a new corpus, sensor, and population.

Table 5.5: Cross-corpus replication of the noise-robustness result on the independent PhysioNet Non-EEG stress corpus [160] (twenty subjects; Affective Q wrist electrodermal activity and temperature plus pulse-oximeter heart rate; cognitive and emotional stress versus relaxation; leave-one-subject-out, linear read-out). As on WESAD, the instantaneous detector collapses under injected sensor noise while the fading-memory banks hold; the heterogeneous-minus-instantaneous advantage at $\sigma = 0.4$ is $+0.130$ (19/20 subjects, $p < 10^{-5}$). Memory, not its spread, is again the active ingredient.

Detector	macro-F1 (clean)	macro-F1 ($\sigma = 0.4$)
Instantaneous (no memory)	0.651	0.516
Homogeneous (memory)	0.682	0.657
Heterogeneous (memory)	0.667	0.646

5.9 Toward a Worn Device: Sensor Site and Operating Budget

The robustness of the previous section is a laboratory statement until two further questions are answered: whether the advantage survives a consumer-grade sensor, and what the device would cost to run. Both are answered from the same measured ingredients — the WESAD wrist recordings and the device constants of chapter 2 — without any new assumption.

Sensor site. WESAD records every session twice: from a chest strap (the clean electrocardiogram and respiration used so far) and from an Empatica E4 wrist band, the form factor of a real affective wearable but with fewer and far noisier channels — skin conductance, skin temperature, and a photoplethysmogram from which heart rate must be recovered, with no respiration. Re-running the continuous stress detector of section 5.8 on the wrist signals (table 5.6) gives a consistent result. The static, instantaneous detector loses ground on the wrist, its clean macro-F1 falling from 0.914

Table 5.6: Continuous binary stress-detection macro-F1 on the WESAD chest strap versus the Empatica E4 wrist band (leave-one-subject-out, linear read-out). Moving to the noisier consumer sensor degrades the instantaneous (static) detector but not the fading-memory reservoir, so the memory advantage is larger on the wrist; σ is the injected sensor-noise level of fig. 5.7.

Detector	Chest		Wrist E4	
	clean	$\sigma = 0.4$	clean	$\sigma = 0.4$
Instantaneous (no memory)	0.914	0.840	0.881	0.847
Memory reservoir (het)	0.920	0.916	0.922	0.916

on the chest to 0.881 as the noisier photoplethysmographic heart rate and the missing respiration channel degrade the instantaneous signal. The fading-memory reservoir is essentially unaffected by the move to the wrist ($0.920 \rightarrow 0.922$): it recovers from temporal context what the static read loses to channel quality. The memory advantage is therefore much larger on the consumer wrist than on the chest (+0.041 on clean streams against +0.006, and a sustained +0.069 under injected noise), which is the right way round for deployment — the harder, more realistic sensor is exactly where the temporal substrate helps most, and fading memory cuts the wrist false-alarm rate by about a third ($0.062 \rightarrow 0.040$).

Operating budget. The reservoir’s appeal for a worn device is that temporal filtering is delegated to the material dynamics: the dynamics are fixed and untrained, and only the linear read-out is fitted, by a single closed-form ridge solve rather than gradient descent. The trained part is therefore one $(N+1) \times K$ weight matrix — 75 weights for the $N = 24$, three-class banks here, 147 for the $N = 48$ banks — against the 10^4 to 10^6 trainable weights of the deep models usually brought to wearable affect, a reduction of roughly two to four orders of magnitude that puts on-device, per-user calibration within reach. The energy budget follows from the

measured device of chapter 2: at the proof-of-concept energy of ≈ 50 nJ per write event (≈ 6 fJ μm^{-2} over the 0.0825 cm² junction), charging each of N nodes once per one-second update draws of order a microwatt (≈ 1.2 μW at $N = 24$), while the read-out is $N K$ multiply-accumulates, well under a nanojoule per step. These are unoptimised, large-area figures; reducing the junction toward the integrated-device regime would cut the per-event energy by orders of magnitude. The estimate is explicitly for the reservoir *core* — device charging plus the read-out multiply-accumulates — and excludes the sensor analogue front-end, the analogue-to-digital conversion, and the microcontroller housekeeping that any wearable carries; it is a statement about the computational substrate, not a full system-power budget, and the same front-end overhead burdens the deep-network alternative it is contrasted with. The point is the order of magnitude and the contrast in kind: the device responds in the sub-kilohertz regime, comfortably above the sub-hertz bandwidth of affective physiology, so the substrate is never the bottleneck, and the temporal feature extraction that a digital pipeline would compute explicitly is performed by the device dynamics during the write/read cycle. The full budget and its assumptions are tabulated in section D.6.

5.10 Design Rule: Match Timescale to Task

Sections 5.4, 5.6 and 5.7 complete the design rule opened in chapter 4. Composition and drive *set* the fading-memory timescale; an application *selects* one timescale, or a set of timescales, and performance follows from how well the device family covers them (fig. 5.4). The composition sweep gives the single-timescale case (fig. 5.2): the lead cell is the memory-capacity optimum because its relaxation time best matches a seconds-scale target. The multi-lag physiological task gives the opposite case (table 5.2): when the target spans bands that no single cell covers, the bank whose

timescales are *spread* to match them is the best single choice.

Memory first, heterogeneity as a generality hedge. The results give a consistent account of what these resources are worth, drawn together in table 5.7. The first-order effect is *fading memory itself*: every memory bank, homogeneous or heterogeneous, far exceeds the memoryless and instantaneous baselines, and the gap widens to as much as +0.08 macro-F1 under the sensor noise of a worn device (sections 5.8 and 5.9). The second-order effect is the *spread* of timescales. Where the task demands memory at many lags at once — random-input memory capacity (a $1.54\times$ broadening, $p = 0.002$; fig. 5.1) and real-physiology context reconstruction (+0.012 in held-out R^2 over the best homogeneous bank, 15/15 subjects, $p < 10^{-4}$; table 5.2) — heterogeneity is a measurable, statistically resolved resource. Where a single slow timescale already dominates, as in the sustained stress labels and the single-channel monitoring task, the best-matched single composition is as good or marginally better, exactly as Demonstration A anticipated. The two readings combine into a deployment rule: because a heterogeneous bank gives up at most about a hundredth of a point on the single-timescale tasks but gains substantially on the multi-timescale ones, it is the rational default for a substrate fabricated once and reused across an unknown mix of affective tasks. For the materials argument of this thesis, that heterogeneity is *low-cost and intrinsic*: it arises naturally in a solution-processed device family through composition spread, device-to-device scatter, and — as further, illustrative controls catalogued in chapter 4 but not required by the quantitative banks here — electrolyte chemistry and drive amplitude.

Operating point and integration. The same parameter cards that drive the simulations also fix the deployment envelope. The devices operate in the sub-kilohertz regime in which affective physiology lives,

Table 5.7: Fading memory versus its spread across the affective workload of this chapter. Every memory bank (homogeneous, heterogeneous) far exceeds the memoryless instantaneous baseline — markedly so under the sensor noise of a worn device — so fading memory is the first-order resource. The heterogeneous bank then wins where several timescales matter at once (physiological context), while the best-matched single composition wins where one timescale dominates (sustained labels, single-channel monitoring); the gap between them never exceeds about 0.01 macro-F1. Bold marks the best bank per row. Scores from tables 5.2 and 5.4 and sections 5.8 and 5.9.

Affective task	Metric	Instantaneous	Homogeneous	Heterogeneous
Physiological context (multi- τ)	R^2	0.673	0.744	0.756
Three-class affect tracking	macro-F1	0.709	0.753	0.758
Stress monitoring, clean	macro-F1	0.914	0.922	0.920
Stress monitoring, $\sigma=0.4$	macro-F1	0.840	0.917	0.916
Wrist stress, clean	macro-F1	0.881	0.930	0.922
Wrist stress, $\sigma=0.4$	macro-F1	0.847	0.918	0.916

so the slow response that disqualifies them for fast logic is exactly what suits them to on-skin sensing. The read-out is assumed sub-threshold and state-preserving, which sets a read-voltage and read-disturb budget that any circuit implementation must respect; and the device-to-device scatter that a memory application would treat as a yield problem is, in the reservoir setting, part of the computational substrate. Figure 5.8 makes this last point quantitative and is perhaps the clearest single statement of the reservoir’s tolerance to the imperfection of solution processing: total memory capacity is reported against the magnitude of injected device-to-device scatter, and the heterogeneous bank holds its advantage across the whole realistic range — including at the *measured* scatter, $\sigma(\ln \tau) \approx 0.85$, estimated from the silver PEO/LiOTf devices within each fixed composition cell and confirmed to be unexplained by

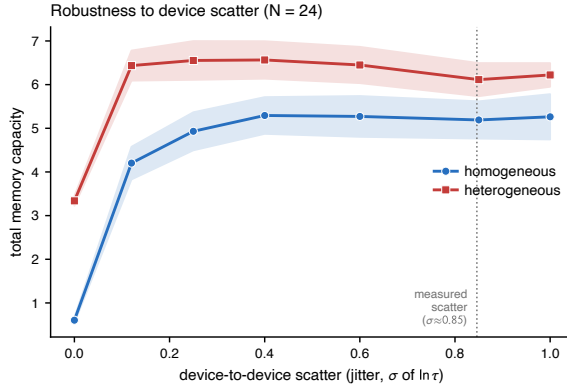


Figure 5.8: Total memory capacity versus the magnitude of injected device-to-device scatter (jitter on τ , β and α), for the homogeneous and heterogeneous banks ($N = 24$, ten seeds; shaded bands ± 1 SD). The heterogeneous bank keeps its advantage across the entire realistic spread, including at the measured device-to-device scatter (dotted line: $\sigma(\ln \tau) \approx 0.85$, derived in section D.3). A bank of perfectly identical nodes (zero scatter) is nearly memory-less because matched nodes are redundant: the scatter that yield engineering fights to remove is what the reservoir computes on.

spin speed, anneal, film thickness, or device age (section D.3). The limiting case is instructive — a bank of *identical* nodes (zero scatter) is nearly memory-less, because perfectly matched nodes are redundant, so the scatter that yield engineering fights to remove is precisely what a homogeneous reservoir lives on, while the heterogeneous bank additionally carries intrinsic timescale spread and therefore degrades most gracefully. These are the design rules that chapter 4 makes available and that an eventual hardware reservoir would inherit.

5.11 Limitations and Scope Conditions

The claims of this chapter are bounded by scope conditions that are stated here in full, in keeping with the claim discipline of chapter 4.

In silico. The demonstrations are simulations. Each node is the measured $\phi \otimes \lambda \otimes f$ behavioural model of section 5.2, not a bench reservoir, and every figure caption names the data behind it; the step from a fitted per-device model to a wired hardware reservoir is left to future work.

The composition assumption. As detailed in section 5.2, the write nonlinearity and the fading-memory law were measured separately, at different write amplitudes and a single inter-pulse cadence. Composing them assumes the two regimes are compatible and probably under-estimates the coupling between drive and retention. Because that assumption underlies every result, its effect is *bounded* rather than only declared: re-running the memory-capacity comparison with an explicit drive–retention coupling deliberately injected (each node’s retention rescaled with its own write strength, swept over a wide range) leaves the central conclusion intact — the heterogeneous bank stays above the homogeneous one across the whole sweep, with its capacity ratio moving only within 1.14–1.53 (section D.5). A single matched-amplitude experiment that sweeps pulse count and inter-pulse interval together would remove the assumption outright; it is the most important bench experiment this chapter calls for.

Operating regime and read-out. The model is identified, and valid, only in the slow, sub-kilohertz regime of the measurements; affective signals fall natively in this band, but fast temporal tasks are out of scope. Every read-out is a single ridge regression — a deliberate constraint, since a nonlinear read-out could solve the tasks by itself and would invalidate the reservoir claim — and the read is assumed sub-threshold and state-preserving.

Heterogeneity axes. The quantitative heterogeneous banks span the replicated PEO/LiOTf composition grid. The chemistry and drive-

amplitude axes of chapter 4, being sample-limited ($n \leq 2$ per matched cell), enter only as illustrative additional controls; Demonstration B is therefore a principle demonstration of composition heterogeneity, not a powered comparison across chemistries. The composition cells that define these banks index *composition*, not film thickness: although spin-coating speed and residual film thickness covary with the PEO fraction, the parameter cards inherited here were shown to be governed by composition rather than thickness (fig. 4.1, section 4.2), so the heterogeneity exploited in this chapter is compositional in origin.

Corpora, and a reported null. The affective classification results rest principally on one corpus (WESAD); the deployment-critical noise-robustness finding is replicated on a second, independent corpus, wrist sensor, and population (the PhysioNet Non-EEG dataset, section 5.8), but a broader multi-corpus study, and the affective dimensions beyond stress such as continuous valence and arousal, remain future work. The central negative result — that timescale heterogeneity does not separably improve final-label classification — is reported with error bars across seeds and sampling intervals rather than omitted, because it is informative: it localises exactly where heterogeneity helps and where it does not.

5.12 Summary and Bridge to the Conclusions

This chapter has shown that compact behavioural models of solution-processed polymer-electrolyte memristors, characterised through the three common measurements of chapter 4 and read out by a single linear regression, can support temporal-computing demonstrations within the stated in-silico scope. The models produce a resolved memory-capacity

gain for heterogeneous banks, give a relative NARMA-10 composition ranking, encode the multi-lag temporal context of real physiological signals, and classify affective state from the WESAD corpus at a leave-one-subject-out macro-F1 of 0.894 on a binary stress task and 0.758 on a three-class streaming task. Beyond accuracy on curated splits, the same models sustain a continuous stress detector under the sensor noise and on the consumer wrist band that a worn device must tolerate — where fading memory holds detection near 0.92 macro-F1 while a static, instantaneous baseline collapses to 0.84 (+0.088, all fifteen subjects, $p < 10^{-4}$). That robustness is the generic virtue of a fading-memory low-pass — a classical moving average matches it — and the device’s contribution is to realise that filter passively in the material, at an estimated always-on budget of order a microwatt and with only the tens-to-a-hundred trained read-out weights of one ridge solve, against the 10^4 – 10^6 gradient-trained weights of a deep model of comparable accuracy.

The results resolve into a single design rule that unifies the two experimental chapters. Chapter 4 establishes that composition and drive set the device timescale; this chapter establishes that the application selects the timescale, that fading memory is the active computational ingredient, and that a spread of timescales is a measurable resource for memory breadth — a $1.54\times$ broadening of memory capacity and a consistent gain in physiological-context encoding, both statistically resolved — even though it is not required when a single slow timescale already dominates a task. The heterogeneity that chapter 4 catalogued as a set of tunable properties is, in this light, what the computation is built from.

The concluding chapter draws together the contributions of the thesis and its limitations. The most consequential next steps follow directly from the scope conditions above: the matched-amplitude, varied-cadence pulse-train experiment that would convert the composition assumption into measured fact; a powered study of the chemistry and drive-amplitude

heterogeneity axes; and a hardware reservoir that replaces the in-silico bank with wired devices, closing the gap between a fitted device model and a physical temporal computer.

Chapter 6

Conclusions and Outlook

This thesis set out to answer whether the volatility of an organic polymer-electrolyte memristive device — the very property that disqualifies it as a non-volatile memory — can be turned into a computational resource. The preceding chapters answered that question along two interlocking axes: a *device-chemistry* axis that built, salvaged, and then deliberately engineered the volatile dynamics of these composites (chapters 2 to 4), and an *application* axis that put those dynamics to work as temporal-computing primitives (chapter 5). This conclusion draws the two together, returns to the thesis questions, places the work against temporal, reservoir, and event-driven neuromorphic hardware rather than digital memory, and defines the limitations and experiments that follow.

6.1 Summary of Contributions

The central thesis is a single design rule, assembled cumulatively across the chapters: *the chemistry of the composite — above all its composition — sets the fading-memory timescale of the device, and a heterogeneous bank of such timescales supplies the substrate for temporal computing.* The

contributions that establish it fall into four layers, each held to a different and explicitly stated standard of evidence.

A fully characterised synaptic exemplar (chapter 2). A reversible two-terminal organic memristive device — a solution-processed *Super Yellow/Hybrane*/LiOTf composite on an ITO/composite/Ag stack — was shown to support the complete synaptic measurement suite in a single platform: analogue I–V hysteresis (an 85 % conductance increase from the first to the second sweep at 1 V, growing ~ 15 -fold over ten cycles), reversible potentiation and depression exceeding 200 % at an energy cost of order 50 nJ per event, two distinguishable retention regimes (an short-term memory regime, $\tau_S \approx 2.5\text{ s} - 3\text{ s}$, and an long-term memory regime, $\tau_L \approx 4.7\text{ s}$, both reported from standard-exponential fits in the published SI), a four-state excitatory post-synaptic current read-out, and an asymmetric Hebbian STDP window with a biologically realistic time constant of 85 ms–90 ms. The two memory regimes were given a mechanistic account in terms of the mobility asymmetry between the weakly coordinated triflate anion and the strongly coordinated lithium cation within the oxygen-rich host, framed by hard–soft acid–base reasoning. This is the one device in the thesis for which the full suite is treated as primary evidence, and it remains validated by independent peer review [6].

A reproducibility diagnosis and a provenance methodology (chapter 3). The scaling of that exemplar into a fabrication programme exposed a reproducibility crisis: successive batches of nominally identical *Hybrane* devices drifted from low-current, high-area, multi-level behaviour toward a high-conductivity response with no usable switching window. Under four mandatory controls — per-device weighting, a standard-protocol corpus, sweep-amplitude matching, and recovery-tail sensitivity — the collapse was established as a quantitative, multi-feature signal anchored to

a documented behavioural inflection (on–off ratio falling from 2.44 to 1.21, normalized hysteresis area from 0.235 to 0.027, and potentiation flipping from +17.9 % to −2.1 % at 3 V), with an accompanying drift toward ohmic conduction at a matched 1.2 V. The failure was localised to the fabrication line over calendar time, not to finished devices — *Hybrane devices* hold their characteristics over weeks — and attributed to aging of the *Hybrane* reagent stock (with *Super Yellow* eliminated), under a moisture-driven hydrolytic-aging hypothesis supported by an annealing-recovery test. The crisis was resolved by switching to a poly(ethylene oxide) host, which both opened a wider window (normalized area $0.26 \rightarrow 0.42$ at matched 3 V) and sustained a usable switching window across roughly two and a half years with no stock-driven collapse. The lasting contribution was methodological: a fabrication-provenance standard, a normalized characterization procedure, an automated feature pipeline, and trend-discovery tooling — the same apparatus that produced every quantitative result in the chapters that followed.

Compositional and chemical control of the dynamics (chapter 4).

On the PEO platform, the volatile dynamics were shown to be controllable through a hierarchy of variables, established through the three common dynamical measurements (I–V hysteresis, variable-number-of-pulses potentiation, variable-delay depotentiation). The strongest and only *replicated* variable is composition: across the *Super Yellow*/PEO/LiOTf/Ag grid the fading-memory time is tunable over roughly 2 s–20 s, with the switching window and potentiation strength moving in concert (more ion-transport polymer yields a weaker, faster-relaxing device). Electrolyte chemistry — host (PEO vs TMPE) and anion (triflate vs TFSI) — extends the range further but is reported only illustratively; the cation, which the chapter 2 mechanism had nominated as the controlling variable, proved the weakest and least legible, and the $\text{Li}^+ > \text{Na}^+ > \text{K}^+$ ordering is reported as a negative result (“not demonstrated in this archive”), confounded with host, anion,

and drive protocol at the available power. Two further results frame how the dynamics must be read: a *methodological* one — the drive amplitude co-sets the apparent timescale, so cross-device comparison demands a matched protocol — and a *microscopic* one — equilibrium impedance falls along the composition axis and tracks the fading-memory time, with ATR, XRD, and UV-Vis localising the host effect to short-range ether order in an X-ray-amorphous, salt-dissolved film while leaving the *Super Yellow* optical gap unshifted. The device-to-device spread is reframed throughout as a *resource*: the per-device design space couples fading memory and input nonlinearity through the PEO fraction, so a substrate must be assembled from a *bank* of compositions rather than tuned to a single optimum.

A temporal-computing demonstration from measured dynamics (chapter 5). Finally, compact behavioural models ($\phi \otimes \lambda \otimes f$) fitted to the chapter 4 measurements, read out by a single linear regression, provide a model-based temporal-computing demonstration under constrained conditions. On the task-agnostic benchmark tier the device banks show reservoir-relevant behaviour: a heterogeneous composition bank raises the total linear memory capacity from 4.2 to 6.4 (a factor of 1.54 ± 0.10 ; Wilcoxon $p = 0.002$ over ten seeds) and the total information-processing capacity from 7.1 to 11.6, with a nonzero nonlinear capacity in even the homogeneous bank indicating that the compressive write supplies a usable nonlinear response. On real physiological streams the heterogeneous bank improves multi-lag temporal-context reconstruction, and on the WESAD affective corpus the reservoir classifies stress at a leave-one-subject-out macro-F1 of 0.894 (binary) and 0.758 (three-class streaming). The chapter is equally disciplined about where heterogeneity does *not* help: timescale diversity does not separably improve final-label classification when a single slow timescale already dominates the task, and that null is reported with error bars rather than omitted. The unifying design rule emerges here:

composition and drive set the device timescale, the application selects it, fading memory is the active ingredient, and a spread of timescales is a measurable resource for memory breadth.

6.2 Answers to the Thesis Questions

The questions posed at the outset can now be answered, with the qualifications the evidence demands.

First — realising and understanding an exemplar. *Can a fully reversible two-terminal organic memristive device be made from a solution-processed polymer-electrolyte composite, supporting the full set of synaptic functions, and can its two memory regimes be explained by cation–anion mobility asymmetry?* Yes, on both counts, for the chapter 2 exemplar: the device is fabricated by solution processing and thermal evaporation alone and demonstrates analogue switching, distinguishable short-term memory/long-term memory, an EPSC response, and STDP; the two-timescale retention is consistently accounted for by the differential displacement and relaxation of the mobile triflate anion and lithium cation. The mechanistic account is a coherent and well-supported *hypothesis* on the strength of a single fully characterised device, not a law established across a population.

Second — composition as the quantitative control, with chemistry as a sample-limited survey. *How do the dynamics depend on composition, and does moving the cation along the Li^+ , Na^+ , K^+ series shift the timescale as coordination chemistry suggests?* The composition dependence is real, replicated, and monotone: the PEO/LiOTf ratio tunes the fading-memory time over roughly an order of magnitude, with window and potentiation strength tracking it. The cation question is answered

in the negative-as-tested: no robust $\text{Li}^+ > \text{Na}^+ > \text{K}^+$ ordering survives the available power and the protocol/electrode confounds, so coordination chemistry is retained as a qualitative organising idea but explicitly not advanced as a transferable quantitative law. Reporting this as a null — rather than presuming the ordering the chapter 2 mechanism would predict — is itself one of the more important outcomes of the comparative chapter.

Third — utility for temporal computing. *Can the volatile response be exploited for temporal computation; can composition supply task-matched timescales while chemistry and device spread populate a heterogeneous bank; and can variability contribute to computation rather than merely degrade it?* Yes, within the in-silico scope of chapter 5: composition supplies a family of fading-memory times that natively overlap the bands of peripheral affective signals, the heterogeneous bank measurably broadens memory and information-processing capacity, and device-to-device variability is shown to be a contributor to memory breadth rather than only a defect. The boundary on this answer is that heterogeneity is a resource *where the task needs timescale breadth*; for a single-timescale task the lead composition alone is optimal, and on the downstream WESAD label task the heterogeneity advantage is not separable. The affirmative answer is therefore specific and bounded, not universal.

6.3 Benchmarking Against the Right Reference Class

A device that forgets in seconds is a poor non-volatile memory and should not be measured against one. The framing decision that runs through the thesis — volatile, heterogeneous temporal elements rather than defective

memory cells — carries a corresponding choice of benchmark: not the retention, endurance, and bit-density figures of DRAM, SRAM, or crossbar ReRAM, but the metrics and substrates of temporal and event-driven neuromorphic computing [5, 26, 27].

Against physical reservoirs. Physical reservoir computing has been demonstrated in dynamic memristors [32, 34], organic electrochemical devices [35], and delay-coupled single nodes [150], and is unified by the requirement of fading memory plus nonlinearity on a single substrate [5, 18]. The devices of this thesis meet that requirement on their own measured terms: they exhibit measured fading memory and IPC-resolved nonlinear capacity, and they do so in the slow, sub-kilohertz band where many physiological and environmental signals natively live — a regime in which fast inorganic reservoirs must artificially slow their input but in which these composites operate without time-rescaling. The distinctive contribution relative to that literature is not a record benchmark score but the *chemical programmability of the timescale*: where most physical reservoirs offer a fixed intrinsic dynamic, here the fading-memory time is set by composition over 2 s–20 s, so a heterogeneous bank can be *designed* rather than merely tolerated. The measured $1.54\times$ memory-capacity broadening quantifies what that programmability provides.

Against organic neuromorphic devices. Within organic electronics the dominant line of work has pursued the opposite goal — linear, symmetric, *non-volatile* conductance updates for in-memory weight storage, as in the electrochemical ENODE family [21–23]. Measured as a weight memory, the present devices are inferior on every axis those works optimise. Measured as temporal elements they occupy a complementary niche: the controlled forgetting that the ENODE programme suppresses is exactly the resource exploited here. The thesis thus contributes a

worked example of the alternative organic-neuromorphic design target — ion-relaxation timescales as compute, not retention to be maximised — alongside the ionic/electrolyte-gating line of organic neuromorphics [35, 113].

Placement in context. Two caveats keep this benchmarking grounded. First, the application-level scores are *in silico*: they establish that the *measured dynamics* are computationally sufficient, not that a wired array reaches them. Second, the reservoir architecture used is deliberately the hard case — a bank of independent, non-recurrent nodes — so the absolute NARMA-10 errors are high and the capacities sit well below the node-count bound; a recurrent or delay-feedback topology would raise them. The claim is therefore one of *demonstrated sufficiency of the device physics under conservative assumptions*, which is the appropriate claim for a thesis whose evidence base is a measured archive rather than a fabricated reservoir chip.

6.4 Limitations of the Evidence Base

The contributions above are bounded by limitations that the body chapters state in place and that are gathered here so that the reader leaves with an accurate map of what is and is not established.

1. **The rich synaptic suite is single-device.** EPSC, separated short-term memory/long-term memory retention, STDP, and impedance-as-primary-evidence exist only for the chapter 2 *Hybrane* device. They function as lithium-device priors and sanity checks in the later chapters and were not lifted to comparative status across the PEO family.

2. **Only composition is replicated.** The quantitative core is the *Super Yellow*/PEO/LiOTf/Ag composition grid. The host, anion, and cation comparisons rest on one or two screened devices per matched cell and are illustrative; the electrode contrast (active silver vs inert gold) is likewise illustrative and partly confounded with fabrication generation. No powered material law — in particular no $\text{Li}^+ > \text{Na}^+ > \text{K}^+$ ordering — is claimed.
3. **Protocol-specific timescales.** Because the drive amplitude co-sets the apparent fading-memory time, absolute retention values are protocol-specific, and the supra-threshold read used in the comparative corpus differs from the sub-threshold read of the proof-of-concept device.
4. **The composition assumption behind the models.** Pulse integration and forgetting were characterised *separately*, at different write amplitudes (potentiation at 3 V, decay at 4 V) and a single inter-pulse cadence. Composing them in the behavioural models assumes the two regimes are compatible; this is an explicit modelling assumption, not a measured fact, and it is the most consequential gap for chapter 5.
5. **The application results are in silico and only partly cross-corpus.** Every temporal-computing claim is a simulation of measured device models read out by linear regression; no wired reservoir was built. The physiological-context and affective-label results rest principally on WESAD, and the central heterogeneity-versus-label result is a reported null. The deployment-critical noise-robustness result is replicated on the independent PhysioNet Non-EEG corpus, but broader multi-corpus validation remains future work.
6. **Material stability remains open.** Devices were characterised unencapsulated on a ~ 300 h ambient shelf scale; cycling endurance,

read-disturb under continuous pulsing, and long-run drift under encapsulation were not systematically studied, and device ageing was not modelled in the comparative corpus.

6.5 Outlook and Future Directions

The limitations map directly onto a forward programme, ordered here from the most immediate experimental closures to the longer-horizon device and application goals.

Close the composition assumption. The single most valuable bench experiment is a matched-amplitude pulse train that sweeps pulse count *and* inter-pulse interval together, on one composition, electrode, and read protocol. It would convert the integration-and-forgetting composition assumption of chapters 4 and 5 from a modelling premise into measured fact, and is the precondition for trusting the behavioural models beyond their present scope.

Power the chemistry landscape. A dedicated cation series in a single host and anion, at a fixed protocol and electrode, with a sub-threshold read and at least three replicates per cell, would convert the illustrative host/anion/cation survey into a powered comparison and settle the $\text{Li}^+ > \text{Na}^+ > \text{K}^+$ question that this thesis could only report as a null. A chemistry- and bias-resolved impedance study, and a generation-controlled silver-versus-gold electrode split, would do the same for the microscopic and electrode comparisons.

Test the degradation chemistry. The moisture-driven hydrolytic-ageing hypothesis for the *Hybrane* stock (chapter 3) remains open, and

may never fully close because it concerns a consumable that no longer exists in its aged state. FTIR, GPC, Karl-Fischer/TGA, and EIS on retained aged-versus-fresh stock are the experiments that would confirm or refute it, and the lesson — pin and archive reagent provenance — is already embedded in the methodology.

Engineer the host and salt for finer timescale control. Chemical tailoring of the polyether host and the salt is the natural route to extending the accessible fading-memory window — in particular toward the minutes-scale tonic band that composition alone does not reach — and to decoupling the memory time from the input nonlinearity that composition currently couples.

Stabilise for long-run inference. Encapsulation strategies and a systematic ageing, endurance, and read-disturb study are required before air-stable, continuously operating temporal inference is credible, and would also remove the uncontrolled ageing variable from the comparative corpus.

Build the hardware reservoir. The most consequential step beyond this thesis is to replace the in-silico bank with wired, composition-diverse PEO/LiOTf devices and an on-chip linear read-out, integrated toward a 1T1M-style array. A task-matched physical reservoir on an affective stream would close the gap between a fitted device model and a physical temporal computer, and would test the heterogeneity-as-resource claim in hardware rather than in simulation.

Toward nervetronic interfaces. Finally, the combination of solution processability, mechanical flexibility, biocompatible chemistry, and native operation in the seconds-to-tens-of-seconds band points to wearable, skin-

conformal sensing-and-inference at the edge, where the device’s controlled forgetting is matched to the timescales of the physiological signals it would process.

6.6 Closing Remarks

The thesis began from a reframing: that a memristive device which forgets at a controlled rate should not be judged as a failed memory but used as a temporal element. Following that reframing through has required building a device and understanding its mechanism (chapter 2), confronting and surviving a reproducibility crisis that forced a measurement methodology into existence (chapter 3), establishing composition as the quantitative control on the volatile dynamics while reporting the chemistry as the sample-limited survey it is (chapter 4), and demonstrating that the measured dynamics are computationally sufficient for real temporal tasks (chapter 5). What ties these together is a single, evidence-graded claim: the chemistry of a polymer-electrolyte composite sets its fading-memory timescale, and a heterogeneous bank of such timescales is a resource for computing on time. The device is no longer vulnerable to the standard objection that it makes a poor non-volatile memory, because being a good non-volatile memory was never the point. The remaining work — a matched-amplitude protocol, a powered chemistry series, and above all a hardware reservoir — is the work of turning a measured sufficiency into a built temporal computer, and it is left, with its methodology in hand, to what comes next.

Appendix A

Supporting Information for Chapter 2

This appendix gathers the detailed experimental information and supporting evidence for the proof-of-concept device of chapter 2. The main chapter retains the evidence needed to assess the device-level claim; the material collected here records the supporting methods, protocol details, and auxiliary measurements.

A.1 Materials and Fabrication Detail

A.1.1 Component Notes

The semiconducting component is *Super Yellow*, the trivial name for the high-molecular-weight PPV copolymer sold as PDY-132 by Merck. The formal copolymer name is:

poly[{2,5-di(3',7'-dimethyloctyloxy)-1,4-phenylenevinylene}-co-
{3-(4'-(3'',7''-dimethyloctyloxy)phenyl)-1,4-phenylenevinylene}-

co-

{3-(3'-(3',7'-dimethyloctyloxy)phenyl)-1,4-phenylenevinylene}}].

The conjugated phenylene-vinylene backbone supplies electronic transport, while the branched alkoxy side chains support solubility and reduce crystallisation-driven phase separation in blends [6, 143]. In ion-containing *Super Yellow/Hybrane* systems, related LEC devices have shown electroluminescence lifetimes exceeding 1600 h, supporting the chemical compatibility of this formulation family [104].

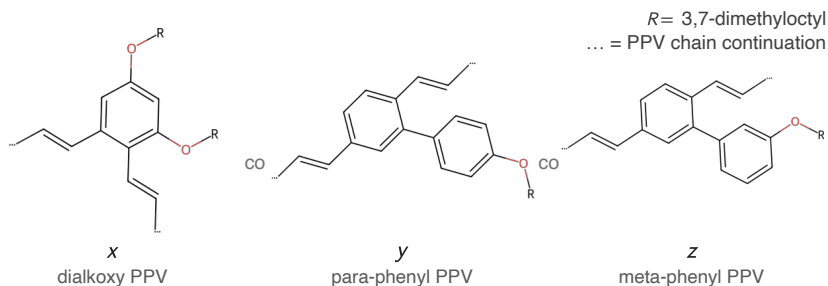
Hybrane is a hyperbranched polyester-amide produced by Polymer Factory (Sweden) [127]. The hyperbranched architecture creates an irregular, amorphous macromolecule rich in ether and ester oxygens. These oxygen atoms provide hard Lewis-base coordination sites for alkali-metal cations. Because ion motion is coupled to polymer segmental motion, transport in this class of host follows the polymer-electrolyte picture described in section 1.6.1, including non-Arrhenius temperature dependence near the glass-transition regime [88, 161].

LiOTf is a polymer-electrolyte salt widely used in batteries and LECs [98, 103]. Its processing role is to dissolve with *Super Yellow* and *Hybrane* in cyclohexanone without visible phase separation at the selected ratio. Its mechanistic role is to provide a small, hard Li^+ cation (0.59 Å–0.76 Å for four- to six-fold oxygen coordination [142]) and a charge-delocalised triflate anion. The resulting cation/anion asymmetry is the chemical basis for the STM/LTM interpretation in section 2.5.1.

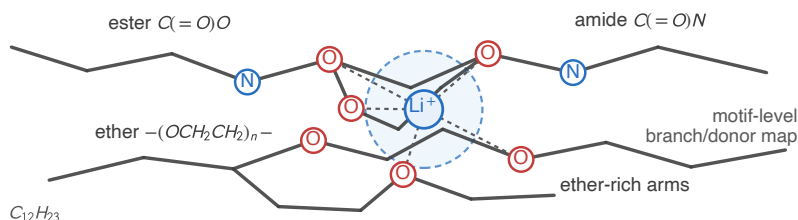
A.1.2 Solution Preparation and Film Deposition

The three stock solutions were prepared separately in cyclohexanone at 15 mg mL⁻¹ (*Super Yellow*), 10 mg mL⁻¹ (*Hybrane*), and 5 mg mL⁻¹ (LiOTf). The *Super Yellow* solution was stirred at 50 °C overnight.

a Super Yellow (PDY-132): statistical PPV copolymer



b Hybrane DEO750 8500: hyperbranched polyester-amide ion host



c Lithium triflate (LiOTf): mobile-ion source



Figure A.1: Chemical components of the proof-of-concept composite. **(a)** *Super Yellow* is the commercial PDY-132 PPV-family statistical copolymer; R denotes the branched 3,7-dimethyloctyl side chain. **(b)** *Hybrane* DEO750 8500 is represented as an oxygen-rich hyperbranched polyester-amide motif, highlighting ether, ester, and amide donor sites rather than a monodisperse molecular formula. **(c)** LiOTf is shown as the lithium salt of triflate, $LiCF_3SO_3$. The selected active-layer formulation is *Super Yellow:Hybrane:LiOTf* = 1:0.30:0.09 by mass.

The solutions were then mixed to give the final mass ratio *Super Yellow:Hybrane:LiOTf* = 1:0.30:0.09, corresponding to final concentrations of 8.72 mg mL^{-1} , 2.62 mg mL^{-1} , and 0.78 mg mL^{-1} , respectively. Solution handling was performed under dry nitrogen to limit moisture uptake.

ITO-patterned glass substrates were ultrasonicated successively in aqueous Mucosol detergent solution, deionised water, and 2-propanol, each for 15 min. After drying under nitrogen, substrates were transferred immediately into a nitrogen glovebox (oxygen and water below 1 ppm). The active layer was spin-coated at 2000 rpm for 60 s, then annealed at $75\text{ }^{\circ}\text{C}$ for 3 h. The anneal removes residual cyclohexanone, allows chain relaxation, and stabilises the mixed polymer-electrolyte morphology.

The top Ag electrode was thermally evaporated in the glovebox through a shadow mask defining the 0.0825 cm^2 junction. Ag was deposited at 0.1 nm s^{-1} – 0.5 nm s^{-1} to a total thickness of 100 nm, monitored by quartz crystal microbalance. The slow initial rate was used to limit metal penetration into the soft polymer layer, which would otherwise risk filamentary shorts [162].

A.2 Electrical Protocol Detail

Electrical measurements used a Keithley 2450 SourceMeter connected to a custom probe station and controlled through Keithley TestScript Builder. Unless noted otherwise, measurements were carried out under nitrogen, with voltage applied to the Ag top electrode relative to the grounded ITO bottom electrode.

A.2.1 Read–Write–Wait Measurement

The variable-pulse and variable-delay measurements use a common read–write–wait sequence (fig. A.2). First, a low reading pulse, $V_{\text{read}} = 0.5 \text{ V}$, establishes the initial conductance G_0 . This voltage is below the independently determined ionic-redistribution threshold of approximately 0.7 V [6]. Second, a train of N_{pulses} write pulses of duration 0.1 s perturbs the ionic configuration. Third, after a waiting time t_{wait} under open-circuit conditions, a second read pulse measures G_f . The reported response is $\Delta G = (G_f/G_0) \times 100\%$. Between experiments, the device was reset by shorting the junction for 300 s , and G_0 was checked before the next measurement.

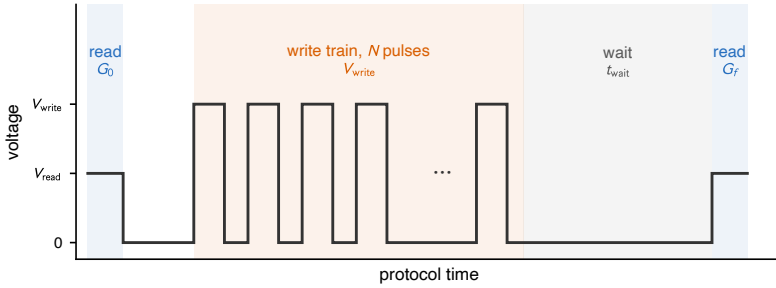


Figure A.2: Read–write–wait protocol used for the pulse-number and retention measurements. A low read pulse establishes G_0 , a train of write pulses perturbs the ionic configuration, a controlled open-circuit waiting time allows relaxation, and a second read pulse measures G_f . Generated from the protocol reported in Ref. [6].

A.2.2 Extended Synaptic Protocols

The EPSC experiment defines states operationally by their signed current response under a common 1 V read-out bias. The device is first held at 1 V for 1 s to define S_0 . Two 2 V , 50 ms pulses generate excited states S_1 and S_2 , and two -2.5 V , 50 ms pulses generate inhibitory states S_3 and

S_4 . The signed ratios $R_n = I_{S_n}/I_{S_0}$ are reported in section 2.4.4.

For STDP, pre- and postsynaptic spikes are emulated by triangular voltage waveforms applied to the two-terminal device. The experimentally applied voltage is the algebraic sum

$$V(t) = V_{\text{pre}}(t) + V_{\text{post}}(t + \Delta t), \quad (\text{A.1})$$

with $\Delta t = t_{\text{post}} - t_{\text{pre}}$. Each spike waveform has maximum amplitude $V_{sp} = 1.55$ V and duration 1.2 s; Δt was varied from -600 ms to 600 ms. Negative Δt produces potentiation and positive Δt produces depression under the sign convention used in section 2.4.5. The applied-voltage logic of both protocols is shown in fig. A.3.

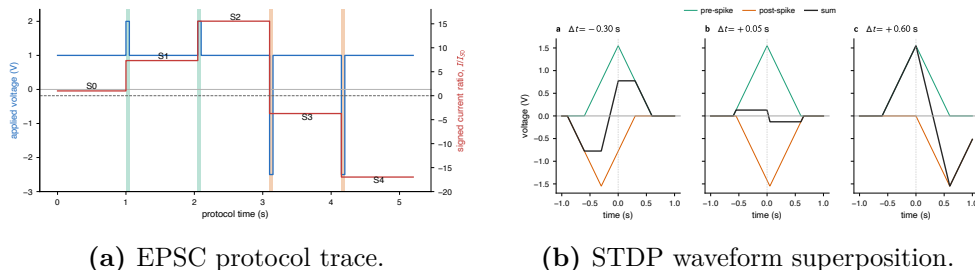


Figure A.3: Supporting protocol panels for the extended synaptic measurements. **(a)** Reconstructed EPSC trace showing the applied-voltage sequence and the signed current ratios reported for states S_0 – S_4 . **(b)** Example pre/post triangular spikes and their algebraic sum for three timing offsets. These panels document the protocol logic; the compact evidence used in the main chapter is reported in fig. 2.5.

A.2.3 Retention-Fit Parameters

Table A.1 lists the standard-exponential fit parameters reported for the proof-of-concept retention traces. The fits use $G_f/G_0 = R_\infty + A \exp(-t_{\text{wait}}/\tau)$, with $\beta = 1$ fixed.

Table A.1: Published standard-exponential fit parameters for the Chapter 2 retention traces. No free stretch exponent, confidence interval, or goodness-of-fit statistic was reported for these proof-of-concept fits.

Condition	τ (s)	R_∞	A
1 V, 10 pulses (STM)	2.57	1.01	0.374
1 V, 50 pulses (STM)	3.01	1.01	0.91
3 V, 10 pulses (LTM)	4.73	1.05	3.30

A.3 Morphology and Stability Support

The spin speed was selected from a 500 rpm–3000 rpm optimisation [6]. Low speeds produced films that were too thick and less uniform; high speeds made the film thin enough for substrate roughness to matter. The main device used 2000 rpm, giving a 209 nm active layer measured by contact profilometry across a masked step edge with an Ambios XP-1 profilometer.

Surface topography was characterised by tapping-mode AFM using a Bruker Dimension Icon instrument, a silicon probe with resonant frequency 300 kHz and force constant 40 N m^{−1}, and a scan rate of 0.6 Hz. The AFM images showed smooth films with RMS roughness of 1 nm–3 nm over the investigated spin-speed range [6]. The absence of large phase-separated features supports the reading of the active layer as a homogeneous polymer-electrolyte composite at the scale relevant to the vertical junction.

Ambient shelf stability was assessed for approximately two weeks (~ 300 h) on unencapsulated devices stored in air at room temperature [6]. The monitored quantities were baseline conductance G_0 , maximum conductance change $\Delta G_{\max}$, and STM retention time τ_S . All three remained within 20 % of their initial values over the measurement window. This is a shelf/calendar stability result, not a cycling-endurance measurement. Figure A.4 summarises the film thickness, AFM roughness, and shelf-stability

metrics together.

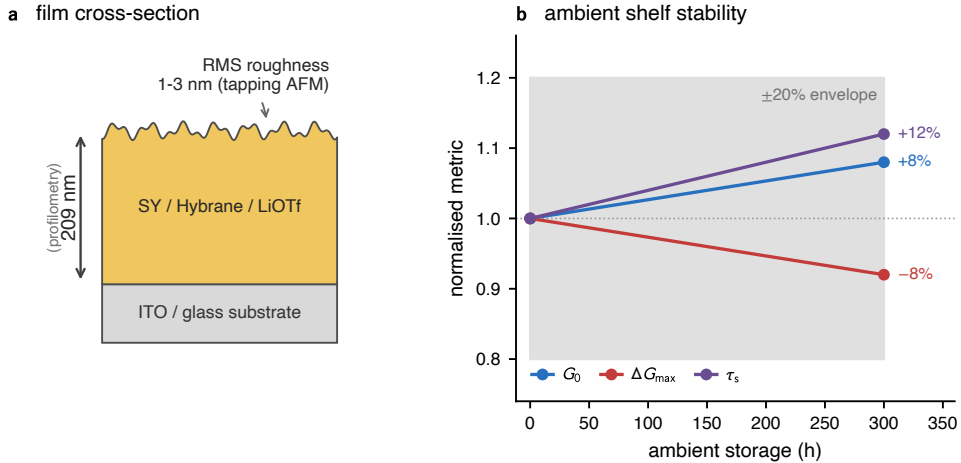


Figure A.4: Supporting morphology and shelf-stability summary for the Chapter 2 device family. The processing condition used in the main device gives a 209 nm active layer; tapping-mode AFM reports 1 nm–3 nm RMS roughness across the investigated spin-speed window; and the ambient shelf-stability metrics remain within a 20 % envelope over approximately 300 h. Compiled from the supporting measurements reported in Ref. [6]; this is not a cycling-endurance result.

Encapsulation outlook. The ambient stability is expected to improve further with encapsulation, standard practice in organic electronics: a thin SiO_2 or Si_3N_4 barrier deposited by atomic-layer or chemical-vapour deposition would suppress the oxygen and water ingress that are the two primary degradation pathways [163]. That even unencapsulated devices survive two weeks establishes the baseline robustness of the composite.

A.4 Ion-Transport Models for the Composite

This section details the injection and bulk-transport descriptions summarised in section 2.5. The two limiting-case models are inherited from the polymer light-emitting electrochemical cell (LEC) literature and reconciled by van Reenen *et al.* [102] as two regimes of the same ion redistribution.

A.4.1 Equilibrium Ion Distribution and Response to an Applied Field

In the absence of an applied voltage, the ions (Li^+ cations and CF_3SO_3^- anions) are distributed homogeneously throughout the active layer. The equilibrium distribution is not strictly uniform at the atomic scale — each Li^+ ion occupies a local coordination site within the Hybrane matrix, and the triflate anions occupy interstitial positions between the coordinated cations — but on the scale of the film thickness (209 nm), the macroscopic charge density is uniform and the net electric field within the film is zero (apart from any built-in field arising from the electrode work function difference), as in the polymer-electrolyte picture inherited from LECs [98, 99, 102].

When a voltage is applied across the device, an electric field $\mathcal{E} = V/d$ (where d is the film thickness) is established. In the absence of free charge carriers that could screen this field, it would be uniform throughout the film. However, the mobile ions respond to the field by migrating: cations drift toward the cathode (ITO at positive bias on Ag), and anions drift toward the anode (Ag). This migration leads to the build-up of a charged double layer at each electrode interface on a timescale determined by the ionic mobility and the film thickness. The resulting non-uniform

ionic charge distribution concentrates the field in a thin screening length near each electrode, while the bulk of the film is field-free [99, 100], with consequences for charge injection described by the two models below.

A.4.2 Electrodynamic Model

In the electrodynamic (ED) model, first formulated by deMello *et al.* [99, 100] to describe LECs, charge injection at the electrode–organic interface is the rate-limiting step for current flow. In this injection-limited regime, the current density is governed by thermionic emission over the Schottky barrier at the metal–polymer interface,

$$J \propto T^2 \exp\left(-\frac{\Phi_B}{k_B T}\right), \quad (\text{A.2})$$

where Φ_B is the effective barrier height, T the absolute temperature, and k_B the Boltzmann constant. The ionic double layer at the electrode interface reduces Φ_B by bending the polymer bands: positive ionic charge near ITO lowers the conduction-band edge there, while negative ionic charge near Ag raises the valence-band edge, both reducing the effective injection barrier. In the memristive context, conductance changes therefore arise from changes in the interface ion concentration: a higher concentration of positive ions near ITO means lower Φ_B , higher hole injection, and higher conductance, and the ions relax back toward equilibrium by thermal diffusion once the field is removed — the physical basis of the memory decay.

A.4.3 Electrochemical Doping Model: Ohmic Injection

The electrochemical doping (ECD) model, formulated by Pei *et al.* [98] and developed by subsequent authors [101, 137], instead takes charge injection to be ohmic (not rate-limiting), so the majority of the voltage drop occurs in the bulk and the current is limited by bulk conductivity. The defining feature is that ion accumulation *electrochemically dopes* the conjugated polymer: cations accumulate near the cathode, stabilising injected electrons (*n*-doping), and anions near the anode, stabilising injected holes (*p*-doping); the doped regions are far more conductive than the undoped polymer. The degree of doping — and hence the conductance — is set by the extent of ionic migration, which is governed by the history of applied voltage, so the conductance records the cumulative ionic-displacement history.

A.4.4 Resolution of the Two Models

The prolonged LEC-community debate over the relative validity of the ED and ECD models was resolved by van Reenen *et al.* [102], who showed both to be correct in different limits: the ED model when injection is strongly barrier-limited (low-conductance semiconductor, large electrode work-function mismatch), the ECD model when injection is ohmic (high-conductance semiconductor, small mismatch). A device may operate in the ED regime at early times, before significant ionic redistribution, and transition toward the ECD regime as the double layers build up. The working interpretation adopted for the present device — ED with the fast triflate-driven response, ECD with the slow Li^+ -driven response — is stated in section 2.5.

A.5 Historical Comparison at the Time of Publication

Table A.2 places the device alongside representative organic and hybrid memristive devices available at the time of the 2022 publication, on the axes of section 1.3.4: device structure, accessible conductance states, short- and long-term memory, STDP, and reversibility. It is a historical positioning, not a 2026 state-of-the-art claim.

Two combinations not represented in the comparison set at the time stand out. The first is the coexistence of short-term memory and long-term memory in a single 2-T device that is also fully reversible. Of the prior reports, Liu and co-workers [94] comes closest, with tunable short- to long-term transitions in a 2-T structure; the redox route, however, modifies the active layer irreversibly at each cycle, precluding practical cyclic operation [116]. The present device reaches comparable short-term memory/long-term memory functionality with full reversibility (conductance returns to G_0 after the reset protocol). The second is STDP in a fully reversible 2-T architecture: the earlier organic STDP demonstrations [90, 105, 115] were all 3-T with a gate supplying the timing rule, whereas here the rule emerges from the superposition of pre- and postsynaptic pulses at the single Ag terminal.

The remaining axes summarise the operating regime rather than a competitive claim. The ambient shelf lifetime (about 300 h unencapsulated) exceeds most comparable unencapsulated organic memristive devices by roughly an order of magnitude [119]; cycling endurance, not characterised here, is where mature inorganic platforms reach $> 10^{10}$ cycles [16]. The energy per synaptic event (about 50 nJ) is consistent with other organic ionic devices [77] but higher than the picojoule budgets of dedicated CMOS neuromorphic hardware [1, 27] and of the best inorganic memristors [68], reflecting the lower switching speed and larger area of this proof-of-concept

Table A.2: Historical comparison of key performance metrics of the present work with representative organic and hybrid memristive devices available at the time of the 2022 publication. “2-T” = two-terminal; “3-T” = three-terminal; “STM” = short-term memory; “LTM” = long-term memory; “STD P” = spike-timing dependent plasticity; “—” = not reported or not applicable.

Reference	Material	Structure	States	STM	LTM	STD P	Reversible
Zakhidov 2010 [97]	$[\text{Ru}(\text{bpy})_3]^{2+}/\text{PF}_6^-$	2-T	2	—	—	No	Partial
Lei 2014 [93]	PVA film	2-T	2	—	—	No	No
Liu 2016 [94]	$[\text{EV}(\text{ClO}_4)]/(\text{BTPA-F})$	2-T	cont.	Yes (tunable)	Yes	No	No
Gkoupidenis 2015 [105]	PEDOT:PSS (OECT)	3-T	cont.	Yes	No	Yes	Yes
Xu 2016 [114]	PEDOT:PSS (OECT)	3-T	cont.	Yes	Yes	No	Yes
Kim 2018 [90]	Elastic organic composite	3-T	cont.	Yes	No	Yes	Yes
Goswami 2017 [95]	Molecular switch	2-T	2	—	Yes	No	Yes
Seo 2019 [115]	P3HT organic FET	3-T	cont.	Yes	No	Yes	Yes
This work [6]	SY/Hybrane/LiOTf	2-T	cont.	Yes (10–15 s)	Yes (>45 s)	Yes	Full

device; area scaling toward the $\sim 100 \text{ nm}^2$ inorganic regime would close much of that gap.

Appendix B

Supporting Information for Chapter 3

This appendix records the methodological backbone of the degradation analysis of section 3.3. The bridge chapter retains the result — the window collapse and the ohmic drift — and the figures; the four controls that make that result defensible against the confounds peculiar to this dataset are stated in full here. The analysis is reproduced by `scripts/bridge_hybrane_peo_reproducibility.py`.

B.1 The Four Methodological Controls

The degradation analysis is built on a methodological backbone of four controls; each was verified to change the conclusion if dropped.

1. **Per-device weighting (mandatory).** The number of pixels measured per device collapses from sixteen to two over the campaign and is strongly anti-correlated with date (Spearman $\rho = -0.80$, $p \approx 1 \times 10^{-15}$, $n = 65$; fig. 3.2a inset). Pixel-pooled statistics are

therefore biased toward the early, densely measured devices. Every feature is reduced to *one value per device* — the median over that device’s freshest-day curves — before any test.

2. **Standard-protocol corpus only** ($n \approx 69$ devices, span 2021-02 to 2022-05): *Super Yellow/Hybrane/LiOTf* on silver, annealed at 75 °C with no second stage, at the lead mass ratios (*Hybrane* 0.3, *LiOTf* 0.09). This excludes the invasive experiments — above all the high-temperature anneal batches that partially recovered devices a year in — which would otherwise contaminate the feature-versus-date story.
3. **Sweep-amplitude matching, with each feature read where it is expressed.** A $0 \rightarrow +X \rightarrow 0$ loop excites the device more at larger X , so it conducts more even when read back at the same voltage; amplitude is confounded with both date and material (fig. 3.2b). The *window* features (on–off ratio, normalized area) only open up at high amplitude — at ~ 1.2 V they sit near unity for every device, early or late, and *cannot* show a collapse — so they are read in the ~ 3 V stratum, while the *conductivity* features are read in a tightly matched ~ 1.2 V band.
4. **Pre/post-inflection contrast and recovery-tail sensitivity.** Because the notes locate the behavioural inflection at NM_v026 (2021-04-22), the window collapse is tested as a step — early ($\leq$ Apr 2021) versus later (Mann–Whitney) — rather than as a whole-campaign correlation. Every conductivity trend is additionally reported with and without the deliberately-reverted recovery batches ($\geq 2022-03$).

The quantitative outcome of applying these controls — the window collapse at ~ 3 V and the matched-amplitude ohmic drift — is reported in table 3.2 and discussed in section 3.3.

Appendix C

Supporting Information for Chapter 4

This appendix gathers the analysis provenance, the confound audits, the fit methodology, and the per-composition-cell quantitative tables that underlie chapter 4. The main chapter retains the result and the figures; the material here records how the numbers were produced and lists the per-cell values behind the heatmaps and design-space plots.

C.1 Analysis Provenance and Aggregation

Data-quality control. Two screening layers are applied before any number is reported, both recorded so that the analysis is reproducible. Individual measurements that are erratic or non-physical are flagged and excluded. In addition, a per-device visual review of the recorded curves was carried out; the resulting verdicts — whether a curve is used as measured, cleaned by removing identified anomalous points, or discarded — together with the exact points retained, are stored in a curation registry and applied automatically by the analysis scripts, rather than being made

Table C.1: Raw-to-summary provenance for the three common measurements used in chapter 4. The same recorded trail underlies the figures of section 4.4 and the behavioural models of chapter 5.

Stage	Artifact	Role in the analysis
Raw measurement	DEVICES_LAB_DATA/.../D1_all.txt or split D1_I/V/T.txt	Keithley current, voltage, and time traces for each device day, measurement type, and crossbar pixel.
Fabrication metadata	DATABASE/UPDATED_DEVICES_LIBRARY.csv	Canonical device composition, electrode, spin speed, annealing, storage, measurement atmosphere, and operator metadata.
Extracted features	DEVICES_HYST_*, DEVICES_PULSES_*, DEVICES_DELAYTIME_*	Per-curve, per-pixel, and per-device feature tables generated by the archive extraction pipeline.
Exclusion and curation	FILTERED_DEVICES.csv; handouts/ch4_png_qa_curation.csv	Red-flag exclusion list plus per-curve visual QA verdicts; retained points are applied by script, not selected separately for each figure.
Recomputed descriptors	handouts/ch4_decay_fits.csv; handouts/ch4_pulse_descriptors.csv	Post-curation descriptors: $t_{1/2}$, identified τ, β where stable, pulse-growth exponent α , peak ratio, and turnover.
Cell summaries and figures	handouts/ch4_decay_by_cell.csv; handouts/ch4_pulses_by_cell.csv; scripts/ch4_comparative_figures.py	Cell medians, device counts, representative curves, heatmaps, swarm plots, and design-space figures.

ad hoc. The fitted timescale obtained directly from the instrument-side pipeline was found not to be robust to such anomalous points, so all timescales reported in chapter 4 are recomputed after this screening. Table C.1 records the full raw-to-summary trail underlying these screened descriptors.

The aggregation ladder. The aggregation is deliberately conservative. HYST raw points are first reduced to sweep-level features in DEVICES_HYST_CURVE_INFO.csv; broken or FILTERED rows are removed; de-

Table C.2: Figure-level provenance for the Chapter 4 figures whose detailed file paths are not repeated in the main captions. Script names are relative to the thesis repository; raw measurements reside in the experimental archive paths listed by the scripts.

Output	Reproduced by	Principal inputs and audit trail
Figure 4.1	<code>scripts/thickness_rpm_audit.py</code>	<code>DEVICES_PROFILOMETRY_STATS.csv</code> joined to the screened Chapter 4 descriptors; audit in <code>handouts/14_thickness_rpm_confound_audit.md</code> .
Figures 4.2 to 4.5 and 4.12	<code>scripts/ch4_comparative_figures.py</code>	<code>handouts/ch4_decay_fits.csv</code> , <code>handouts/ch4_pulse_descriptors.csv</code> , <code>handouts/ch4_decay_by_cell.csv</code> , and <code>handouts/ch4_pulses_by_cell.csv</code> ; curve curation in <code>handouts/ch4_png_qa_curation.csv</code> . Chemistry-landscape values from the screened descriptor tables; the protocol overlay uses the same-device v114 matched-amplitude comparison recorded in the Chapter 4 audit trail.
Figures 4.7 and 4.9	<code>scripts/ch4_comparative_figures.py</code>	ATR band extraction in <code>handouts/ch4_iratr_bands.csv</code> ; assessments in <code>handouts/16_iratr_assessment.md</code> and <code>handouts/17_xrd_assessment.md</code> .
Figure 4.6	<code>scripts/ch4_iratr.py</code> ; <code>scripts/ch4_xrd.py</code>	2025 <i>Super Yellow</i> /TMPE cation-series UV-Vis spectra; assessment in <code>handouts/18_uvvis_assessment.md</code> .
Figure 4.8	<code>scripts/ch4_uvvis.py</code>	0 V-DC impedance descriptors from the experimental database, joined to <code>handouts/ch4_decay_by_cell.csv</code> ; per-cell output in <code>handouts/ch4_eis_by_cell.csv</code> .
Figure 4.10	<code>scripts/ch4_eis.py</code>	Silver-versus-gold metrics in <code>handouts/ch4_electrode_by_cell.csv</code> ; audit in <code>handouts/19_electrode_a_u_assessment.md</code> .
Figure 4.11	<code>scripts/ch4_electrode.py</code>	On-state conductance from <code>DEVICES_HYST_PIXEL_INFO</code> ; gradient/factorial/bootstrap, EIS timescale bridge, reproducibility and heterogeneity-resource statistics in <code>handouts/ch4_gradient_stats.csv</code> , <code>ch4_cell_ci.csv</code> , <code>ch4_factorial.csv</code> , <code>ch4_percolation.csv</code> , <code>ch4_eis_timescale.csv</code> , <code>ch4_cycling.csv</code> , <code>ch4_heterogeneity_resource.csv</code> .
Figure C.2	<code>scripts/ch4_rigor.py</code>	

vice summaries are medians across the surviving sweeps and pixels; and composition-cell values are medians across devices. PULSES and DELAY-TIME follow the same curve/pixel/device/cell logic, but with protocol-specific descriptors: the pulse curves are summarised by α , peak ratio, and turnover, while the decay curves are summarised primarily by $t_{1/2}$, with τ and β retained only when the fit passes the identification thresholds in `scripts/ch4_dynamics_fits.py`. The representative curves in fig. 4.2 are selected by the figure script from devices near the relevant cell medians, so they are illustrative examples of the same summarized population rather than hand-picked evidence outside the pipeline.

Stretched-exponential identification. The depotentiation decays are fitted with the Kohlrausch form of eq. (4.1). The eight-to-thirteen-point decay curves often under-constrain β and τ jointly, so the fitted τ is reported only where the fit is well identified, and the model-free half-enhancement time $t_{1/2}$ is used as the primary timescale; where both are available they agree (table C.3). The stretch exponent is constrained to the physical Kohlrausch range $0 < \beta \leq 1$ of eq. (4.1): an unconstrained fit occasionally returns $\beta > 1$ (a *compressed* exponential, which does not correspond to a distribution of activation barriers), and such cells are instead pinned at $\beta = 1$, i.e. a near-undispersed simple-exponential relaxation, rather than reported as a spurious stretched fit. Fitted β values then lie predominantly in 0.6–0.9, i.e. genuinely stretched relaxation consistent with the heterogeneous distribution of activation barriers invoked in chapter 2, with the most ion-rich cells (high PEO, high salt) approaching $\beta = 1$ — a narrower barrier distribution, as expected when ionic transport dominates. The intra-device disorder implied by the stretch and the device-to-device heterogeneity $\sigma(\ln \tau)$ of section D.3 are related rather than independent: the relaxation-time spread implied by the median stretch is comparable in magnitude to the measured inter-device scatter (section C.11), so the same barrier-distribution width underlies both.

C.2 Film-Thickness and Fabrication Confound Audits

Film thickness is a controlled covariate. In most batches, the spin-coating speed was raised for higher-PEO cells to partly equalise thickness (spanning 1500 rpm–3500 rpm). The compensation was deliberate but incomplete: median film thickness still rises from ≈ 227 nm at PEO 0.3 to ≈ 298 nm at PEO 1.2, giving a Pearson $r \approx 0.68$ between PEO fraction and thickness (fig. 4.1). Four observations show that thickness is nonetheless not the operative variable:

1. The reported metrics are dimensionless ratios and timescales (on–off ratio, log–log slope, relaxation time), in which an absolute series-resistance or geometric effect of thickness cancels.
2. The activation voltage is flat (2.2 V–2.4 V, $r \approx -0.06$ against thickness) across a $2.6\times$ thickness range, so the switching threshold is an intrinsic ionic property rather than a bulk-field effect that would scale with film thickness.
3. Within the single lead cell (PEO 0.3/salt 0.09) a 38% device-to-device thickness spread (196 nm–271 nm) leaves the fading-memory time essentially unchanged ($t_{1/2} \approx 18$ s–22 s).
4. Most directly, once composition is held fixed the thickness correlation with $\log t_{1/2}$ vanishes (partial $r \approx +0.05$), while the PEO effect survives after controlling for thickness (partial $r \approx -0.42$); the same holds for the potentiation metrics.

Thickness covaries with PEO as a downstream consequence of polymer loading, but the dynamics track composition. The full join and statistics are reproduced by `scripts/thickness_rpm_audit.py` and `handouts/14_thickness_rpm_confound_audit.md`.

Other fabrication variables are held constant or crossed. The full per-device fabrication record was audited to confirm that the composition trends are not driven by some other processing variable. Across the silver composition grid the principal process variables are held *constant*: the annealing temperature (75 °C, no second stage), the electrode (100 nm silver, evaporated by the same operator), the semiconductor loading, the solvent, and — which matters for a hygroscopic polyether — storage *and* measurement under a glovebox atmosphere, so ambient humidity is controlled out. The variables that do change across the grid (fabrication batch and date, the operator who prepared the solutions, the semiconductor lot, and the device age at measurement) are *crossed* with composition rather than confounded with it: each composition cell is populated from several independent batches, and within any single batch every composition level was fabricated together and measured on the same day. The composition trends are accordingly reproduced *within* individual single-day batches — including one in which the spin speed was held fixed across all compositions, so that the trend appears with no thickness compensation at all. The full column-by-column audit is recorded in `handouts/15_fabrication_confound_audit.md` and reproduced by `scripts/fabrication_confound_audit.py`.

C.3 Per-Composition-Cell Summaries

Tables C.3 and C.4 list the per-cell values behind the heatmaps of section 4.4 and the design space of fig. 4.12. Each row is a composition cell; values are medians over the screened devices in the cell, with the device count n given so that the replicated Li^+ composition series is distinguished from the single-device Na^+/K^+ chemistry points.

Table C.3: Fading-memory (DELAYTIME) descriptors per composition cell, silver electrode, matched 4 V/2 V protocol. $t_{1/2}$: model-free half-enhancement time (s). R_{60} : retention at 60 s. τ, β : Kohlrausch parameters (median over the n_{id} devices with an identified fit). Em-dash: not identified.

Cation	PEO/salt	n	$t_{1/2}$ (s)	R_{60}	n_{id}	τ (s)	β
Li ⁺	0.15/0.18	1	22.4	0.27	0	—	—
Li ⁺	0.3/0.045	4	15.8	0.20	3	25.8	0.64
Li ⁺	0.3/0.09	3	19.2	0.28	2	23.9	0.83
Li ⁺	0.6/0.045	3	3.3	0.06	1	3.0	0.97
Li ⁺	0.6/0.09	3	3.5	0.02	1	3.0	0.61
Li ⁺	0.6/0.18	2	5.5	0.04	2	4.6	0.61
Li ⁺	1.2/0.045	3	2.7	0.13	0	—	—
Li ⁺	1.2/0.09	3	6.0	0.09	2	6.7	0.91
Li ⁺	1.2/0.18	3	9.0	0.13	3	10.0	0.87
Na ⁺	0.3/0.09	1	24.0	0.42	0	—	—
K ⁺	0.3/0.09	1	8.0	0.22	0	—	—

C.4 Realised Device Palette

The quantitative analyses of chapter 4 are deliberately confined to the replicated silver PEO/LiOTf composition series. It is nonetheless useful to see the full set of behaviours the device family actually realises, because it is this realised spread — not any single optimised cell — that the temporal chapter treats as a computational resource. Figure C.1 places every silver device that carries both a measured switching window and a measured fading timescale ($n = 54$, all hosts, cations and anions; FILTERED-flagged and broken pixels removed; device medians) in a single window-versus-retention map. The map is descriptive only: the chemistry axes are illustrative (section 4.5) and no comparative claim is drawn from it. Two features are worth noting. The populated region spans more than an order of magnitude in both axes, so the substrate offers a broad palette of (window, τ) operating points. And the lead composition is not an isolated point but a cluster of replicate devices spread over $\tau \approx 0.4 \text{ s} - 20 \text{ s}$, the visual counterpart of the within-cell device-to-device

Table C.4: Potentiation (PULSES) descriptors per composition cell, silver electrode, 3 V/1.5 V protocol. α : log–log growth exponent. Peak: peak conductance ratio ($\times$). N_{peak} : pulse count at peak. Turnover: fraction of devices whose response turns over within $N \leq 1000$.

Cation	PEO/salt	n	α	Peak ($\times$)	N_{peak}	Turnover
Li ⁺	0.15/0.045	1	0.06	1.4	100	100%
Li ⁺	0.3/0.045	4	1.11	482.6	300	100%
Li ⁺	0.3/0.09	4	0.90	149.9	1000	0%
Li ⁺	0.3/0.18	2	0.18	3.8	1000	0%
Li ⁺	0.6/0.045	3	0.57	14.8	100	100%
Li ⁺	0.6/0.09	3	0.83	202.3	300	100%
Li ⁺	0.6/0.18	3	0.63	76.9	1000	0%
Li ⁺	1.2/0.045	3	0.33	4.4	100	100%
Li ⁺	1.2/0.09	2	0.49	42.6	1000	0%
Li ⁺	1.2/0.18	3	0.37	8.0	1000	0%
Na ⁺	0.3/0.09	1	0.27	10.0	1000	0%
K ⁺	0.3/0.09	1	0.86	103.2	300	100%

scatter quantified in section D.3.

C.5 Impedance: Equivalent Circuit and Per-Cell Values

Equivalent-circuit model. Beyond the model-free Nyquist-apex resistance used in section 4.7, each spectrum was also summarised by the bulk ionic resistance R_{ion} of an equivalent circuit adapted from the polymer light-emitting electrochemical cell analysis of Munar *et al.* [149], in which a series cable term is placed in parallel with (i) the semiconductor electronic branch (effectively open at 0 V), (ii) a geometric constant-phase element, and (iii) an ionic branch of two interfacial double layers in series with R_{ion} . Two cautions apply. First, the archive holds two automated fits of this circuit; the freely-fitted one (with per-parameter error estimates) tracks composition appreciably better than a more tightly bounded variant, and

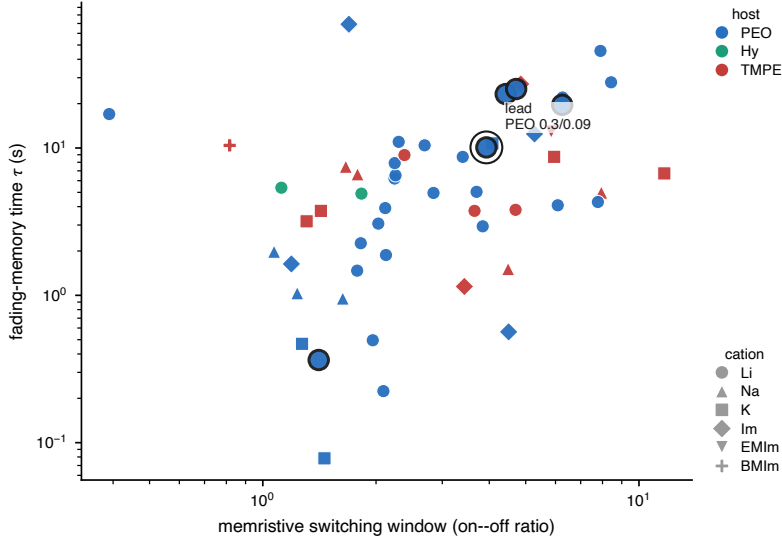


Figure C.1: Measured window-versus-retention palette of the silver device library ($n = 54$ devices with both a clean hysteresis window and a fading-memory fit, all chemistries). Colour encodes the ion-conductor host, marker encodes the salt cation; the lead PEO 0.3/0.09 cell is ringed. The cloud spans over a decade in each axis, and the lead-cell replicates alone span $\tau \approx 0.4\text{s} - 20\text{s}$. Descriptive overview only; the quantitative claims of chapter 4 rest on the replicated composition series, not on this map.

only the former is used. Second, per-spectrum circuit fits are genuinely ill-conditioned — an independent refit confirms that several do not converge robustly — so R_{ion} is reported only as corroboration of the model-free quantity, never as a precise number. Both observables nonetheless fall steeply and monotonically with PEO fraction (table C.5): the model-free $Z_{\text{real}}^{\text{apex}}$ at rank correlation -0.83 against PEO on cell medians, the fitted R_{ion} at -0.84 .

Table C.5: Equilibrium (0 V-DC) impedance per composition cell, silver *Super Yellow*/PEO/LiOTf. $Z_{\text{real}}^{\text{apex}}$: model-free real impedance at the Nyquist apex. R_{ion} : freely-fitted bulk ionic resistance (corroborative only). $t_{1/2}$: independently measured fading-memory time, repeated from table C.3 for the co-decline of fig. 4.10b. Em-dash: no matched decay.

PEO	salt	$Z_{\text{real}}^{\text{apex}} (\Omega)$	$R_{\text{ion}} (\Omega)$	$t_{1/2} (\text{s})$
0.15	0.045	2.5×10^8	2.3×10^7	—
0.15	0.09	1.7×10^8	8.7×10^6	—
0.15	0.18	1.0×10^8	8.4×10^6	22.4
0.3	0.045	5.3×10^7	4.0×10^5	15.8
0.3	0.09	1.3×10^8	1.0×10^6	19.2
0.3	0.18	9.9×10^7	4.9×10^5	—
0.6	0.045	5.3×10^6	1.4×10^5	3.3
0.6	0.09	2.6×10^7	4.6×10^4	3.5
0.6	0.18	1.9×10^7	8.9×10^4	5.5
1.2	0.045	4.1×10^5	1.9×10^2	2.7
1.2	0.09	5.6×10^5	1.1×10^4	6.0
1.2	0.18	3.8×10^7	1.5×10^5	9.0

C.6 Electro-Contrast Per-Cell Values

Table C.6 lists the per-cell behavioural metrics behind the silver-versus-gold contrast of section 4.8, including the host-free and salt-free negative controls that confirm both the ion-transport polymer and the dissolved salt are required for the memristive response.

C.7 Significance of the Composition Gradient

The composition trends of section 4.4 are reported in the main text as cell medians. This section attaches a significance test and a non-parametric uncertainty to them, on the replicated Ag/Li *Super Yellow*/PEO/LiOTf series. For each descriptor we compute the Spearman rank correlation with the PEO fraction across individual devices, with a two-sided permutation

Table C.6: Per-cell behavioural metrics by top electrode (active silver vs inert gold), at the lead composition 0.3/0.09. On-off and normalised loop area from HYST; V_{act} activation voltage (V); peak ratio and α from PULSES; $t_{1/2}$ from DELAYTIME (s). Em-dash: no clean decay/fit at that cell. “none” rows are host-free or salt-free negative controls.

Chemistry	Elec.	on-off	area	V_{act}	peak	α	$t_{1/2}$
PEO/Li/OTf	Ag	3.25	0.33	2.38	123.5	0.89	19.2
PEO/Li/OTf	Au	2.00	0.12	2.74	29.4	0.45	87.2
TMPE/Li/OTf	Ag	3.82	0.32	4.18	53.7	0.33	3.7
TMPE/Li/OTf	Au	1.29	0.07	3.61	1.9	0.17	—
TMPE/Na/OTf	Ag	5.54	0.30	3.26	497.9	1.23	3.9
TMPE/Na/OTf	Au	1.47	0.11	6.39	—	—	—
TMPE/K/OTf	Ag	7.01	0.41	3.10	237.5	1.19	5.2
TMPE/K/OTf	Au	1.43	0.06	5.86	—	—	—
TMPE/Na/TFSI	Ag	1.76	0.10	2.38	25.1	0.46	—
Hy/Li/OTf	Ag	1.41	0.06	1.61	25.2	0.39	—
none/Li/OTf	Ag	0.99	0.00	1.07	—	—	—
TMPE/none	Ag	2.95	0.20	3.18	73.0	0.96	—
none/none	Au	1.03	0.01	1.29	2.8	—	—

p -value (20 000 relabellings; table C.7). The fading-memory time $t_{1/2}$ — the descriptor that matters for chapter 5 — carries a clearly significant PEO gradient ($\rho = -0.52$, $p = 0.009$, $n = 25$). The remaining descriptors fall in the same direction but are only marginal at this sample size ($p \approx 0.06$ – 0.07): the per-device gradient is genuinely weaker and noisier than the cell medians alone suggest, and we report it as suggestive rather than established. This is the honest counterpart to the median heatmaps — the timescale trend is statistically secure, the window and potentiation trends are indicative.

A non-parametric bootstrap (20 000 device-level resamples per cell) gives a 95 % confidence interval on every cell median; the full set is in `handouts/ch4_cell_ci.csv`. The intervals are tight where the cell is well replicated and consistent (e.g. PEO 0.3/salt 0.09: $t_{1/2} = 19.2 [18.6, 22.0]$ s) and honestly wide where a cell mixes a long- and a short-retention device (e.g. PEO 0.3/salt 0.045: $15.8 [3.3, 36.7]$ s), which is the right-skew already

Table C.7: Per-device Spearman rank correlation of each descriptor with the PEO mass fraction (Ag/Li composition series), with a two-sided permutation p -value. Only the fading-memory time clears $p < 0.05$.

Descriptor	n	Spearman ρ (vs PEO)	permutation p
Fading-memory time $t_{1/2}$	25	−0.52	0.009
Growth exponent α	28	−0.36	0.065
Peak conductance ratio	28	−0.34	0.073
On–off ratio	47	−0.28	0.060

Table C.8: Two-way decomposition of each log-descriptor into $\log PEO$, $\log$ salt, and their interaction (Ag/Li series; centred predictors; permutation p). PEO is the only significant axis, on the fading-memory time, and no interaction term is significant — the two axes act near-orthogonally.

Descriptor	$b_{PEO}(p)$	$b_{\text{salt}}(p)$	$b_{\text{int}}(p)$	R^2	n
$\log t_{1/2}$	−0.92 (0.001)	+0.38 (0.26)	+0.61 (0.22)	0.47	25
α	−0.15 (0.14)	−0.10 (0.38)	+0.19 (0.30)	0.19	28
$\log(\text{peak})$	−0.89 (0.14)	−0.13 (0.84)	+1.02 (0.33)	0.15	28

flagged in section 4.4 made quantitative.

Two composition axes act near-orthogonally. To test the main-text reading that PEO sets the strength and memory while salt sets the saturation, each log-descriptor was regressed on $\log PEO$, $\log$ salt, and their (centred) interaction, with permutation p -values per term (table C.8). The fading-memory time is governed by PEO alone ($b_{PEO} = -0.92$, $p = 0.001$) with no significant salt or interaction term, so the two axes do not interact detectably at this power: a clean, near-orthogonal control. The potentiation descriptors are noisier ($R^2 \leq 0.2$) and no term is individually significant, consistent with the salt axis acting on *saturation* — the turnover fraction and N_{peak} of table C.4, a categorical effect not captured by a log-linear slope — rather than on the growth magnitude.

C.8 Absolute Conductance Tests the Dilution Picture

The working mechanism of section 4.9 initially read the window contraction at high PEO as a *dilution* of the *Super Yellow* electronic-percolation network. That picture makes a direct, falsifiable prediction on a quantity the peak-normalised dynamics discard: if added ion-transport polymer simply interrupted the electronic network, the *absolute* on-state conductance should *fall* with PEO. It does the opposite. The on-state conductance (HYST conductance at the sweep maximum, device medians, $n = 47$ Ag series devices) *rises* steeply and monotonically with PEO — Spearman $\rho = +0.65$ ($p < 10^{-4}$), a log-log slope of $\approx +3.6$, and more than three orders of magnitude from PEO 0.15 to 1.2 (fig. C.2). Because film thickness rises with PEO (section C.2), a geometric effect would push the conductance the other way, so the rise is if anything understated.

The dilution reading is therefore wrong in its literal form, and the corrected mechanism is the one carried in the revised section 4.9: the ion-rich phase contributes a parallel, weakly switchable conduction path that *raises the conductance floor*. Since the on-off ratio nonetheless falls with PEO, the off-state conductance must rise even faster than the on-state, so the window contracts not because the network is diluted but because its baseline is lifted toward the on-level. The fading-memory and potentiation trends are unaffected by this correction — they were always read in terms of ionic relaxation, not electronic percolation.

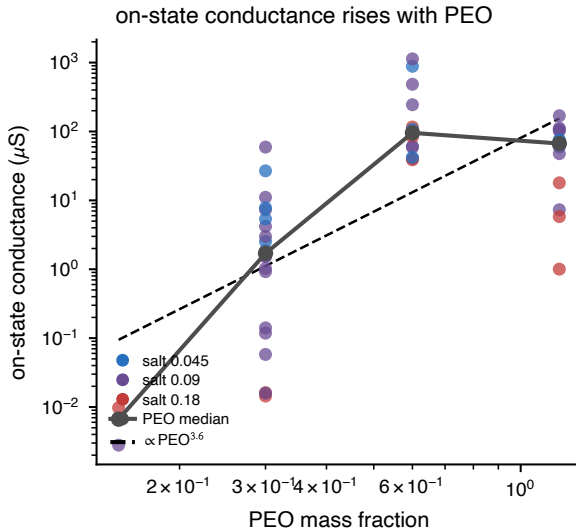


Figure C.2: Absolute on-state conductance (HYST, conductance at the sweep maximum; device medians, Ag series, $n = 47$, coloured by salt) against PEO fraction, log-log. The conductance rises steeply with PEO ($\rho = +0.65$, $p < 10^{-4}$, slope $\approx +3.6$, dashed) — the opposite of the prediction of an electronic-percolation *dilution*, and the basis for the rising-floor reading of the window contraction in section 4.9.

C.9 Reconciling the Impedance and Fading-Memory Timescales

Section 4.7 notes that the EIS resistance *scale* tracks the fading-memory time across composition, while the small-signal apex *frequency* does not: the Nyquist-apex RC time is $\tau_{RC} = 1/(2\pi f_{\text{apex}}) \approx 0.12\text{ s}$ (Ag series, 0 V DC, $n = 26$), more than an order of magnitude shorter than the seconds-long $t_{1/2}$. This section shows the gap is not a contradiction but a small-signal/large-signal capacitance ratio, so the two land at the same *magnitude*.

The apex RC is set by the geometric capacitance $C_{\text{geo}} = 1/(2\pi f_{\text{apex}} Z_{\text{real}}^{\text{apex}})$. The forgetting of a *written* state, by contrast, is the relaxation of a dis-

placed ionic profile and is governed by the much larger chemical/double-layer capacitance C_{chem} of the ion-rich film. Measuring C_{chem} independently from the low-frequency branch of the same spectra ($C_{\text{lf}} = -1/(2\pi f Z_{\text{imag}})$ at the lowest measured frequency) gives a capacitance enhancement $C_{\text{lf}}/C_{\text{geo}} \approx 33$ (median, $n = 26$) — and since the lowest measured frequency does not reach the full DC plateau, this is a *lower bound*. The predicted large-signal relaxation is then $\tau_{\text{large}} \approx \tau_{RC} (C_{\text{chem}}/C_{\text{geo}}) \gtrsim 3.9\text{ s}$, squarely inside the observed $t_{1/2}$ range of 2 s–22 s and within a factor of two of the series median (6.6 s). The independently measured capacitance enhancement thus accounts for the timescale separation quantitatively, upgrading the impedance corroboration of section 4.7 from same-direction to same-magnitude.

C.10 Within-Device Reproducibility

A physical reservoir requires elements that are diverse *across* the bank but *individually reproducible*. The chapter documents the diversity at length; this section quantifies the reproducibility. The fading-memory protocol measured a single junction per device, so a population-level repeat-*decay* reproducibility cannot be estimated — only one series device carries two junctions, and there the model-free $t_{1/2}$ agrees to $\sigma(\ln t_{1/2}) \approx 0.19$; broader repeat-decay statistics are a stated gap (section 4.10). The steady-state switching window, however, is recorded over several sweeps per junction (median ≈ 5), giving a genuine population-level within-device reproducibility: the sweep-to-sweep scatter of the on-off ratio is $\sigma(\ln R_{\text{on/off}}) \approx 0.14$ across 165 series junctions, i.e. each junction reproduces its switching window to $\approx 15\%$. This within-device scatter is far smaller than the across-composition span the bank exploits ($\Delta \ln t_{1/2} \approx 2.1$, section C.11), so a device’s behaviour is a stable signature relative to the diversity the substrate spans — the premise chapter 5 relies on.

C.11 Heterogeneity as a Computational Resource

The chapter treats the spread of fading-memory times as a resource rather than noise. This section makes the claim a number. The cell-median $t_{1/2}$ spans $\Delta \ln t_{1/2} \approx 2.1$ across the composition series (from ≈ 2.7 s to ≈ 22 s), while the device-to-device scatter *within* a cell is $\sigma_{\text{within}}(\ln t_{1/2}) \approx 0.29$ on the robust model-free timescale, rising to ≈ 0.85 on the fit-noisier single-exponential τ cited in section D.3. Taking a two- σ separation as the criterion for two timescales to be distinguishable, the bank provides $N_{\text{eff}} = 1 + \Delta \ln t_{1/2} / (2\sigma_{\text{within}}) \approx 2.2$ (using the conservative $\sigma = 0.85$) to ≈ 4.6 (using the model-free $\sigma = 0.29$) resolvably distinct memory bands selectable by composition. The resource is therefore modest but real: a handful of cleanly separated timescales, not a continuum, which is why chapter 5 assembles a *bank* of compositions rather than tuning a single optimum.

Single-device and inter-device disorder are linked. The stretched exponent provides an independent read on dispersion: the median single-device stretch corresponds to a barrier-distribution width $1 - \beta \approx 0.36$, of the same order as the within-cell device-to-device $\sigma(\ln t_{1/2})$. The disorder seen *within* one device’s relaxation and the spread *between* sibling devices are thus comparable in magnitude, consistent with both arising from the same heterogeneous distribution of local activation barriers rather than from independent causes.

Appendix D

Supporting Information for Chapter 5

This appendix documents how the behavioural-model parameter cards of section 5.2 are sourced from the measured archive, and lists the simulation and read-out settings used across the three evidence tiers. Together with the per-cell tables of section C.3, it makes the in-silico results reproducible without consulting the analysis scripts (`scripts/ch5_figures.py`, `scripts/ch5_physio_context.py`, `scripts/ch5_onset.py`, `scripts/ch5_deployment.py`, `scripts/ch5_noneeg.py`).

D.1 Parameter-Card Sourcing

Each composition cell is reduced to a parameter card $(\tau_c, \beta_c, \alpha_c, \text{peak}_c)$ that drives one reservoir node type. The values are taken directly from the comparative fits, with no free re-tuning:

- the fading-memory law $\lambda_c(\Delta t) = \exp[-(\Delta t/\tau_c)^{\beta_c}]$ uses the per-cell (τ_c, β_c) of table C.3; for the few cells without an identified stretched

fit, a single exponential reproducing the model-free half-life is used instead ($\tau_c = t_{1/2}/\ln 2$, $\beta_c = 1$);

- the write nonlinearity uses the per-cell growth exponent α_c and peak enhancement of table C.4;
- the device-to-device spread injected into each bank is the within-cell scatter of those same tables.

The heterogeneous banks of sections 5.4 and 5.6 span the replicated PEO/LiOTf composition cells of table C.3 (τ from ≈ 3 to ≈ 26 s); the homogeneous control banks are jittered copies of the lead cell PEO 0.3/0.09 alone.

D.2 Simulation and Read-out Settings

Table D.1 collects the settings used throughout chapter 5. All read-outs are a single linear ridge regression; no nonlinear decoder is used, so any demonstrated computation is attributable to the device nodes rather than the read-out. The read-out ridge penalty ($\lambda = 10^{-3}$), the causal label-smoothing horizon (15 s), the bank size N , and the reservoir step Δt are fixed *a priori* and held identical across all banks and all held-out subjects; none is selected on the held-out subject, so the leave-one-subject-out protocol carries no hyperparameter leakage.

D.3 Measured Device-to-Device Scatter

The robustness analysis of section 5.9 (fig. 5.8) injects device-to-device variation as a multiplicative log-normal *jitter* on each node’s timescale, $\tau_i = \tau e^\varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \sigma^2)$; the jitter magnitude σ is, by construction,

Table D.1: Simulation and read-out settings across the three evidence tiers of chapter 5. All banks are spatial banks of independent (non-recurrent) leaky nodes driven by randomly-masked input; all read-outs are linear ridge regression.

Setting	Value
Node model	$x_n^{(i)} = \lambda_i x_{n-1}^{(i)} + [\max(w_i u_n, 0)]^{\alpha_i}$ (section 5.2)
Read-out	single linear ridge regression
Bank size N	24 (benchmarks); 48 (physiology/affect)
Seeds	10 random input/bank seeds; means $\pm$ SD reported
Significance tests	paired Wilcoxon signed-rank over seeds (or subjects)
Memory capacity	MC_k summed over lags, random uniform input [151]
NARMA	NARMA-10 system identification, NRMSE [152]
Information-processing capacity	degree-1 (linear) + degree-2 (nonlinear), bounded by N [80]
Affective corpus	WESAD chest signals, leave-one-subject-out over 15 subjects [157]
Streaming cadence	$\Delta t = 1$ s; reconstruction delays 1, 3, 8, 20, 45 s
Affect scoring	class-balanced macro-F1

the standard deviation of $\ln \tau$ across nominally identical devices. That quantity is measured directly here, so the dotted “measured scatter” line in fig. 5.8 is data-derived rather than assumed.

The estimate uses the fading-memory time constants of every silver PEO/LiOTf device ($n = 31$). A naive spread taken across the whole SY, PEO, LiTr group conflates the *deliberate* PEO $\times$ salt tuning sweep of chapter 4 with random scatter and over-states it ($\sigma \approx 1.59$); the meaningful quantity is the spread *within* each fixed PEO/salt cell, $\sigma(\ln \tau) \approx 0.85$ (table D.2). This residual is genuine device-to-device variation: it is essentially unchanged after regressing out spin-coating speed and annealing (≈ 0.91), and within-cell residuals show no monotonic dependence on film thickness (Spearman $\rho \approx 0.14$, $n = 31$, complete profilometry) or on device age ($\rho \approx 0.05$). The measured value is roughly seven times the figure previously annotated on the robustness plot, and the heterogeneous bank retains its memory-capacity advantage at it (fig. 5.8), so the realised material diversity is, if anything, larger than the simulations assume. The

Table D.2: Variance decomposition of $\ln \tau$ for the silver PEO/LiOTf series ($n = 31$ devices). The within-cell value is the measured jitter σ ; the group-level value is reported only to show the confound it would introduce.

Estimate of $\sigma(\ln \tau)$	Value
Across SY, PEO, LiTr group (conflates designed sweep)	1.59
Within fixed PEO/salt cell (= measured jitter)	0.85
after regressing out spin speed + anneal	0.91
Within-cell residual vs. film thickness (Spearman ρ)	0.14
Within-cell residual vs. device age (Spearman ρ)	0.05
Value previously annotated on fig. 5.8	0.12

audit is reproduced by `scripts/ch5_scatter_audit.py`.

D.4 Continuous-Monitoring Stress Test

The deployment-robustness result of section 5.8 (`scripts/ch5_onset.py`) is a continuous binary stress detector: at every $\Delta t = 1$ s step the read-out predicts stress (WESAD label 2) versus not-stress (baseline and amusement pooled), leave-one-subject-out, with the same single linear ridge read-out and 15-second causal label smoothing used in the streaming task. Four matched banks of $N = 24$ nodes are compared on identical input masks: the instantaneous input (no memory), a memoryless bank (leak set to zero, dimensionality only), the homogeneous lead-only bank, and the heterogeneous composition bank.

Sensor corruption is modelled by adding, to the per-subject scaled streams (channels in $[0, 1]$), independent Gaussian noise of standard deviation σ at every sample, plus sparse motion-artefact bursts (short windows of amplified noise at a 1% per-sample rate), clipped back to $[0, 1]$; the same realisation is applied to the training and held-out streams so the comparison isolates each architecture’s intrinsic noise tolerance. Robustness curves are seed-averaged over three bank/noise seeds; the per-subject

significance test at $\sigma = 0.4$ is a paired Wilcoxon signed-rank over the fifteen subjects. Reported metrics are the binary macro-F1, the pooled false-alarm rate (fraction of not-stress steps called stress), and — for completeness, though they do not separate the banks — the transition-window macro-F1 (steps within ± 20 s of a label change) and the onset detection latency (time to the first ≥ 5 s sustained stress call after each onset).

The cross-corpus replication (table 5.5, `scripts/ch5_noneeg.py`) applies the *same* detector and noise model to the PhysioNet Non-EEG corpus [160] (open access; obtainable with `wfdb.dl_database('noneeg', 'data/noneeg')`). The 8 Hz electrodermal and temperature record and the 1 Hz heart-rate record are resampled onto a common timeline and per-subject scaled exactly as the WESAD streams; the phase annotations map cognitive and emotional stress to the stress class and relaxation to not-stress, with the physical-stress (exercise) phase excluded.

Real-motion corruption. To confirm the robustness is not an artefact of the Gaussian noise model, the corruption is also driven by the *measured* chest accelerometer (`add_motion_noise`, `load_motion` in the scripts). A per-step motion index is the magnitude of the gravity-removed acceleration, RMS-smoothed over 1 s and robustly scaled to $[0, 1]$ per subject; the injected noise standard deviation at each step is $\text{scale} \times (0.05 + \text{motion})$, with an amplified artefact burst where the index exceeds 0.6, so corruption lands where the wearer actually moves. Across a corruption scale of 0 to 0.8 the instantaneous detector falls from macro-F1 0.914 to 0.898 while the memory bank holds near 0.92; at the strongest gating the memory-minus-instantaneous advantage is +0.032 (paired Wilcoxon $p = 0.04$, $r_{\text{rb}} = +0.6$) and the single- τ moving average matches it (+0.036). The advantage is smaller than under the synthetic injection because the chest strap is mechanically stable and the protocol largely seated; the wrist site

(table 5.6) is where motion artefact dominates.

D.5 Control Experiments

Three controls underpin the claims of chapter 5 and are tabulated here. They are reproduced by `scripts/ch5_reservoir.py` (the generic-ESN and coupling controls) and `scripts/ch5_onset.py` (the classical-filter control); each significance test reports, alongside the paired Wilcoxon p , the matched-pairs rank-biserial effect size $r_{\text{rb}} \in [-1, 1]$, since the Wilcoxon p floors at 2×10^{-3} for the ten-seed comparisons.

Classical-filter control. Table D.3 is the control behind the deployment-robustness reframing of section 5.8: an ordinary exponential moving average of the raw channels, fed to the same linear read-out with no reservoir, recovers the noise robustness, so the robustness is the generic property of a fading-memory low-pass rather than anything specific to the device or its nonlinearity. The device’s contribution is to *be* that filter, passively and at the budget of section D.6. The cross-corpus and wrist robustness results (tables 5.5 and 5.6) share this character: there too the gain is the low-pass advantage of temporal memory over an instantaneous read.

Generic-ESN control (matching thesis). Table D.4 asks whether the *particular* measured timescales matter or only their spread, by replacing the device bank with an echo-state network of the same size whose leak times are drawn at random. A random bank over the same range ties the device bank, and a broader random range exceeds it: the resource is the timescale *spread*, which the composition grid supplies intrinsically, and extending the range (the drive-amplitude variable of chapter 4) is where further headroom lies.

Table D.3: Classical-filter control for the continuous stress monitor (binary macro-F1, leave-one-subject-out, σ the injected sensor noise of fig. 5.7). A single-time-constant moving average of the raw channels recovers the robustness an instantaneous read loses, and a linear moving-average bank spread over the device timescale range adds little; the device reservoir’s compressive nonlinearity adds nothing further on this single-timescale task. Paired at $\sigma = 0.4$ over the fifteen subjects: EMA(single)–instantaneous = +0.091 ($p < 10^{-4}$, $r_{\text{rb}} = +1.0$); EMA(bank)–EMA(single) = +0.005 (*ns*); reservoir–EMA(bank) = −0.008 (*ns*).

Detector (linear read-out on ...)	clean	$\sigma = 0.4$
Instantaneous channels (no memory)	0.914	0.840
EMA, single τ (lead cell)	0.937	0.927
EMA bank, τ spread (linear reservoir)	0.926	0.930
Device reservoir (nonlinear, heterogeneous)	0.920	0.916

Table D.4: Total linear memory capacity ($N = 24$, ten seeds, means $\pm$ SD) of the measured device heterogeneous bank versus random echo-state networks of equal size. Paired against the device bank: same-range ESN +0.28 ($p = 0.05$, $r_{\text{rb}} = -0.7$, marginally favouring the random bank); broad-range ESN +1.14 ($p = 2 \times 10^{-3}$, $r_{\text{rb}} = -1.0$).

Reservoir ($N = 24$)	total MC
Measured device heterogeneous bank	6.4 ± 0.3
Random ESN, same τ range (3–26 s)	6.7 ± 0.3
Random ESN, broad τ range (0.5–60 s)	7.6 ± 0.3

Coupling-sensitivity control. Table D.5 bounds the composition assumption of section 5.2: each node’s retention time is rescaled with its own write-nonlinearity exponent by a factor $e^{\kappa z}$ (z the standardised exponent), modelling the drive–retention coupling the separate-amplitude measurement cannot exclude. The heterogeneous bank stays above the homogeneous one for every coupling strength swept, so the heterogeneity conclusion does not rest on the assumption being exact.

Chance baselines. Under the streaming three-class protocol of section 5.7, a majority-class predictor scores macro-F1 0.23 and a stratified-

Table D.5: Memory capacity of the heterogeneous and homogeneous banks under an injected drive–retention coupling of strength κ ($N = 24$, ten seeds; $\kappa = 0$ is the nominal composed model). The heterogeneous advantage holds across the whole range (paired Wilcoxon $p = 2 \times 10^{-3}$, $r_{\text{rb}} = +1.0$ at every κ).

κ	het MC	hom MC	ratio
−0.50	6.27	5.50	1.14
−0.25	6.22	4.65	1.34
0.00	6.44	4.20	1.53
+0.25	6.60	4.64	1.42
+0.50	6.78	5.43	1.25

random predictor 0.33 (leave-one-subject-out, per step), so the reservoir’s 0.758 is well above chance. Significance of the bank comparisons is assessed across the fifteen independent held-out subjects, so within-subject temporal autocorrelation does not inflate it.

Drive-boosted tonic-node extension. The minutes-scale tonic gap of fig. 5.4 is probed by extending the heterogeneous bank with drive-boosted nodes at $\tau = 120$ and 300 s — a labelled extrapolation of the drive-amplitude variable, copying the lead cell’s write nonlinearity and stretch and overriding only τ (`tonic_cards` in `scripts/ch5_model.py`). On memory capacity (DT = 5 s, k to 40) the extension nearly doubles the long-lag tail beyond 75 s ($\text{MC}_{k>15}$ from 0.73 to 1.37, $p = 2 \times 10^{-3}$) and raises total capacity from 6.4 to 7.0. On the physiological-context reconstruction, with the target delays extended to 90 and 120 s so the genuinely tonic context is present, the slow-band held-out R^2 rises from 0.672 to 0.690 (+0.019, $p = 2 \times 10^{-3}$, $r_{\text{rb}} = +1.0$ over ten seeds) and the overall R^2 by +0.009 (`tonic_extension` in `scripts/ch5_physio_context.py`). The extension is not part of the measured composition banks anywhere else in the chapter.

Table D.6: Deployment envelope for an N -node device reservoir read out by a linear classifier of K classes, from the measured device constants of chapter 2, contrasted with a typical gradient-trained deep model. Parameter counts are exact; energy figures are order-of-magnitude estimates (see text).

Quantity	Device reservoir (this work)	Deep model (typical)
Trainable parameters	$(N+1)K$: 75 ($N=24$) / 147 ($N=48$), $K=3$	10^4 – 10^6
Reservoir dynamics	physical, untrained (0 parameters)	—
Training method	one closed-form ridge solve	gradient descent / backpropagation
Energy per update	$\approx N \times 50$ nJ (≈ 1.2 μ J at $N=24$)	—
Average power (1 Hz, always-on)	≈ 1.2 μ W– 2.4 μ W ($N=24$ – 48)	—
Read-out compute	NK MACs/step (< 1 nJ)	—
Operating regime	sub-kHz (affect is sub-Hz)	—

D.6 Deployment Envelope

Table D.6 collects the operating budget quoted in section 5.9 (`scripts/ch5_deployment.py`). The trainable-parameter count is exact: the reservoir dynamics are fixed physical devices, and the read-out is a single $(N+1) \times K$ ridge-regression weight matrix. The energy figures are order-of-magnitude estimates built from the *measured* proof-of-concept device of chapter 2 (chapter A): an energy per write event of ≈ 50 nJ (a 1 V, 0.1 s pulse), equivalently ≈ 6 fJ μm^{-2} over the 0.0825 cm^2 junction (the two agree to within 1%, 6 fJ $\mu\text{m}^{-2} \times 0.0825$ $\text{cm}^2 \approx 49.5$ nJ). The estimate conservatively charges each node one full write event per one-second update; the sub-threshold, state-preserving read at 0.5 V is cheaper, and these are unoptimised large-area figures, so the absolute energy should be read as an upper bound that integration would lower. The read-out compute is NK multiply–accumulates per step at an assumed 1 pJ per operation.

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